Artificial Intelligence in the Knowledge Economy*

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Abstract

Artificial Intelligence (AI) can transform the knowledge economy by automating non-codifiable work. To analyze this transformation, we incorporate AI into an economy where humans form hierarchical organizations: Less knowledgeable individuals become "workers" doing routine work, while others become "solvers" handling exceptions. We model AI as a technology that converts computational resources into "AI agents" that operate autonomously (as co-workers and solvers/co-pilots) or non-autonomously (solely as co-pilots). Autonomous AI primarily benefits the most knowledgeable individuals; non-autonomous AI benefits the least knowledgeable. However, output is higher with autonomous AI. These findings reconcile contradictory empirical evidence and reveal tradeoffs when regulating AI autonomy.

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1 Introduction

Recent developments in Artificial Intelligence (AI) are driving a new wave of automation, enabling machines to perform sophisticated knowledge work such as coding, research, and problem-solving. While the potential of this technology to reshape the landscape of work is undeniable, its precise implications remain the subject of growing controversy (Brynjolfsson, 2022; Johnson and Acemoglu, 2023; Acemoglu, 2024; Autor, 2024; Korinek and Suh, 2024). This controversy arises from two factors. First, it is unclear whether lessons from previous waves of automation can inform our understanding of AI's effects (Muro et al., 2019; Agrawal et al., 2019). Second, because AI is still in its infancy and individuals and firms are experimenting with its use (McElheran et al., 2023; Humlum and Vestergaard, 2024), current empirical evidence cannot yet fully capture its equilibrium effects.

In this paper, we develop a framework for studying the equilibrium effects of AI on the knowledge economy.[1] Our approach is grounded in two insights. First, the defining characteristic of knowledge work is that production know-how is predominantly tacit (or non-codifiable)—developed through repeated observations of practical successes and failures—and therefore inherently embodied in individuals (Polanyi, 1966; Garicano, 2000; Garicano and Wu, 2012). As a result, individuals' time and knowledge are critical bottlenecks, making firms central to organizing production (Garicano and Rossi-Hansberg, 2015). Second, unlike traditional automation technologies, AI learns by example, uncovering patterns that cannot be codified into explicit rules (Brynjolfsson and Mitchell, 2017; Autor, 2024; Brynjolfsson et al., 2025). This ability allows AI systems to acquire tacit knowledge and perform cognitive, non-codifiable tasks (see Table 1). By automating such work, AI has the potential to alleviate knowledge bottlenecks, opening the door to a fundamental reorganization of production.

[Insert Table 1 Here]

More precisely, we incorporate AI into a canonical model of the knowledge economy: The knowledge hierarchies first introduced by Garicano (2000). This model emphasizes the central role of tacit knowledge in shaping organizations and labor market outcomes. Within this framework, we study the equilibrium effects of AI on occupational choice, organizational structure, and the distribution of labor income. Our analysis highlights that introducing AI agents is fundamentally different from expanding the human labor force, and that AI's impact crucially depends on its problem-solving capabilities and its degree of autonomy.

Our starting point—the pre-AI economy—is the baseline model of Garicano and Rossi-Hansberg (2004), Antràs et al. (2006), and Fuchs et al. (2015). Labor and knowledge are the sole inputs in production. Humans are endowed with one unit of time and are heterogeneous in terms of knowledge.

[1] According to data from the U.S. Bureau of Labor Statistics, as of December 2024, over 100 million people in the U.S. were employed in cognitive or knowledge-based roles (St. Louis Fed, 2024).

Individuals use their time to pursue production opportunities that require cognitive work—such as handling customer support, drafting contracts, or designing products—but encounter problems of varying difficulty during the production process. Output is produced when an individual successfully solves the problem she confronts, which occurs when her knowledge exceeds the problem's difficulty. If a human cannot solve a problem on her own, she may seek help from another human. Help, however, is costly in terms of time. Knowledge is tacit because matching problems to those who can solve them is difficult, and experts cannot outline a plan of action in advance. In this sense, individuals "know more than they can tell" (Polanyi, 1966).

In the competitive equilibrium of the pre-AI economy, humans either pursue production opportunities on their own or join hierarchical firms to optimize the use of their time and knowledge. These firms exhibit *management by exception*: Less knowledgeable individuals become "workers" who pursue production opportunities, while more knowledgeable individuals become "solvers" specializing in assisting workers with the exceptional problems they are unable to solve. This structure effectively shields the most knowledgeable individuals from routine work, enabling them to focus on their comparative advantage: solving complex problems.[2]

Our innovation is to incorporate AI into this otherwise canonical setting. We base our model on three key developments in modern AI, documented in Section 2. First, unlike earlier automation technologies, AI—like humans—can acquire tacit knowledge. However, unlike humans—who are constrained by their individual time—firms can leverage computational resources ("compute") to use AI's knowledge at scale. Second, the rise of flexible, general-purpose foundation models—adaptable to diverse tasks with minimal additional training—enables firms to rely on a shared AI model. Third, technology firms are developing and starting to deploy AI agents that can do cognitive work autonomously. These agents do not just passively answer questions but can pursue projects independently.

Guided by these developments, we model AI as a technology available to all firms that converts compute into AI agents, all endowed with the same exogenously fixed level of knowledge. As a benchmark, we first consider the case where these agents can do exactly the same as humans. These AI agents are "autonomous" because they can function both as "co-workers" (pursuing production opportunities) and as "solvers/co-pilots" (providing advice). A key contribution of our analysis is to contrast this benchmark with the case of "non-autonomous" AI, where—for technological or regulatory reasons—AI agents are restricted to acting solely as co-pilots.

In the post-AI economy, firms decide their organizational structure and whether to integrate AI agents into their production processes. The total amount of compute available in the economy is

[2]There is both anecdotal and systematic empirical evidence showing the emergence of such "knowledge hierarchies" (see, e.g., Garicano and Hubbard, 2012; Caliendo and Rossi-Hansberg, 2012; Caliendo et al., 2015, 2020). For instance, Alfred Sloan (1924), a former head of General Motors (GM), once wrote: "We do not do much routine work with details. They never get up to us. I work fairly hard, but on exceptions."

exogenous, while its price is endogenously determined in equilibrium. We assume that compute is abundant relative to human time, making humans the binding constraint in human-AI interactions. This assumption reflects the exponential growth in computational capacity over the past two centuries (Nordhaus, 2007) and the fact that current AI models process information and generate outputs 10 to 100 times faster than humans (Amodei, 2024).

Regarding autonomous AI, we show that its impact critically depends on whether it is "basic" or "advanced." The introduction of basic autonomous AI—which creates AI agents with the knowledge of pre-AI workers—induces the most knowledgeable pre-AI solvers to focus on supporting AI-driven production. This reallocation forces the least knowledgeable human workers to seek assistance from less knowledgeable individuals, pushing the marginal pre-AI workers to take specialized problem-solving roles. In other words, basic autonomous AI displaces humans from routine work towards complex problem-solving. In this scenario, the least knowledgeable individuals are harmed by the technology—they face competition from AI in production work and must rely on less knowledgeable individuals for support—whereas the most knowledgeable individuals benefit by using AI to leverage their expertise at a low cost.

In contrast, the introduction of an advanced autonomous AI—which creates AI agents with the knowledge of pre-AI solvers—induces the least knowledgeable humans to seek assistance from AI instead of humans. Consequently, the marginal pre-AI solvers are pushed into routine work, displacing humans from complex problem-solving. In this scenario, both the least and the most knowledgeable individuals benefit from the technology. The least knowledgeable gain access to relatively cheap AI agents that can help them solve complex problems, while the most knowledgeable continue to thrive by leveraging advanced AI agents for routine work.

Recent anecdotal evidence from professional services industries illustrates these reorganizations. For example, Beane and Anthony (2024) document that junior analysts in investment banking are increasingly distanced from senior partners as AI takes over tasks that once required junior involvement. Similarly, a 2022 survey of M&A lawyers found that AI, by automating routine knowledge work such as document review, has allowed junior lawyers to shift their focus to more complex responsibilities, including client engagement and legal analysis (Litera, 2022).

The impact of AI on labor outcomes also critically depends on its autonomy. We show that non-autonomous AI tends to benefit the least knowledgeable individuals the most, in contrast to autonomous AI, which mainly benefits the most knowledgeable. However, overall output is higher with autonomous AI than with non-autonomous AI.

Intuitively, non-autonomous AI primarily benefits the least knowledgeable because it does not compete with them for production work and provides a means of solving complex problems without human assistance. Its lower opportunity cost relative to autonomous AI—stemming from its inability to perform production work independently—reinforces this advantage, allowing the least knowledgeable to capture a larger share of AI-generated benefits. For the most knowledgeable individuals, the effect is reversed. Non-autonomous AI is less advantageous than autonomous AI because it lacks the capacity to handle routine work and competes with them for advisory roles.

Our findings reveal a trade-off between output and inequality when regulating AI autonomy. They also reconcile seemingly contradictory empirical findings on AI's impact on work. On the one hand, some evidence suggests that AI disproportionately benefits the least knowledgeable individuals and reduces performance inequality (e.g., Dell'Acqua et al., 2023; Noy and Zhang, 2023; Peng et al., 2023; Wiles et al., 2023; Caplin et al., 2024; Brynjolfsson et al., 2025). On the other hand, there is also evidence that AI complements high-skilled knowledge workers while substituting for their low-skilled counterparts (Berger et al., 2024; Alekseeva et al., 2024). Although these findings may seem contradictory, they can be rationalized by our framework as we explain in Section 7: They align with the distinct impacts of non-autonomous AI and basic autonomous AI, respectively.

Related Literature

This paper builds on the literature on knowledge hierarchies—which investigates how tacit (or non-codifiable) knowledge shapes organizational structures and labor market outcomes—to deepen our understanding of the effects of automation. Specifically, we explore the implications of AI, a new technology that can automate work traditionally reliant on tacit knowledge.

The literature on knowledge hierarchies originates with Garicano (2000), who introduces the model and shows that knowledge hierarchies are optimal when production requires the application of tacit knowledge. Garicano and Rossi-Hansberg (2004, 2006) extend this model to include heterogeneous agents, exploring how the endogenous organization of knowledge work affects labor income inequality. Fuchs et al. (2015) further develop this literature by characterizing the equilibrium contractual arrangements when there is asymmetric information about knowledge.[3] Building on this foundation, we develop a framework to examine how modern AI can transform knowledge work.

In the context of knowledge hierarchies, the most closely related paper is Antràs et al. (2006). They study the effects of offshoring by comparing the equilibrium of a closed economy with one in which firms can form international teams. Our paper differs from theirs in two key respects. First, while offshoring gives firms access to a human population with varying knowledge levels, AI allows firms to automate knowledge work at scale. As discussed in Section 7, this leads to qualitatively different outcomes. Second, we introduce AI-specific dimensions—such as AI's knowledge, autonomy, and computational resources—to explore how AI's impact is shaped by these dimensions. These considerations, absent from Antràs et al. (2006), are crucial for understanding AI's unique effects.[4]

[3]Other important contributions to this literature include Garicano and Hubbard (2007, 2009), Garicano and Rossi-Hansberg (2012), Bloom et al. (2014), Garicano and Hubbard (2018), Caicedo et al. (2019), Gumpert et al. (2022), Adhvaryu et al. (2023), and Carmona and Laohakunakorn (2024).

[4]Another related contribution is Pancs's (2018, ch. 5) textbook example, which shows how superintelligent machines—matching the abilities of the most capable humans—eliminate labor income inequality. In contrast, we demon-

With regard to the existing literature on automation, the first important recent contribution is Zeira (1998), who shows how automation can lead to a decline in the labor share as the economy develops. Acemoglu and Restrepo (2018) contend that, by depressing wages, automation also encourages the creation of new tasks in which labor has a comparative advantage. Further developing these ideas, Acemoglu and Restrepo (2022) and Acemoglu et al. (2024) study how different types of technological advances—such as automation and labor-augmenting technologies—affect wages, inequality, and productivity. Complementary research by Autor et al. (2003) and Acemoglu and Autor (2011) emphasizes how the automation of codifiable tasks—driven by the adoption of computers and robots—explains employment and wage polarization, as these tasks are predominantly performed by middle-skilled workers.[5]

Our work contributes to this literature by focusing on AI, a fundamentally new form of automation. Unlike earlier technologies that primarily automated codifiable tasks, AI can acquire and apply tacit knowledge, allowing it to perform cognitive, non-codifiable work. As discussed above, by automating such work, AI has the potential to ease knowledge bottlenecks and enable a fundamental reorganization of production.

Traditional models of automation, while useful for understanding earlier automation waves, have a hard time capturing this transformation. These models often overlook firms' central role in structuring production—an essential factor for assessing AI's impact on the knowledge economy. In contrast, our framework explicitly accounts for the organization of knowledge work, showing how AI reshapes production and how shifts in firm composition create new pathways for AI to influence labor outcomes.[6] Moreover, we introduce AI-specific dimensions absent from existing models, leading to new insights. For example, we find that non-autonomous AI disproportionately benefits the least knowledgeable individuals, while autonomous AI primarily benefits the most knowledgeable. As far as we know, these results have no parallel in the existing literature on automation or AI.

Finally, we also contribute to the broader literature on the economics of knowledge and ideas (e.g., Romer, 1990, 1992; Jones, 2005), the knowledge-based view of the firm (e.g., Nonaka, 1991; Grant, 1996; Garicano, 2000; Garicano and Wu, 2012), and the economics of data (e.g. Jones and Tonetti, 2020; Farboodi and Veldkamp, 2023, 2024). Specifically, we highlight how modern AI systems blur the conventional boundary between codifiable and non-codifiable knowledge.

Both types of knowledge are, in principle, non-rival—one person's use does not reduce availability for others—but their practical application depends on rival resources such as time, effort, or compu-

strate that AI's effects on labor income distribution are nuanced, depending critically on AI's proficiency and autonomy.

[5]Other important contributions include Aghion et al. (2017), Moll et al. (2022), Korinek and Suh (2024), Acemoglu and Loebbing (2024), Acemoglu (2024), and Jones (2024).

[6]For instance, we show that in the case of a basic autonomous AI, the most knowledgeable human solvers switch toward using AI for routine production work. As a result, a lower-quality pool of solvers is left to assist those who continue as workers in the post-AI world, reducing their productivity. As discussed in Section 5.1, this outcome is consistent with anecdotal evidence from sectors such as investment banking and legal services.

tational power. The key distinction between codifiable and non-codifiable knowledge lies in their transferability. Codifiable knowledge, such as information stored in databases or manuals, is highly transferable because it can be expressed as explicit rules and procedures. In contrast, non-codifiable knowledge has limited transferability, spreading primarily through direct experience. Consequently, individuals who possess tacit knowledge often become critical bottlenecks in its application.

Modern AI systems do not fit neatly into this classification. These systems learn from examples, identifying patterns that cannot always be reduced to explicit, formal descriptions. Once trained, however, they can apply these insights at scale using compute. In this sense, AI effectively encodes tacit-like knowledge into model parameters—making it codifiable to machines but not humans. This creates a hybrid form of "embodied" knowledge that is neither purely human nor strictly codified, challenging conventional wisdom on how knowledge is transferred, scaled, and applied.

Roadmap

The rest of the paper is organized as follows. Section 2 documents three key developments in modern AI that shape the assumptions of our baseline model, introduced in Section 3, which focuses on autonomous AI. Section 4 characterizes the equilibrium with autonomous AI, while Section 5 examines the impact of this technology relative to the pre-AI equilibrium. Section 6 explores the implications of non-autonomous AI. Finally, Section 7 concludes by discussing additional implications of our analysis and linking our findings to emerging empirical evidence about AI's effects on the labor market. We relegate all formal mathematical proofs to the Online Appendix.

2 Three Key Developments in Modern AI

In this section, we document three key developments in modern AI that are essential for understanding the excitement and concerns surrounding its impact on labor and organizational outcomes: (i) AI's ability to scale the use of tacit knowledge, (ii) the advent of foundation models, and (iii) the rise of AI agents capable of performing cognitive work autonomously. These three developments guide the assumptions of our baseline model in Section 3.

1. AI Can Use Tacit Knowledge at Scale.— AI represents a fundamentally new form of automation. Unlike earlier technologies, AI—like humans—can acquire tacit knowledge, enabling it to perform cognitive, non-codifiable work. As Autor (2024, p. 7) observes:

> Pre-AI, computing's core capability was its faultless and nearly costless execution of... procedural tasks. Its Achilles' heel was its inability to master... tasks requiring tacit knowledge. Artificial Intelligence's capabilities are precisely the inverse [...] Rather than relying on hard-coded procedures, AI learns by example, gains mastery without explicit instruc-

> tion and acquires capabilities that it was not explicitly engineered to possess [...] Like a human expert, AI can weave formal knowledge (rules) with acquired experience to make — or support — one-off, high-stakes decisions.

However, unlike humans—who can only apply their tacit knowledge within the limits of their own time—firms can leverage AI across all available compute in the economy. This reflects the nonrival nature of digital information, which has a negligible marginal cost of reproduction (Brynjolfsson and McAfee, 2016; Goldfarb and Tucker, 2019). As Waisberg and Hudek (2021, p. 177) explain:

> You can ask a question at a seminar or webinar, but you can only get the data as provided by the host or speaker, and a given speaker can only be in one seminar at a time... [In contrast, AI] scales; no longer are you limited by the time of a single domain expert. If their knowledge and behavior are encoded into an AI, you can have as many copies of that AI working at the same time as you want.

2. The Advent of Foundation Models.— Foundation models represent a transformative shift in AI. These are models characterized by their ability to handle a wide variety of tasks through training on large-scale datasets using self-supervised learning techniques (Bommasani et al., 2022; Council of Economic Advisers, 2024). Models like GPT-4, Gemini, and Claude Sonnet exemplify this approach, serving as flexible, general-purpose platforms that can be adapted to diverse downstream tasks with minimal additional training. This adaptability enables firms to leverage shared AI models, removing the need for costly development of proprietary, domain-specific systems.[7] As a result, a small number of powerful AI models can function across multiple domains, driving significant economic and industry-wide impact.[8]

Foundation models have demonstrated their versatility across various industries, from customer service automation to legal research and financial analysis. For instance, Klarna, a fintech company, automated over 700 customer service roles using a version of OpenAI's models. In the legal field, Harvey leverages OpenAI's technology to automate research and document analysis. Similarly, Claude Sonnet powers applications at Lonely Planet for itinerary creation and at Bridgewater Associates for tasks like research, coding, and data visualization.

Moreover, evidence suggests that general-purpose models often outperform specialized AI systems. For instance, ChatGPT—a fine-tuned version of GPT-4—has demonstrated superior performance to BloombergGPT—a Large Language Model (LLM) trained specifically on financial data—in

[7] For instance, Andreessen Horowitz, a venture capital firm, surveyed around 100 Fortune 500 companies across industries such as technology, telecom, banking, healthcare, and energy. The survey found that 94% of these firms were using existing foundation models, while only 6% were creating their own specialized models (Wang and Xu, 2024).

[8] Using the language of computer scientists, foundation models exhibit *emergence*—the ability to develop unexpected capabilities that enable them to perform tasks beyond their explicit training. Combined with scalability, emergence then leads to *homogenization*, defined as the reliance on a few powerful models across multiple domains (Bommasani et al., 2022).

various finance-related tasks (Li et al., 2023). Similarly, Nori et al. (2023) demonstrate that GPT-4 surpasses specialist models across medical benchmarks and argue that this advantage extends to other domains, including accounting, law, nursing, and clinical psychology. This shows that even in fields that traditionally relied on highly specialized models, general-purpose foundation models offer better performance, increased flexibility, and lower costs.

3. The Rise of AI Agents.—The adaptability, efficiency, and scalability of foundation models are driving a new wave of automation: the rise of AI agents capable of performing cognitive work autonomously. Technology firms such as DeepMind and OpenAI are actively developing and starting to deploy these AI agents. As Dario Amodei, co-founder and CEO of Anthropic, explains (Amodei, 2024):

> I am not talking about AI as merely a tool to analyze data [...] It does not just passively answer questions; instead, it can be given tasks that take hours, days, or weeks to complete, and then goes off and does those tasks autonomously, in the way a smart employee would, asking for clarification as necessary.

The emergence of autonomous AI agents raises the crucial question of whether current AI models are sophisticated enough for such responsibilities. Sarah Tavel, a general partner at the venture capital firm Benchmark, is optimistic (Stebbings, 2024b):

> There are certainly use cases that can be automated right now, and we see, we see a lot of them... beyond that, there is no question that as the foundation models improve, we are going to see the ability to take on more and more complex tasks.

This optimism is grounded in the fact that much of today's economy is driven by knowledge work, an area where AI excels compared to industries reliant on manual labor. For instance, Eloundou et al. (2024) estimate that 46% of jobs could have more than half of their tasks affected by AI foundation models in the near future. Similarly, Goldman Sachs (2023) projects that two-thirds of U.S. occupations are at least partially exposed to AI-driven automation, with approximately a quarter of all work potentially automatable when weighted by employment share.

These developments call for new research into AI's economic implications. As Hadfield (2025) remarked during a panel at the ASSA 2025 meetings in San Francisco:

> All of this is about autonomy [. . .] Think about what that means [...] [AI agents would be] engaging in transactions, entering into contracts, hiring people, designing products, setting prices for products, advertising, doing all of those things. So I think it's really important for us to be shifting our thinking a little bit. [...] Not just thinking about it as a technology or a tool but rather as a novel economic agent.

3 Autonomous AI

3.1 The Model

The Pre-AI Economy.— There is a unit mass of humans, each endowed with one unit of time and exogenous knowledge $z \in [0,1]$. The distribution of knowledge in the population is given by a cumulative distribution function G with density g. The only assumption we impose on this distribution is that g is continuous and strictly positive for all z. The knowledge of each individual is perfectly observable.

There is a large measure of identical competitive firms. Production occurs inside firms, which are the residual claimants of all output. Labor and knowledge are the sole inputs in production. Firms have no fixed costs and, for simplicity, two layers at most.[9]

Single-layer firms rely on a single individual to carry out production. This "independent producer" dedicates her entire time to pursuing a single production opportunity. These opportunities can be thought of as work requiring cognitive ability, such as handling customer support, drafting contracts, hiring employees, and designing products. Each production opportunity is linked to a problem whose difficulty x is ex-ante unknown and distributed uniformly on $[0,1]$, independently across problems. If the knowledge of the human engaging in production exceeds the problem's difficulty, she solves the problem and produces one unit of output. Otherwise, no output is produced.[10]

Two-layer firms hire one "solver" and multiple "workers," where all workers have the same knowledge. The restriction that all workers share the same knowledge is without loss because—as we show below—the equilibrium matching arrangement between (human) workers and (human) solvers exhibits strict positive assortative matching.

As in single-layer firms, each worker in a two-layer firm devotes her time to a single production opportunity. The difference is that if the worker cannot solve the problem independently, she can ask the solver for help. If the solver's knowledge exceeds the problem's difficulty, she communicates the solution to the corresponding worker, who then produces a unit of output. Otherwise, the worker fails to produce. However, communication is costly in that each request for help consumes $h \in (0,1)$ units of the solver's time, regardless of whether the solver knows the solution. Hence, a two-layer organization optimally hires exactly $n(z)$ workers of knowledge $z < 1$ to fully exploit its solver's time, where $n(z)$ satisfies $h \times n(z) \times (1-z) = 1$.

Figure 1 depicts the two possible firm configurations of the pre-AI world, where the letter attached to each human corresponds to her knowledge. In this context, knowledge is tacit because solvers

[9] In the Online Appendix, we show that many insights we uncover—such as the conditions under which the least and most knowledgeable individuals benefit from AI—naturally generalize to settings with an arbitrary number of layers.

[10] The assumption that $x \sim U[0,1]$ is without loss, as the distribution of knowledge in the population is arbitrary. Under this assumption, an individual's knowledge z is interpreted as the fraction of problems she can solve on her own.

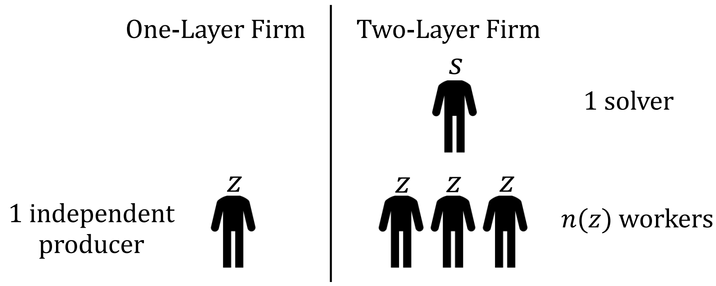

Figure 1: The Two Possible Firm Configurations in the Pre-AI World

cannot prescribe a plan of action to workers in advance, and a worker must interact with her solver to determine whether the solver knows the solution to a problem.

Autonomous AI.—We model AI as a technology that converts compute into "AI agents," all with the same exogenously fixed level of knowledge, $z_{\text{AI}} \in [0, 1)$.[11] Each AI agent requires one unit of compute and serves as a perfect substitute for a human with equivalent knowledge. These AI agents are "autonomous" because they can function both as "co-workers" (pursuing production opportunities) and as "co-pilots" (providing advice). In Section 6, we explore the case where the AI agents can only be co-pilots, rendering them non-autonomous.

All firms have access to AI. Post-AI, firms decide their organizational structure and whether to integrate AI agents into their production processes. The total amount of compute in the economy, denoted by μ, is exogenous. We further assume that production opportunities exceed the combined capacity of humans and AI agents to pursue them, meaning opportunities are abundant relative to available resources. As we discuss below, this implies that there is no unemployment in equilibrium.

Figure 2 illustrates the five possible post-AI firm configurations. In addition to single- and two-layer firms that hire only humans (the only possible pre-AI firms), there are three additional possibilities: Single-layer automated firms (which use AI agents as independent producers), bottom-automated firms (which use AI agents as workers), and top-automated firms (which use AI agents as solvers). Note that a firm will never use AI agents in both layers of the organization because an AI solver knows the solution to the same set of problems as an AI worker.

Our modeling approach is motivated by the three AI developments described in Section 2. First, unlike humans—who can only apply their knowledge within the limits of their own time—AI is scalable, meaning that once knowledge z_{AI} is encoded into the technology, such knowledge can be used simultaneously across all available compute. Second, foundation models—flexible and general-purpose—allow firms to access AI on equal terms through a shared AI model. Third, firms can combine AI with compute to deploy autonomous AI agents capable of performing cognitive work independently.

[11] We assume $z_{\text{AI}} < 1$ because the equilibrium has a discontinuity at $z_{\text{AI}} = 1$. See the Online Appendix for the case $z_{\text{AI}} = 1$.

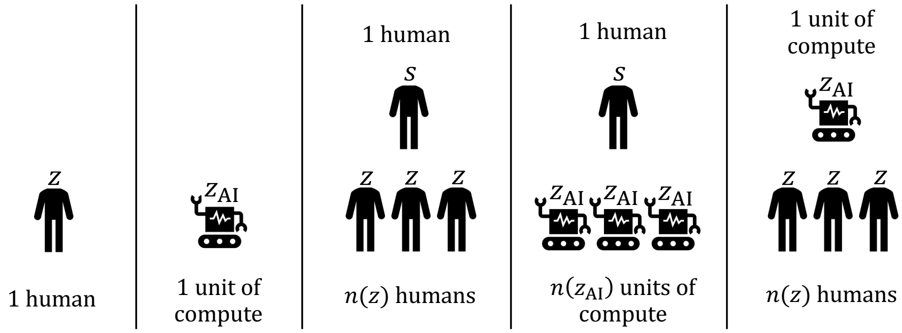

Figure 2: The Five Possible Firm Configurations in the Post-AI World

Wages, Prices, and Profits.—Let $w(z)$ be the wage of a human with knowledge z and denote by r the rental rate of one unit of compute (i.e., the rental rate of an AI agent). All humans in this economy are risk neutral and maximize their income. In particular, the owners of compute rent it to the highest bidder. We normalize the value of each unit of output to one.

The profit of a single-layer organization is:

$$\Pi_1 = \begin{cases} z - w(z) & \text{if the firm hires a human with knowledge } z \\ z_{\text{AI}} - r & \text{if the firm uses an AI agent} \end{cases}$$

In other words, it is equal to the expected output of its independent producer minus the hiring cost.

The profit of a two-layer organization, in turn, depends on whether it uses AI as a solver (i.e., automates the top layer, a "tA" firm), uses AI as a worker (i.e., automates the bottom layer, a "bA" firm), or it does not use AI (an "nA" firm):

$$\begin{aligned} \Pi_2^{tA}(z) &= n(z)[z_{\text{AI}} - w(z)] - r && \text{(where } z \leq z_{\text{AI}}) \\ \Pi_2^{bA}(s) &= n(z_{\text{AI}})[s - r] - w(s) && \text{(where } z_{\text{AI}} \leq s) \\ \Pi_2^{nA}(s, z) &= n(z)[s - w(z)] - w(s) && \text{(where } z \leq s) \end{aligned}$$

where z and s denote the knowledge of a human worker and a human solver, respectively, and we use the fact that no two-layer firm hires a solver who is less knowledgeable than its workers. In all three cases, the profit of a firm is its expected output minus the cost of the resources it uses. For instance, in the case of a tA firm that hires workers with knowledge z, its total expected output is $n(z)z_{\text{AI}}$, while the cost of resources is $n(z)w(z) + r$.

Competitive Equilibrium.— A competitive equilibrium comprises the following objects: (1) an allocation of compute across its various potential uses; (2) a partition of the human population into distinct occupations; (3) a description of the matching between different types of agents; and (4) a wage schedule, $w(z)$, and a rental rate of compute, r.

Let μ_i, μ_w, and μ_s be the amount of compute used to create AI independent producers, workers, and solvers, respectively. We denote by I the set of humans hired as independent producers, and by

W_p and W_a the set of human workers assisted by humans and AI agents, respectively.[12] Similarly, let S_p be the set of humans who assist other humans, and S_a the set of humans assisting AI agents in production. We assume that the sets I, W_p, W_a, S_p, and S_a are measurable with respect to G. Since a tA firm that hires human workers with knowledge $z \in W_a$ rents one AI agent, $\mu_s = \int_{W_a} h(1-z)dG(z)$. Similarly, since a bA firm that hires a human solver with knowledge $s \in S_a$ rents $n(z_{\text{AI}})$ AI agents, $\mu_w = n(z_{\text{AI}}) \int_{S_a} dG(z)$.

Finally, let $m : W_p \to S_p$ represent the function describing the pointwise matching arrangement generated by the hiring decisions of nA firms. Specifically, $m(z)$ denotes the knowledge of the human solver assisting human workers with knowledge z.[13] As mentioned earlier, the equilibrium matching arrangement between human workers and human solvers exhibits strict positive assortative matching, so m is strictly increasing. Additionally, it must satisfy the following resource constraint:

$$\int_Y h(1-u)dG(u) = \int_{m(Y)} dG(u), \text{ for any measurable set } Y \subseteq W_p \tag{1}$$

where $m(Y)$ denotes the set of solvers assigned to the set of workers $Y \subseteq W_p$. This condition ensures that the total time required to consult on problems left unsolved by the workers in Y equals the total time available from the solvers matched to these workers, $m(Y)$. In essence, it reflects that nA firms optimally hire $n(z)$ workers with knowledge z to fully use the time of their solvers.

Definition (Competitive Equilibrium). An equilibrium consists of nonnegative amounts (μ_i, μ_w, μ_s), measurable sets (I, W_p, W_a, S_p, S_a), a function $m : W_p \to S_p$ satisfying (1), a wage schedule $w : [0,1] \to \mathbb{R}_{\geq 0}$ and a rental rate of compute $r \in \mathbb{R}_{\geq 0}$, such that:

1. Firms optimally choose their structure (while earning zero profits).
2. Markets clear: (i) $\mu_i + \mu_w + \mu_s = \mu$, and (ii) the union of the sets (I, W_p, W_a, S_p, S_a) is $[0,1]$ and the intersection of any two of these sets has measure zero.

Note that the human workers in W_p are endogenously matched with the human solvers in S_p according to the pointwise matching function m. In contrast, all the humans in W_a are matched with an AI solver (which has knowledge z_{AI}), while all the humans in S_a are matched with AI workers (whose knowledge is z_{AI}).

Total output in this economy is $\int_I z dG(z) + z_{\text{AI}}\mu_i + \int_{W_p} m(z)dG(z) + \int_{S_a} n(z_{\text{AI}})z dG(z) + \int_{W_a} z_{\text{AI}} dG(z)$, where each term represents the total output of each of the five firm configurations depicted in Figure 2 (in the same order). Total labor income, in turn, is equal to $\int_0^1 w(z)dG(z)$, whereas total capital income is μr. Since firms earn zero profits in equilibrium, total output equals the sum of total labor income and total capital income.

[12] We use the subscript "p" (for people) instead of "h" (for humans), to avoid any confusion with the helping cost h.

[13] As discussed in Fuchs et al. (2015), $m(z)$ is an approximation of a firm that hires a small interval of human workers $(z - \varepsilon_1, z + \varepsilon_1)$ and a small interval of human solvers $(m(z) - \varepsilon_2, m(z) + \varepsilon_2)$ with the requirement that the mass of solvers in the firm $\int_{m(z)-\varepsilon_2}^{m(z)+\varepsilon_2} dG(u)$ is equal to h times the mass of problems left unsolved by the workers in the firm $\int_{z-\varepsilon_1}^{z+\varepsilon_1} (1-u)dG(u)$. The latter requirement captures that nA firms optimally hire $n(z)$ workers of knowledge z.

Compute is "Abundant" Relative to Human Time.— We focus on the case in which compute is sufficiently abundant so that the binding constraint in human-AI interactions is human time, not compute. In other words, there are more AI agents than can be matched with humans, implying that some AI agents must engage in independent production.[14]

We consider this the most relevant case due to the exponential growth in computational capacity over the past two centuries (Nordhaus, 2007) and the fact that current AI models process information and generate outputs 10 to 100 times faster than humans (Amodei, 2024). However, to explore the effects of this assumption, we also examine the case of limited compute in the Online Appendix.

Some Notation.— For future reference, we define $W \equiv W_a \cup W_p$ and $S \equiv S_a \cup S_p$ as the overall set of human workers and solvers of the economy, respectively. We also denote by $e : S_p \to W_p$ the inverse of the matching function m. That is, e is the "employee matching function" denoting the knowledge of the human worker matched with a human solver with knowledge $s \in S_p$. This function always exists given that, as argued above, the matching function m is strictly increasing in equilibrium.

Finally, for any arbitrary set B, we denote by $\text{int}B$ and by $\text{cl}B$ the interior and closure of B, respectively. We also use $B \preceq B'$ to indicate that the set $B \subseteq [0,1]$ "lies below" the set $B' \subseteq [0,1]$. Formally, $B \preceq B'$ if $\sup B \leq \inf B'$. For example, $W_a \preceq W_p$ means that the most knowledgeable human worker assisted by AI is weakly less knowledgeable than the least knowledgeable human worker assisted by another human.

3.2 Discussion of the Model

Before proceeding with the analysis, we map our model to real-world applications. We also clarify how to interpret compute in our framework, justify our focus on a single AI technology, and explain why we do not consider the possibility of unemployment. The Online Appendix discusses some limitations of our model and suggests avenues for future research.

Interpretation.— In our benchmark model, AI agents are autonomous in that they can operate both as co-workers—capable of pursuing production opportunities—and as co-pilots—providing assistance.[15] A real-world example of an AI co-worker is Klarna, which automated over 700 customer service agents using OpenAI's models (see Section 2). Another is Agentforce by Salesforce, an AI agent that not only identifies business leads using company data but also contacts prospects and schedules meetings on behalf of salespeople (Rosenbush, 2024). Similarly, All Day TA, a startup founded by

[14] A sufficient condition for this to hold is $\int_0^{z_{\text{AI}}} n(z)^{-1} dG(z) + n(z_{\text{AI}})\big(1 - G(z_{\text{AI}})\big) < \mu$. For any distribution G and helping cost $h \in (0,1)$, there exists a finite μ that satisfies this condition for all $z_{\text{AI}} \in [0,1)$.

[15] The model is silent about where production opportunities come from. The role of organizations is to decide who to assign to routine work and who to assign to specialized problem solving. Hence, workers and independent producers—human or AI—should not be thought of as being responsible for generating production opportunities; they are only responsible for pursuing them.

University of Toronto Professors Kevin Bryan and Joshua Gans, leverages foundation models to fully automate the work of university teaching assistants.

In contrast, Brynjolfsson et al. (2025) provide an example of an AI co-pilot in the customer support industry: their study highlights the productivity effects of equipping human agents with a real-time, AI-based conversational assistant to manage customer inquiries. Another example is GitHub Copilot, which aids developers by suggesting code snippets, autocompleting functions, and recommending improvements to code quality. Similarly, BrainTrust AIR assists HR professionals by reviewing candidates, grading applications, and generating detailed scorecards to streamline recruitment decisions.

Regarding AI autonomy, for simplicity, we focus on the two extreme cases. First, we consider a scenario in which AI agents are fully autonomous and capable of performing all tasks that humans possessing knowledge z_{AI} can accomplish. Then, in Section 6, we examine the opposite case, where AI agents are entirely non-autonomous and can only provide advice. Most AI applications, however, operate between these extremes, with autonomy levels varying by sector. For example, tools like Harvey automate many routine legal tasks, such as document review and contract analysis. Yet, they remain unable to engage directly with clients or argue cases in court.

Because our framework models the broader knowledge economy, we interpret our results as approximations of AI's impact based on its average level of autonomy across the different relevant sectors. Accordingly, the practical implications of our analysis will likely fall between the predictions of these two benchmark cases. To determine which extreme better reflects current and future realities, the critical question is: How many tasks traditionally performed by humans can AI agents like Harvey handle autonomously, and how will this capability evolve over time?

On Compute.— Our main goal is to analyze how AI affects human labor outcomes. For this reason, we take AI technology as given. Thus, in our framework, "compute" refers to the resources required during the inference phase—the stage in which an AI system generates outputs based on new input data after it has already been trained.

To facilitate comparison between AI output and human labor, we express compute in terms of human time. In practice, however, compute is often measured in FLOPs or, for LLMs, in tokens. In an earlier version of this paper (Ide and Talamàs, 2025), we explain how compute in our model translates into these real-world units.

Why Only One AI?— A key property of AI is that it relies on compute, which is general-purpose hardware that can be rented on demand. This implies that competition among AI providers drives compute toward the highest-performing AI systems—a phenomenon that computer scientists and venture capitalists call the commoditization of foundation models (Navani, 2023; Stebbings, 2024a).[16]

[16]Compute can be rented on demand through cloud computing providers and easily redeployed across various AI foundation models. According to a recent survey by Andreessen Horowitz, "Most enterprises are designing their applications so that switching between models requires little more than an API change. Some companies are even pre-testing prompts

As Aravind Srinivas, founder of PerplexityAI, explains (Stebbings, 2024a):

> [It] is a losing game... every time you end up finishing a large training run, you burn a lot of money, you have a great model, and then you watch it be destroyed by the next update... So where are you recovering all that money back? (...) Nobody wants to use the API [i.e., the model] if somebody else is offering a better model at a cheaper price.

This competition for compute underpins our assumption of focusing on a single AI technology. Indeed, in the Online Appendix, we show that if multiple AI technologies are available and all of them require similar compute to create AI agents, only the most advanced AI will be used in equilibrium. However, if less advanced AI technologies require substantially less compute, they may coexist alongside more advanced models. This result is consistent with the development of smaller foundation models, such as GPT-4o Mini and Claude 3 Haiku.

Furthermore, this extension highlights a fundamental distinction between human and AI heterogeneity. While humans with varying levels of knowledge can coexist in equilibrium despite requiring the same resources—time—to perform knowledge work, AI heterogeneity emerges only when models differ significantly in both knowledge and resource requirements. Hence, introducing a population of AI agents is fundamentally different from expanding the human labor force.

No Unemployment.— In defining our competitive equilibrium, we excluded the possibility of unemployment. This is without loss of generality, as we assume that total available resources—human time and compute—are scarce relative to production opportunities (even though compute is abundant relative to time). This scarcity ensures that, in equilibrium, it is always worthwhile for every human to be employed in some capacity, even if AI is more knowledgeable than a fraction of the human population.

In the Online Appendix, we explore an extension where compute is so abundant that it outstrips production opportunities. In that case, we show that AI leads to technological unemployment: All humans who are less knowledgeable than AI become unemployed. Organizations, however, still display a hierarchical structure in that the most knowledgeable humans specialize in tackling the problems that AI cannot solve.

3.3 Benchmark: The Pre-AI Equilibrium

We begin by presenting a partial characterization of the equilibrium without AI. This is also the equilibrium when the amount of compute μ is zero, and was originally described by Fuchs et al. (2015).[17] Note that in this case, $W_a = S_a = \emptyset$, so $W = W_p$ and $S = S_p$.

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so the change happens literally at the flick of a switch" (see Wang and Xu, 2024).

[17] An economy with $\mu = 0$ is different from one with $\mu > 0$ and $z_{\mathrm{AI}} = 0$. This is because, even if AI cannot solve any problems, it can still pursue production opportunities, thereby expanding the economy's production possibility frontier.

Proposition 1. *In the absence of AI, there is a unique equilibrium. The equilibrium is efficient (i.e., it maximizes total output) and has the following features:*

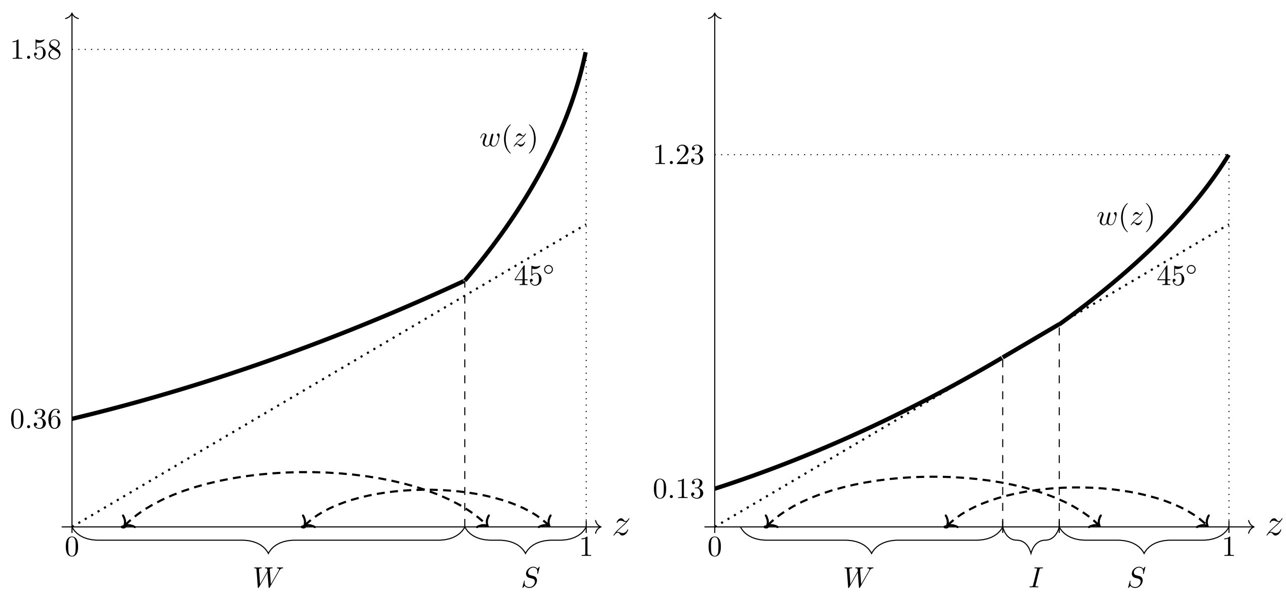
- *Occupational stratification:* $W \preceq I \preceq S$.
- *Positive assortative matching: The function* $m : W \to S$ *is strictly increasing.*
- $W \neq \emptyset$ *and* $S \neq \emptyset$*. However,* $I \neq \emptyset$ *if and only if* $h > h_0 \in (0, 1)$.

Moreover, the wage function w *is continuous, strictly increasing, and convex (strictly so when* $z \in W \cup S$*), and is given by:*

- $w(z) = m(z) - w(m(z))/n(z)$ *for all* $z \in W$.
- $w(z) = z$ *for all* $z \in I$.
- $w(z) = C + \int_{\inf S}^{z} n(e(u))du > z$ *for all* $z \in S$*, where* $C > \inf S$ *when* $h < h_0$ *(and* $C = \inf S$ *otherwise).*

In particular, $w(z) > z$ *for all* $z \notin \mathrm{cl} I$ *(so* $w(z) > z$ *for all* $z \in [0, 1]$ *when* $h < h_0$*).*

The equilibrium without AI—which we illustrate in Figure 3—has several salient features. First, it is efficient, as the First Welfare Theorem holds in this economy (with perfect competition, complete information, and only pecuniary externalities). Moreover, the wage of an individual with knowledge z corresponds to her marginal product, defined as the increase in output from introducing one additional human with this knowledge into the economy.



(a) Pre-AI Equilibrium when $h < h_0$

(b) Pre-AI Equilibrium when $h > h_0$

Figure 3: Illustration of the Pre-AI Equilibrium.

Notes. Distribution of knowledge: $G(z) = z$. Parameter values: For panel (a), $h = 1/2$ ($< h_0 = 3/4$), while for panel (b), $h = 0.8125$ ($> h_0 = 3/4$). The thick line depicts the equilibrium wage function. The dashed arrows illustrate the matching between workers and solvers.

Second, the equilibrium exhibits occupational stratification: Workers are less knowledgeable than independent producers, who are, in turn, less knowledgeable than solvers. Intuitively, more knowledgeable individuals have a comparative advantage in specialized problem solving, as this allows them to leverage their knowledge by applying it to more than one problem. Hence, $(W \cup I) \preceq S$. Similarly, less knowledgeable individuals have a comparative advantage in assisted rather than independent production work, as they are less likely to succeed on their own. Consequently, $W \preceq I$.

Third, there is strict positive assortative matching: More knowledgeable workers in W match with more knowledgeable solvers in S. The reason is that worker and solver knowledge are complements: For a given team of workers, a more knowledgeable solver increases expected output, while for any given solver, more knowledgeable workers increase team size.

Fourth, the set of workers and the set of solvers are always nonempty, but the set of independent producers is empty when the communication cost h is below a threshold $h_0 \in (0, 1)$. Intuitively, lower values of h make two-layer organizations more attractive than single-layer ones, leading all humans to be employed by two-layer organizations.

Fifth, workers and solvers earn strictly more than their expected output as independent producers (except in the case of the most knowledgeable worker and the least knowledgeable solver when $h \geq h_0$). The reason for this is that the marginal value of knowledge is strictly lower than 1 for workers (as their knowledge is used to free up solver time),[18] exactly equal to 1 for independent producers (as their expected output equals their knowledge), and strictly greater than 1 for solvers (as they can use their knowledge in more than one problem). Hence, if solvers earned their expected output as independent producers, then all two-layer firms would try to hire the most knowledgeable individuals as solvers. Similarly, if workers earned their expected output as independent producers, then all two-layer firms would try to hire the least knowledgeable individuals as workers.

Finally, the equilibrium wage function w is continuous, strictly increasing, and convex (strictly so when $z \in W \cup S$). The wages for the different occupations are obtained as follows. For independent producers, $w(z) = z$ is an immediate implication of the zero-profit condition of single-layer firms. For workers and solvers, consider the problem of a two-layer organization that recruited $n(z)$ workers with knowledge $z \in W$ and is deciding which solver $s \in S$ to hire:

$$\max_{s \in S} \Big\{ n(z)[s - w(z)] - w(s) \Big\}$$

The corresponding first-order condition evaluated at $s = m(z)$ implies that $w'(m(z)) = n(z)$, or, equivalently, $w'(z) = n(e(z))$ for any $z \in S$. Thus $w(z) = C + \int_{\inf S}^{z} n(e(u))du$ for any $z \in S$, where the constant C is chosen so that the wage function is continuous, as the latter condition is necessary for market clearing. The wages of workers are then determined by the zero-profit condition of two-layer

[18] A marginal increase in the knowledge of a worker with knowledge z frees $h < 1$ units of her solver's time. This allows her firm to hire $1/(1 - z)$ extra workers, with an expected net output gain of $(m(z) - w(z))/(1 - z) < 1$ (as the wage $w(z)$ of that worker is strictly greater than z in the interior of W).

organizations: $w(z) = m(z) - w(m(z))/n(z)$ for any $z \in W$.

For simplicity, we restrict attention to $h < h_0$. This implies that there are no independent producers in the pre-AI equilibrium. In a previous version of this paper (Ide and Talamàs, 2024), we show that virtually all of our results—except one, discussed in Section 5.2—extend to the case where $h \geq h_0$.

4 The Equilibrium with Autonomous AI

Our objective is to understand the effects of autonomous AI by comparing the pre- and post-AI equilibrium. In this section, we provide a partial characterization of the post-AI equilibrium, focusing on the essential information needed for the comparison (which is carried out in Section 5).

For future reference, we index the post-AI equilibrium using the superscript "$*$" (note that the pre-AI equilibrium has no superscript). Furthermore, recall that $W^* \equiv W_a^* \cup W_p^*$ is the overall set of human workers and that $S^* \equiv S_a^* \cup S_p^*$ is the overall set of human solvers.

Proposition 2. *In the presence of an autonomous AI, there is a unique equilibrium. The equilibrium is efficient and has the following features:*

- *Occupational stratification: $W^* \preceq I^* \preceq S^*$.*
- *No worker is more knowledgeable than AI, and no solver is less knowledgeable than AI: $W^* \preceq \{z_{\mathrm{AI}}\} \preceq S^*$.*
- *Positive assortative matching: $m^* : W_p^* \to S_p^*$ is strictly increasing and $W_a^* \preceq W_p^*$ and $S_p^* \preceq S_a^*$.*

Furthermore, AI is always used for independent production, and whether it is also used as a worker or as a solver depends on its knowledge level ***relative to the pre-AI equilibrium****:*

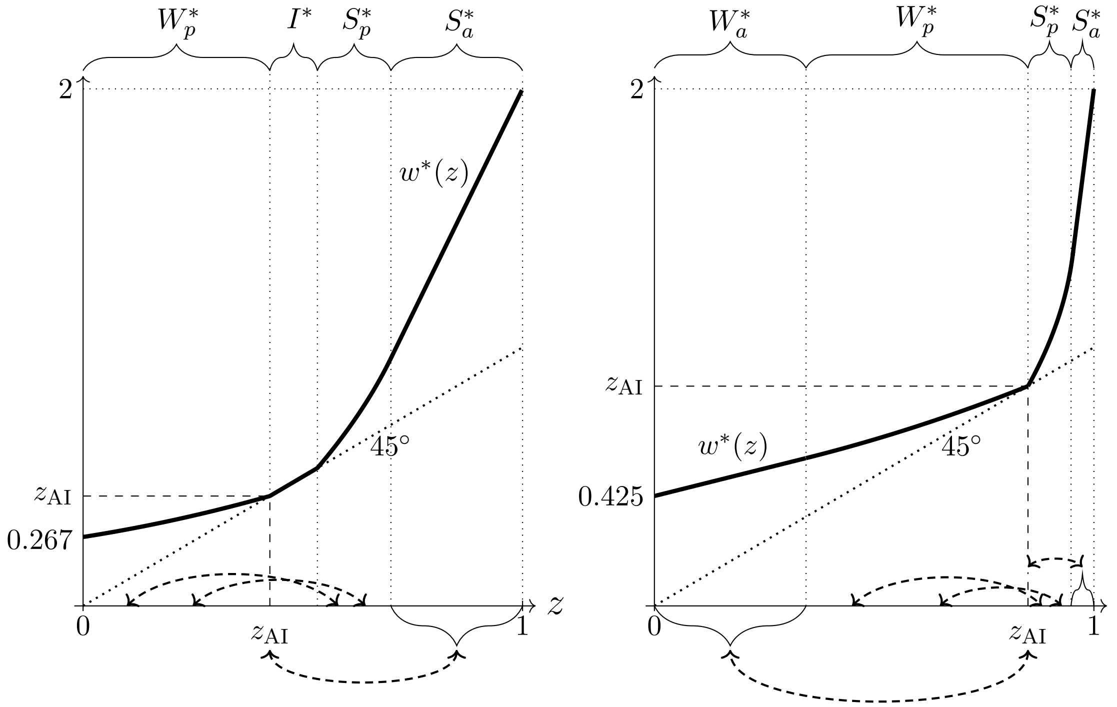
- *If $z_{\mathrm{AI}} \in W$, then AI is necessarily used as a worker (and possibly also as a solver).*
- *If $z_{\mathrm{AI}} \in S$, then AI is necessarily used as a solver (and possibly also as a worker).*

Finally, the rental rate of compute r^* is equal to z_{AI}, and the wage function w^* is continuous, strictly increasing, and convex (strictly so when $z \in W_p^* \cup S_p^*$), and is given by:

- *$w^*(z) = z_{\mathrm{AI}}(1 - 1/n(z)) > z$ for all $z \in W_a^*$.*
- *$w^*(z) = m^*(z) - w^*(m^*(z))/n(z)$ for all $z \in W_p^*$.*
- *$w^*(z) = z$ for all $z \in I^*$.*
- *$w^*(z) = C^* + \int_{\inf S_p^*}^{z} n(e^*(u))du$ for all $z \in S_p^*$, where $C^* = \inf S_p^*$.*
- *$w^*(z) = n(z_{\mathrm{AI}})(z - z_{\mathrm{AI}}) > z$ for all $z \in S_a^*$.*

In particular, $w^*(\sup W^*) = \sup W_p^*$, $w^*(z_{\mathrm{AI}}) = z_{\mathrm{AI}}$, and $w^*(\inf S^*) = \inf S_p^*$.

Figure 4 illustrates the post-AI equilibrium in two scenarios. In panel (a), AI agents serve only as workers and independent producers but not as solvers. In panel (b), AI agents perform all three



(a) AI is used as a worker but not as a solver (b) AI is used as a worker and a solver

Figure 4: Illustration of the Post-AI Equilibrium for Two Different Values of z_{AI}

Notes. Distribution of knowledge: $G(z) = z$. Parameter values: Both panels have $h = 1/2$. For panel (a), $z_{\mathrm{AI}} = 0.425$, while for panel (b), $z_{\mathrm{AI}} = 0.85$. The thick line depicts the equilibrium wage function. The dashed arrows illustrate the matching between workers and solvers, both humans and AI. Human workers in W_p^* are endogenously matched with the human solvers in S_p^* according to m^*. All the humans in W_a^* are matched with an AI solver, while all humans in S_a^* are matched with AI workers.

roles. In both cases, AI agents are the most knowledgeable workers, so they are assisted by the most knowledgeable human solvers. In panel (b), AI agents are also the least knowledgeable solvers, so they assist the production work of the least knowledgeable human workers. Regardless of the scenario, humans with knowledge z_{AI} earn their output as independent producers because they are perfect substitutes for AI agents (each priced at $r^* = z_{\mathrm{AI}}$).

Proposition 2 has three parts. The first part states the basic properties of the equilibrium. The second part describes how firms use AI agents as a function of their knowledge. The third part characterizes the equilibrium prices.

Let us consider part one. As in the pre-AI world, the post-AI equilibrium is unique and efficient, exhibits occupational stratification, and features positive assortative matching. The remaining properties follow from occupational stratification, positive assortative matching, and AI's scalability. Indeed, because compute is abundant relative to time, some AI agents must engage in independent production. Hence, occupational stratification implies no worker is more knowledgeable than AI, and no solver is less knowledgeable than AI (i.e., $W^* \preceq \{z_{\mathrm{AI}}\} \preceq S^*$). Moreover, by positive assortative matching, if AI is used as a worker, then it is supervised by the most knowledgeable solvers (i.e., $S_p^* \preceq S_a^*$), and if AI is used as a solver, then it assists the least knowledgeable workers (i.e., $W_a^* \preceq W_p^*$).

For part two, the result that AI agents are necessarily used as workers when $z_{\text{AI}} \in W$ is intuitive. Indeed, AI agents must be employed either as workers or solvers; otherwise, they would not interact with humans, so humans would match exactly as in the pre-AI equilibrium, preserving pre-AI wages. This would imply $w^*(z_{\text{AI}}) = w(z_{\text{AI}}) > z_{\text{AI}}$, a contradiction.

Hence, if AI agents are not workers, they must be solvers. However, this would imply that, post-AI, the economy has more solvers and weakly fewer workers than before AI (as $W^* \preceq \{z_{\text{AI}}\}$), violating market clearing. A similar logic explains why AI agents are necessarily used as solvers in equilibrium when $z_{\text{AI}} \in S$. Determining whether AI agents are simultaneously used in all three roles when $z_{\text{AI}} \in W$ or $z_{\text{AI}} \in S$ is not fundamental for our main results. For this reason, we relegate such details to the Online Appendix.

Finally, the third part of the proposition addresses equilibrium prices. The result that $r^* = z_{\text{AI}}$ follows because some compute must be used for independent production, and single-layer automated firms must break even. The wages of the humans employed by tA and bA firms—those in W_a^* and S_a^*—are determined by the zero-profit conditions of these firms, along with the fact that $r^* = z_{\text{AI}}$:

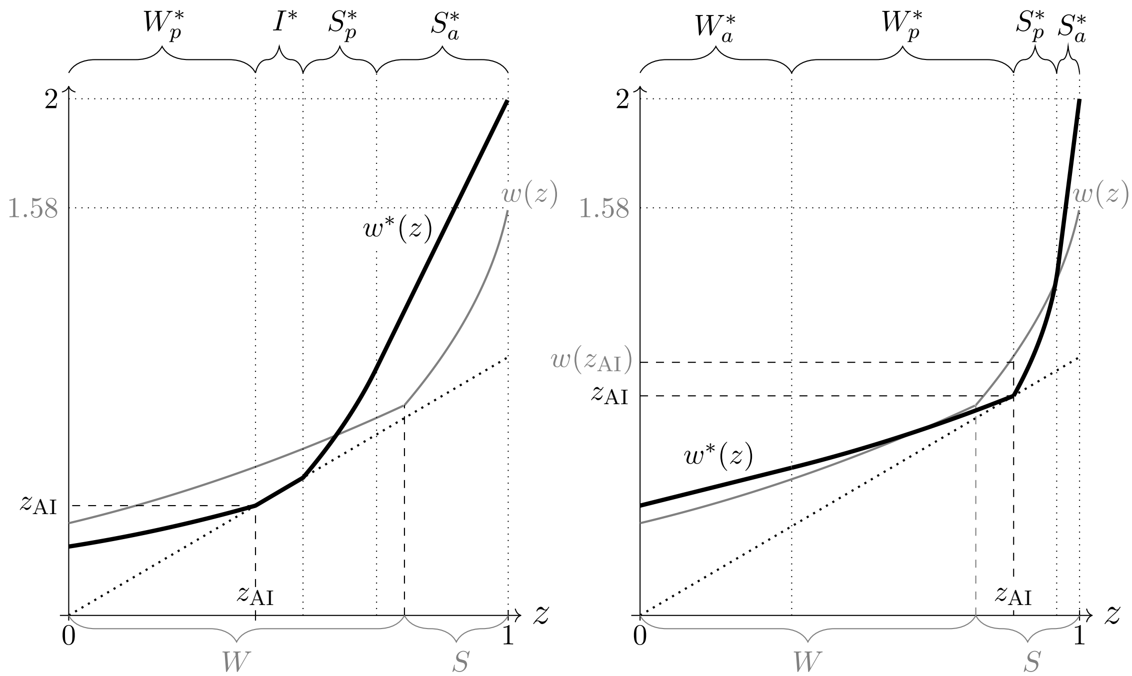
$$n(z)[z_{\text{AI}} - w^*(z)] - r^* = 0 \Longrightarrow w^*(z) = z_{\text{AI}}(1 - 1/n(z)), \text{ for any } z \in W_a^*$$
$$n(z_{\text{AI}})[z - r^*] - w^*(z) = 0 \Longrightarrow w^*(z) = n(z_{\text{AI}})(z - z_{\text{AI}}), \text{ for any } z \in S_a^*$$

The wages of the human workers and solvers hired by nA firms—those in W_p^* and S_p^*—are derived following a similar logic as in the pre-AI equilibrium. The only difference is that the most knowledgeable workers and least knowledgeable solvers earn their expected output as independent producers (while they earn strictly more than that in the pre-AI equilibrium). The reason for this is that the equilibrium wage function must be continuous, and there is always some independent production in the post-AI equilibrium (done by AI agents and possibly also humans).

5 The Impact of Autonomous AI

We now analyze the impact of an autonomous AI by comparing the pre- and post-AI equilibrium. Figure 5 illustrates this comparison, showing substantial changes in three areas: (i) occupational choice, (ii) the matching of workers and solvers, and (iii) the distribution of labor income. In the Online Appendix, we further examine how these changes influence the distribution of firm size, productivity, and decentralization.

For brevity, we focus on two cases: (i) ”basic” AI, which creates AI agents with the knowledge of pre-AI workers ($z_{\text{AI}} \in \text{int}W$), and (ii) ”advanced” AI, which creates AI agents with the knowledge of pre-AI solvers ($z_{\text{AI}} \in \text{int}S$). The results for the knife-edge case where AI has the knowledge of both



(a) Basic AI (i.e., $z_{\text{AI}} \in \text{int}W$)

(b) Advanced AI (i.e., $z_{\text{AI}} \in \text{int}S$)

Figure 5: Comparison of the Pre- and Post-AI Equilibrium

Notes. Distribution of knowledge: $G(z) = z$. Parameter values: For both panels, $h = 1/2$ ($< h_0 = 3/4$). For panel (a), $z_{\text{AI}} = 0.425$, while for panel (b), $z_{\text{AI}} = 0.85$. In the post-AI equilibrium depicted in panel (a), AI is used as a worker and independent producer. In panel (b), AI is used in all three possible roles.

($z_{\text{AI}} \in W \cap S$) are in the Online Appendix.

5.1 Occupational Choice and Worker-Solver Matching

We begin by analyzing the effects of AI on occupational choice.

Proposition 3. *If $z_{\text{AI}} \in \text{int}W$, AI displaces humans from routine knowledge work to specialized problem solving, i.e., $W^* \subset W$ and $S^* \supset S$. In contrast, if $z_{\text{AI}} \in \text{int}S$, AI displaces humans in the opposite direction, i.e., $W^* \supset W$ and $S^* \subset S$.*

Intuitively, when AI is basic ($z_{\text{AI}} \in \text{int}W$), it serves as a relatively inexpensive technology for routine work, lowering workers' wages. This increases demand for solvers to match with the less expensive, more abundant workers (both humans and AI), drawing marginal pre-AI workers into specialized problem solving. Conversely, when AI is advanced ($z_{\text{AI}} \in \text{int}S$), it serves as a relatively inexpensive technology for specialized problem solving. This increases the demand for workers to match with the less expensive, more abundant solvers (both humans and AI), inducing the marginal pre-AI solvers to shift to routine knowledge work.

Next, we examine how AI affects the productivity of pre-AI workers who continue as workers and the span of control of pre-AI solvers who continue as solvers. We define a worker's (average) productivity as her expected output per unit of time. Thus, it is equal to the knowledge of the solver with whom she is matched. Similarly, a solver's span of control is equal to the number of workers (human or AI) under her supervision. Hence, a solver's span of control is increasing in the knowledge of the workers with whom she is matched. For the following result, recall that $e(z)$ is the pre-AI equilibrium employee function:

Proposition 4.

- *If $z_{\mathrm{AI}} \in \mathrm{int}W$, then the productivity of $z \in W^* \subset W$ is strictly lower post-AI than pre-AI. Moreover, the span of control of $z \in S \subset S^*$ is strictly larger post-AI than pre-AI if $e(z) < z_{\mathrm{AI}}$, and strictly smaller post-AI than pre-AI if $e(z) > z_{\mathrm{AI}}$.*
- *If $z_{\mathrm{AI}} \in \mathrm{int}S$, then the productivity of $z \in W \subset W^*$ is strictly higher post-AI than pre-AI if $z < e(z_{\mathrm{AI}})$, and strictly lower post-AI than pre-AI if $z > e(z_{\mathrm{AI}})$. Moreover, the span of control of $z \in S^* \subset S$ is strictly larger post-AI than pre-AI.*

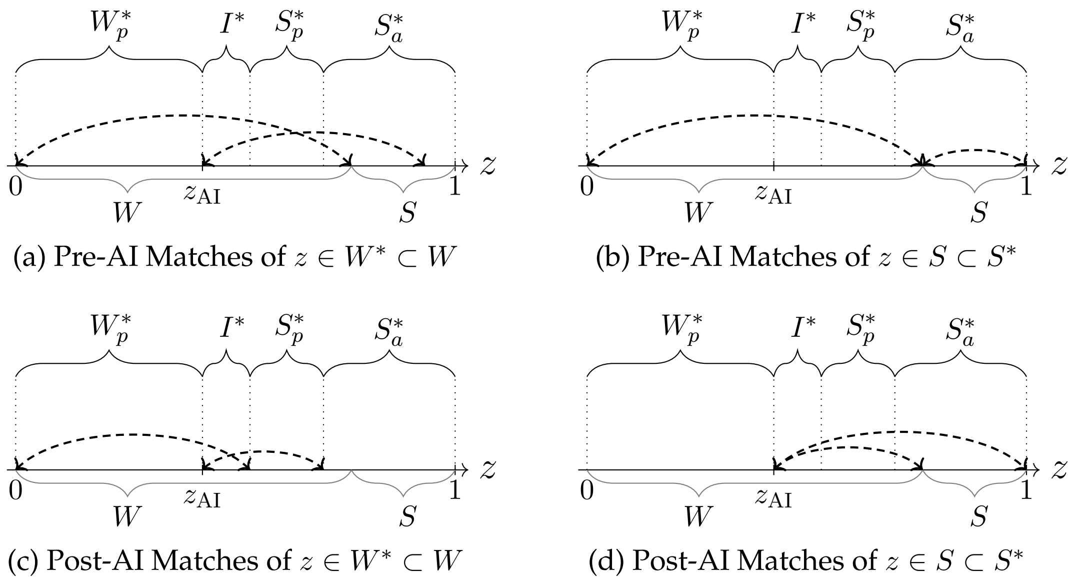
Proposition 4 is illustrated in Figure 6 when $z_{\mathrm{AI}} \in \mathrm{int}W$ and AI is used as a worker and as an independent producer (but not as a solver). As the figure shows, AI worsens the matches of all pre-AI workers who remain workers (i.e., those humans in W_p^*), reducing their productivity. Moreover, AI improves the match of the least knowledgeable solvers who are solvers both pre- and post-AI—increasing their span of control—but worsens the match of the most knowledgeable solvers.

For intuition, consider the case $z_{\mathrm{AI}} \in \mathrm{int}W$ (the argument for $z_{\mathrm{AI}} \in \mathrm{int}S$ is similar). In this case, (i) AI agents are the best workers in the post-AI economy and are therefore assisted by the most knowledgeable solvers (Proposition 2), and (ii) the knowledge of the marginal solvers decreases with AI's introduction (Proposition 3). Both effects worsen the pool of solvers available for those pre-AI workers who remain workers post-AI, thus reducing their productivity.

Regarding the solvers who are not occupationally displaced, the match of the most knowledgeable solvers worsens with AI's introduction because post-AI, they assist production by AI, while pre-AI, they assist humans who are more knowledgeable than AI. However, the match of the least knowledgeable solvers improves, as post-AI, the least knowledgeable workers are assisted by the newly appointed solvers (humans and possibly AI), who are less knowledgeable than the least knowledgeable pre-AI solvers.

There is anecdotal evidence of these reorganizations in professional services. For example, a 2022 survey of M&A lawyers found that AI, by automating routine knowledge work such as document review, has allowed junior lawyers to focus on more complex tasks, including client engagement and legal analysis (Litera, 2022). This aligns with Proposition 3, as the introduction of basic AI pushes marginal human workers into problem-solving roles.

Similarly, in investment banking, Beane and Anthony (2024) document that junior analysts are



(a) Pre-AI Matches of $z \in W^* \subset W$ (b) Pre-AI Matches of $z \in S \subset S^*$

(c) Post-AI Matches of $z \in W^* \subset W$ (d) Post-AI Matches of $z \in S \subset S^*$

Figure 6: An Illustration of Proposition 4 - $z_{\text{AI}} \in \text{int}W$

Notes. Distribution of knowledge: $G(z) = z$. Parameter values: $h = 1/2$ and $z_{\text{AI}} = 0.425$. In the post-AI equilibrium shown in the figure, AI is used as a worker and independent producer. The dashed arrows illustrate the matching between workers and solvers.

becoming increasingly distanced from senior partners as AI takes over tasks that once required junior involvement. This is consistent with Proposition 4, which indicates that basic AI forces those workers who remain in their positions to rely on less knowledgeable individuals for support, as the most knowledgeable pre-AI solvers shift their focus to assisting AI-driven production.

5.2 Labor Income

We now analyze the effects of AI on total labor income and its distribution. To begin, note that total labor income increases with AI's introduction: if this were not the case, then the set of firms that existed pre-AI could collectively earn strictly positive aggregate profits post-AI—a contradiction. This also immediately implies that total output increases with the introduction of AI, as it is the sum of labor income and capital income, and pre-AI, capital income is zero in this knowledge economy.

Next, we examine AI's impact on the distribution of labor income. Analyzing these distributional effects is challenging due to the presence of two potentially opposing forces. On one hand, AI alters firm composition and, therefore, the quality of matches (as shown by Proposition 4). On the other hand, by automating humans with knowledge z_{AI}, AI shifts the relative scarcities of different knowledge levels, influencing how each firm's output is divided between workers and solvers.

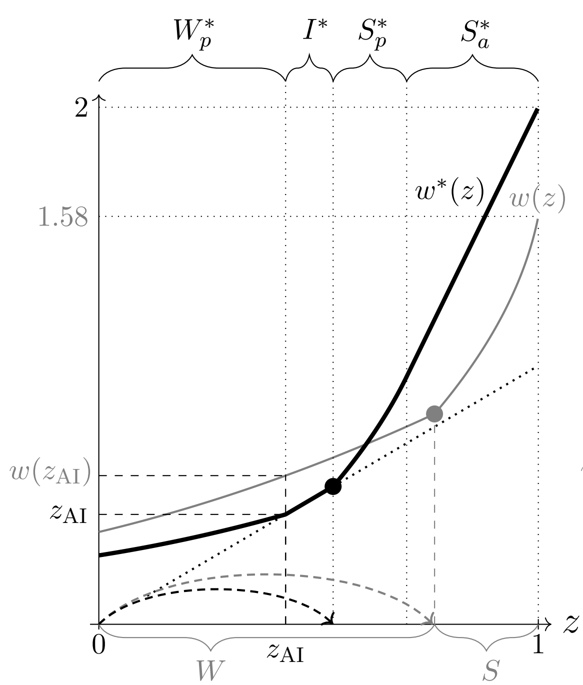
To begin, note that the individual with knowledge z_{AI} always loses from AI, i.e., $w^*(z_{\text{AI}}) < w(z_{\text{AI}})$. Define then B and T as the sets of individuals below and above z_{AI} who benefit from AI, respectively, i.e., $B \equiv \{z \in [0, z_{\text{AI}}] : w^*(z) > w(z)\}$ and $T \equiv \{z \in [z_{\text{AI}}, 1] : w^*(z) > w(z)\}$. As we show in the Online

Appendix, B takes the form $[0, z_b)$, while T takes the form $(z_t, 1]$ for some $z_b \geq 0$ and $z_t \leq 1$. In other words, the individuals who gain from AI are necessarily located at the extremes of the knowledge distribution. The next proposition characterizes when B and T are non-empty:

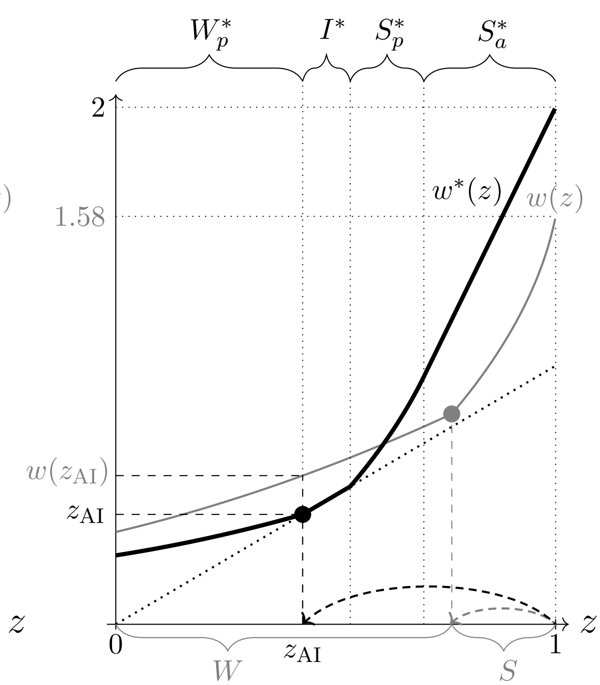
Proposition 5. *There are winners at the bottom if AI is good enough, i.e., $B \neq \emptyset$ if and only if $z_{\text{AI}} > \bar{z}_{\text{AI}}$, where $\bar{z}_{\text{AI}} \in \text{int}W$. In contrast, there are always winners at the top, i.e., $T \neq \emptyset$ for all $z_{\text{AI}} \in [0, 1)$.*

Figure 5, at the beginning of this section, provides an illustration of Proposition 5. As shown in panel (a) of the figure, only the most knowledgeable humans benefit when z_{AI} is relatively low. In contrast, as shown in panel (b), both the least and the most knowledgeable humans benefit when z_{AI} is relatively high.

We now provide intuition for this proposition. We do this using Figure 7, which builds on Figure 5(a) and illustrates the pre- and post-AI scenarios when $z_{\text{AI}} \in \text{int}W$. Since B takes the form $[0, z_b)$ and T takes the form $(z_t, 1]$, to verify the existence of winners below or above z_{AI}, it suffices to analyze AI's impact on the wages of the least and most knowledgeable individuals. Accordingly, Figure 7(a) shows how AI affects the match of individuals with $z = 0$ and the wages of their match before and after AI, while Figure 7(b) provides the same analysis for individuals with $z = 1$.



(a) The Least Knowledgeable Individuals



(b) The Most Knowledgeable Individuals

Figure 7: Decomposing the Effects of AI on Wages

Notes. Distribution of knowledge: $G(z) = z$. Parameter values: $h = 1/2$ ($< h_0 = 3/4$) and $z_{\text{AI}} = 0.425$, so AI is used as a worker and independent producer. Panel (a) shows how AI affects the match of $z = 0$ and the wages of their match before and after AI, while panel (b) provides the same analysis for $z = 1$.

Consider Figure 7(a) first. AI's introduction has two opposing effects on the wages of the least knowledgeable individuals. On the one hand, AI reduces their wages by lowering their productivity, as they are now matched with a less knowledgeable solver (a negative "match" effect). On the other hand, AI increases their wages by allowing them to appropriate a larger share of the firm's output (a positive "share" effect). This occurs because the least knowledgeable solvers (with whom they are matched) now earn their expected output as independent producers—their wages fall on the 45° line—whereas pre-AI, they earned strictly more.

Thus, when $z_{\text{AI}} \in \text{int}W$, AI benefits the least knowledgeable individuals if the positive share effect outweighs the negative match effect. This happens when z_{AI} is sufficiently high (i.e., when $z_{\text{AI}} > \bar{z}_{\text{AI}}$, where $\bar{z}_{\text{AI}} \in \text{int}W$). Conversely, when $z_{\text{AI}} \in \text{int}S$, the least knowledgeable individuals always benefit from AI's introduction because both the match and share effects are positive: AI always improves the match of the least knowledgeable individuals in this case (see Proposition 4).

Now consider Figure 7(b). As in the case of the least knowledgeable individuals, AI generates both a negative match effect and a positive share effect for the most knowledgeable individuals. However, in this case, the positive share effect always dominates, explaining why there is always a set of winners above z_{AI}. The key driver behind this result is that $h < h_0$, meaning the size of the team each solver assists is large. Consequently, even a small decrease in the wages of the most knowledgeable workers (with whom the most knowledgeable individuals are matched) is amplified by team size, helping the positive share effect outweigh any negative match effect. In a previous version of this paper (Ide and Talamàs, 2025), we show that AI can sometimes harm the most knowledgeable individuals when $h \geq h_0$.[19]

6 Non-Autonomous AI

In our baseline model, AI is a technology that transforms units of compute into AI agents that can do exactly the same as humans with knowledge z_{AI}. This type of AI is "autonomous" because it creates agents that function both as co-workers (pursuing production opportunities) and as co-pilots (providing advice). We focus on this case first, as it generates the most anxiety and debate, driven by technology firms developing and deploying AI agents capable of independent cognitive work.

However, for technical or regulatory reasons, many existing AI systems lack the ability to operate autonomously. In this section, we analyze the impact of non-autonomous AI—a technology that creates agents that operate only as co-pilots. Our main finding is that non-autonomous AI tends to benefit the least knowledgeable individuals the most, whereas autonomous AI primarily benefits the most knowledgeable individuals. However, autonomous AI generates higher overall output than

[19]The finding that the most knowledgeable individuals always benefit from AI's introduction depends not only on $h < h_0$ but also on the assumption that $z_{\text{AI}} \in [0, 1)$, meaning AI is not "superintelligent." As we show in the Online Appendix, when $z_{\text{AI}} = 1$, the most knowledgeable humans necessarily lose from AI's introduction.

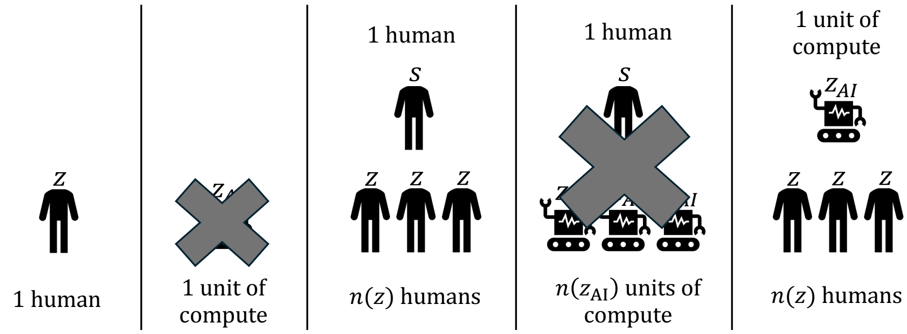


Figure 8: The Three Possible Firm Configurations with Non-Autonomous AI

non-autonomous AI.

Formally, the model is the same as in Section 3.1, except that AI agents cannot pursue production opportunities. As a result, single-layer and bottom-automated firms do not emerge in equilibrium since they produce no output (see Figure 8). Thus, $S_a = \emptyset$ and $\mu_w = 0$. In this context, we interpret the amount of compute rented for independent production, μ_i, as the compute that remains idle (i.e., not productive).

To ensure a meaningful comparison of equilibrium outcomes between autonomous and non-autonomous AI, we maintain the assumption that firms have at most two layers. Additionally, we assume that compute availability is identical and abundant relative to time in both scenarios. In the Online Appendix, we extend this analysis to different compute levels. To distinguish between cases, we use the superscript "$\star$" for equilibrium outcomes with non-autonomous AI and "$*$" for those with autonomous AI.

6.1 The Impact of Non-Autonomous AI

Since compute is abundant relative to time and AI is non-autonomous, the equilibrium rental rate of compute is zero, $r^\star = 0$. The reason is that some compute must remain idle. However, in contrast to the case of an autonomous AI, the wage of an individual with knowledge z_{AI} is no longer equal to the equilibrium rental rate of compute ($w^\star(z_{\text{AI}}) \neq r^\star = 0$). This difference occurs because an AI agent is no longer a perfect substitute for a human with knowledge z_{AI}.

For the following result, recall that $w(z)$ denotes the wage schedule of the pre-AI equilibrium:

Proposition 6. *In the presence of a non-autonomous AI, there is a unique equilibrium. The equilibrium is efficient, maximizes labor income, and the rental rate of compute is $r^\star = 0$. Moreover,*

- *If $z_{\text{AI}} \leq w(0)$, then AI is not used in equilibrium, i.e., $W_a^\star = \emptyset$. Hence, human occupations and wages coincide with the pre-AI ones.*

- *If $z_{\mathrm{AI}} > w(0)$, then only the least knowledgeable individuals use AI as a solver, i.e., $W_a^\star \preceq (W_p^\star \cup I^\star \cup S_p^\star)$, where all these sets are not empty except possibly $I^\star$.*

In any case,

1. *Overall output is strictly higher with autonomous than non-autonomous AI.*
2. *Non-autonomous AI creates losers:*
   $\exists\, z \in (0, 1]$ *such* $w^\star(z) \leq w(z)$ *(with strict inequality if* $z_{\mathrm{AI}} > w(0)$*).*
3. *The least knowledgeable benefit more from non-autonomous AI than from no AI or autonomous AI:*
   $\exists\, \epsilon > 0$ *such that, for all* $z \in [0, \epsilon)$, $w^\star(z) \geq \max\{w(z), w^*(z)\}$ *(with strict inequality if* $z_{\mathrm{AI}} > w(0)$*).*
4. *The most knowledgeable benefit more from autonomous AI than from non-autonomous AI:*
   $\exists\, \epsilon > 0$ *such that, for all* $z \in (1 - \epsilon, 1]$, $w^\star(z) \leq w^*(z)$ *(with strict inequality if* $z \neq 1$*).*

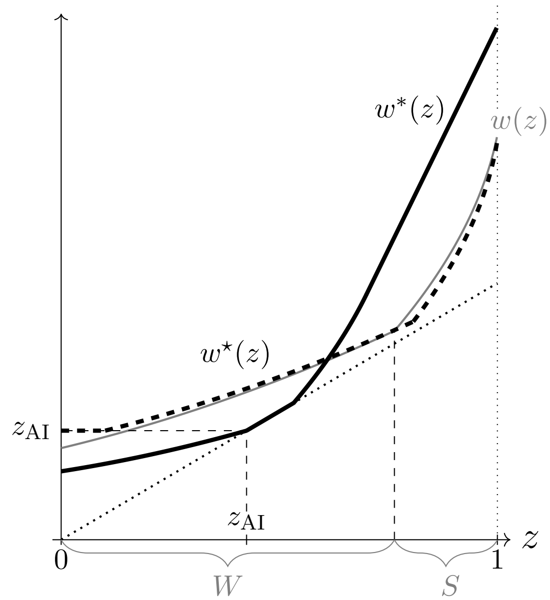
Figure 9 illustrates Proposition 6.[20] According to this proposition, a non-autonomous AI is used in equilibrium only if it is sufficiently advanced (i.e., if $z_{\mathrm{AI}} > w(0)$). In this case, individuals with knowledge below a certain threshold work in automated firms—using AI as a solver or co-pilot—while those above the threshold work in non-automated firms. Moreover, total output with non-autonomous AI is strictly lower than with autonomous AI because non-autonomy imposes a binding constraint on compute use, leaving some of it idle.

In terms of wages, the introduction of non-autonomous AI creates both winners and losers, similar to autonomous AI. However, the distribution of gains and losses changes dramatically. Non-autonomous AI always benefits the least knowledgeable individuals relative to the pre-AI equilibrium. Moreover, these benefits exceed those of autonomous AI. In contrast, non-autonomous AI may lead to gains or losses compared to the pre-AI scenario for the most knowledgeable individuals. In any case, their wages with non-autonomous AI are always lower than with autonomous AI.

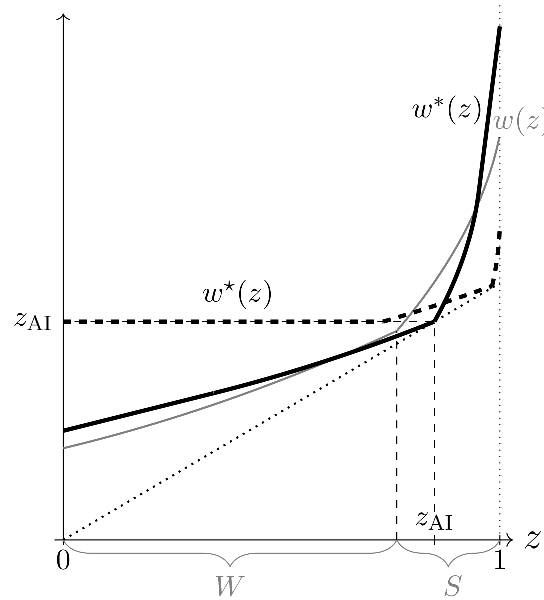
Intuitively, non-autonomous AI benefits the least knowledgeable individuals the most because it functions solely as a solver. This eliminates competition between these individuals and AI for production work and enables them to solve complex problems without human assistance. Moreover, non-autonomous AI's lower opportunity cost—stemming from its inability to produce independently—enhances this advantage by allowing the least knowledgeable to capture a larger share of AI-generated benefits. For the most knowledgeable individuals, the opposite occurs because non-autonomous AI cannot handle routine work and may compete with them in advisory roles.

As noted earlier, we limit firms to a maximum of two layers to ensure a meaningful comparison of equilibrium outcomes between autonomous and non-autonomous AI. This restriction provides

[20] In the Online Appendix, we also examine how non-autonomous AI affects (i) human occupational choices, (ii) the distribution of firm size, productivity, and decentralization, and (iii) the productivity and span of control of workers and solvers who remain in their positions, compared to the pre-AI equilibrium.



(a) Basic AI (i.e., $z_{\text{AI}} \in \text{int}W$)



(b) Advanced AI (i.e., $z_{\text{AI}} \in \text{int}S$)

Figure 9: Non-Autonomous AI vs. Autonomous AI vs. No AI

Notes. Distribution of knowledge: $G(z) = z$. Parameter values: $h = 1/2$. Moreover, for panel (a), $z_{\text{AI}} = 0.425$, while for panel (b), $z_{\text{AI}} = 0.85$. The thick gray line represents the pre-AI equilibrium wage function w. The thick black dashed line depicts the equilibrium wage function with non-autonomous AI, $w^\star$. The thick black line represents the equilibrium wage function with autonomous AI, w^*.

an additional benefit: it implies that problems within the organization can be escalated only once, effectively introducing a cost of seeking help beyond the helping cost h (which is borne solely by the recipient of the question). Notably, this implies that even with free compute, individuals with slightly less knowledge than AI prefer to seek help from another human rather than from AI (see Figure 9), as the probability that AI provides them with helpful assistance is low.[21]

We believe this is a reasonable outcome, as a similar situation would arise if problems could be escalated multiple times but the agents seeking assistance bore part of the time cost associated with requesting help. Without these constraints—such as with an arbitrary and endogenous number of layers and no time cost for the asking party—all individuals less knowledgeable than AI would always rely on the system for help. In this scenario, introducing AI effectively shifts the knowledge of all agents with $z < z_{\text{AI}}$ towards z_{AI}. Nevertheless, the key result of this section—that the least knowledgeable individuals prefer non-autonomous AI while the most knowledgeable favor autonomous AI—remains valid even under this alternative case.[22]

[21] The equilibrium is continuous with respect to the availability of compute, so all our results hold when the price of compute is strictly positive but sufficiently close to zero. In the Online Appendix, we also examine the case where compute is significantly less abundant. The results of this section extend to that scenario as well, albeit with certain qualifications.

[22] With more than two layers, an advanced non-autonomous AI could further amplify the expertise of the most knowl-

7 Discussion and Final Remarks

To conclude, we discuss additional implications of our analysis and link our findings to emerging evidence and stylized facts about AI's effects on the labor market.

Co-pilots vs. Co-workers.—Our analysis uncovers distinct labor market implications when comparing AI co-pilots—which provide problem-solving assistance—to AI co-workers—which independently execute production work. Broadly speaking, AI co-pilots primarily benefit less knowledgeable individuals, reducing inequality in performance and earnings. In contrast, AI co-workers offer the greatest advantages to highly knowledgeable individuals, amplifying the value of their expertise. Co-pilots emerge in equilibrium when AI is either (i) not autonomous or (ii) autonomous but sufficiently advanced, whereas co-workers can arise for any knowledge level but only in the case in which AI is autonomous.

These findings are particularly relevant given the emerging evidence on AI's labor market effects. Several studies suggest that AI disproportionately benefits individuals with the least knowledge and reduces performance inequality (e.g., Dell'Acqua et al., 2023; Noy and Zhang, 2023; Peng et al., 2023; Wiles et al., 2023; Caplin et al., 2024; Brynjolfsson et al., 2025).[23] However, these studies focus on AI co-pilots. While our framework supports these conclusions, it also highlights that they may not extend to autonomous AI that can perform independent knowledge work, such as conducting research, drafting documents, or performing data analysis.

Moreover, non-experimental evidence suggests that firms perceive current iterations of generative AI more as a basic co-worker than as a co-pilot. For instance, using high-frequency job posting data from LinkUp, Berger et al. (2024) document a marked decline in job postings for white-collar roles requiring low levels of knowledge, such as entry-level office jobs, following the introduction of ChatGPT in November 2022. At the same time, job postings for high-knowledge roles, such as executive positions, increased. This trend indicates that firms view generative AI as a technology that complements high-skilled knowledge workers while substituting for their low-skilled counterparts.

Further empirical studies that disentangle and estimate the distinct labor market effects of AI co-workers and AI co-pilots represent an important avenue for future research.

AI vs.Traditional Robots and Physical AI.—Automation technologies that replace manual labor, such as traditional robots and physical AI, differ fundamentally from those that automate cognitive work,

edgeable individuals by giving them access to workers that require less help (as workers would first seek AI assistance before turning to them). However, the key insight remains: the most knowledgeable individuals still prefer autonomous AI over non-autonomous AI, as the former can also perform routine work for them.

[23]For instance, Brynjolfsson et al. (2025) reports that deploying an AI-based conversational assistant significantly improved productivity for less-skilled agents in a Fortune 500 customer support center, with minimal impact on experienced ones. Similarly, Caplin et al. (2024) reported that AI assistance was more beneficial for lower-ability individuals than for higher-ability ones when tasked with estimating whether people in photographs appeared to be over 21 years old.

such as AI and enterprise software. This difference arises because physical machines must be tailored to their specific operating environments, making them best described as a "putty-clay" technology. In contrast, digital machines—such as AI agents—exhibit "putty-putty" characteristics because they rely on compute, which is general-purpose and can be rented on demand. This allows software improvements to be immediately deployed across all available hardware.[24]

A key implication of this distinction is that traditional robots and physical AI are likely to exist in vintages with varying capabilities, while compute is allocated to the most advanced and highest-performing digital machines.

AI vs.Offshoring and Immigration.—The shock induced by AI is different from classical labor shocks such as offshoring or immigration. Non-autonomous AI, for instance, has no clear analog in these frameworks. Moreover, even for autonomous AI, the introduction of the technology diverges significantly from offshoring or immigration. The key distinction lies in AI's scalability. Once knowledge is encoded into an AI system, it can be deployed across all units of compute, which are widely available. This scalability amounts to the introduction of a large population of homogeneous agents.

In contrast, large human populations are inherently heterogeneous in terms of tacit knowledge—different individuals undoubtedly have different experiences. Consequently, offshoring or immigration can involve the introduction of a relatively small population of homogeneous agents or the introduction of a large population of heterogeneous agents, but not the introduction of a large population of homogeneous agents. As we formally show in the Online Appendix, this difference leads to qualitatively different outcomes.

AI vs. ICTs.— A significant implication of our model is that not all firms adopt AI to *automate knowledge work*. This characteristic makes AI, as an automation technology, fundamentally different from earlier waves of digitalization—such as the introduction of computers or email—which saw near-universal adoption across firms. Partial adoption is a common characteristic of automation technologies. Both empirical observations and standard models of automation (e.g., Acemoglu and Loebbing, 2024) suggest that it is often not cost-effective to automate all tasks that can potentially be automated.

For autonomous AI, partial adoption occurs because deploying AI involves a strictly positive opportunity cost: the foregone output that an autonomous AI agent could generate as an independent producer. As a result, adopting autonomous AI may not be economically viable for all firms, consistent with the evidence presented by Svanberg et al. (2024). For non-autonomous AI, partial adoption arises even when the opportunity cost is zero. This is because AI's advice is useless in firms with very knowledgeable human workers.

However, the value of AI extends beyond automating knowledge work. Thus, firms may adopt AI technologies for purposes unrelated to automation, such as reducing communication costs or lower-

[24]Putty-clay models assume that capital and technology are flexible only prior to investment decisions, whereas putty-putty models allow for flexibility both before and after investments are made (Atkeson and Kehoe, 1999; Martinez, 2025).

ing knowledge acquisition costs. The effects of such technologies have been extensively studied (see, e.g., Garicano and Rossi-Hansberg, 2006; Antràs et al., 2008; Bloom et al., 2014; Garicano and Rossi-Hansberg, 2015), and are fundamentally different from the ones of AI as an automation technology.[25]

AI in Advanced vs. Developing Economies.— An important implication of our analysis is that AI's effects depend on its capabilities relative to human pre-AI occupations. Since these occupations are endogenous—shaped by the knowledge of the population and the level of communication technologies—the same AI technology can lead to different outcomes in advanced and developing economies.

Indeed, in economies with superior communication technologies or a more knowledgeable population, the knowledge threshold required to become a solver in the pre-AI equilibrium is higher (see the Online Appendix for details). As a result, the same autonomous AI technology might be considered "basic" in an advanced economy—displacing humans from routine knowledge work to complex problem-solving—while being regarded as "advanced" in a developing economy, thereby displacing humans in the opposite direction.

Autonomy vs. No Autonomy: Policy Implications.— The rise of AI and its potential to transform labor markets has captured policymakers' attention, as evidenced by the recent report of Council of Economic Advisers (2024) and the EU Artificial Intelligence Act (2024). Our findings reveal key tradeoffs in regulating AI autonomy. While autonomous AI generates higher overall output than non-autonomous AI, it also exacerbates labor income inequality.

Using this framework to quantify the tradeoff between output and inequality associated with autonomous and non-autonomous AI is an important direction for future research.

[25]For instance, better communication technologies—modeled as a reduction in the communication cost h—always displace humans from specialized problem-solving towards routine work as the very best solvers can now supervise larger teams. In contrast, the displacement generated by AI as an automation technology depends on its autonomy and its problem-solving capabilities.

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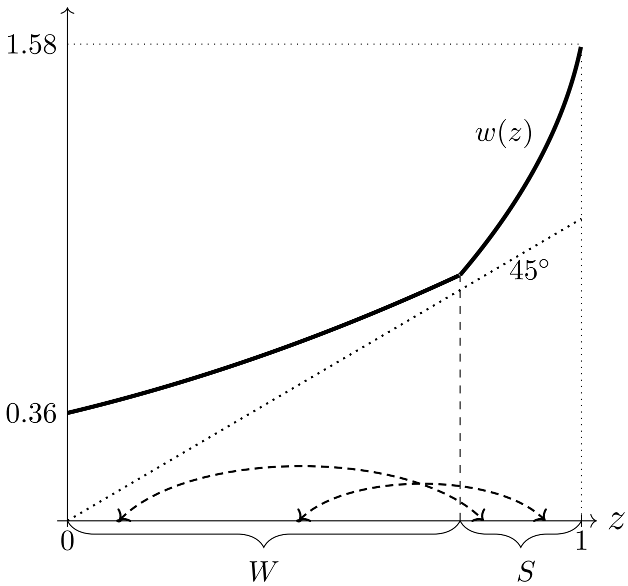
Tables

	Codifiable	**Non-Codifiable**
Manual	Robots	Physical AI
Cognitive	Enterprise Software	AI

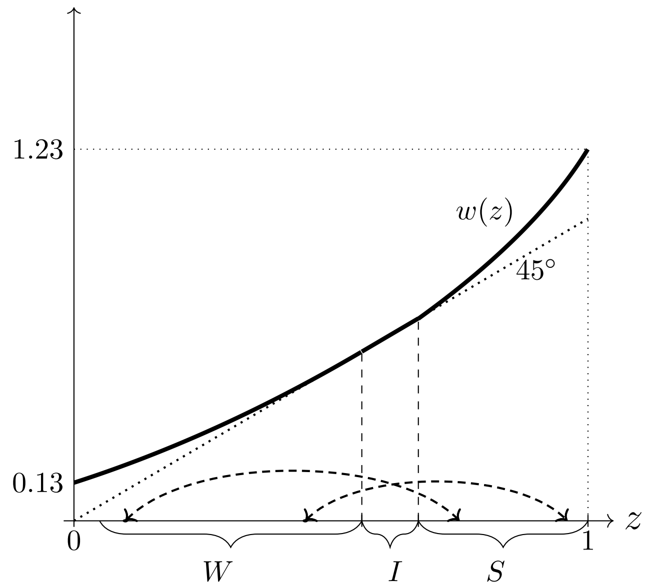
Table 1: The Distinguishing Feature of AI

Notes. The defining feature of AI is its ability to perform cognitive work that relies on non-codifiable knowledge. This distinguishes it from enterprise software and robots, which automate codifiable cognitive and manual work, respectively. AI also differs from "Physical AI" (Nvidia, 2025), a concept that envisions robots powered by AI, which may, in the future, be capable of automating manual work that requires non-codifiable knowledge, such as caregiving or construction.

Figure 3



(a) Pre-AI Equilibrium when $h < h_0$

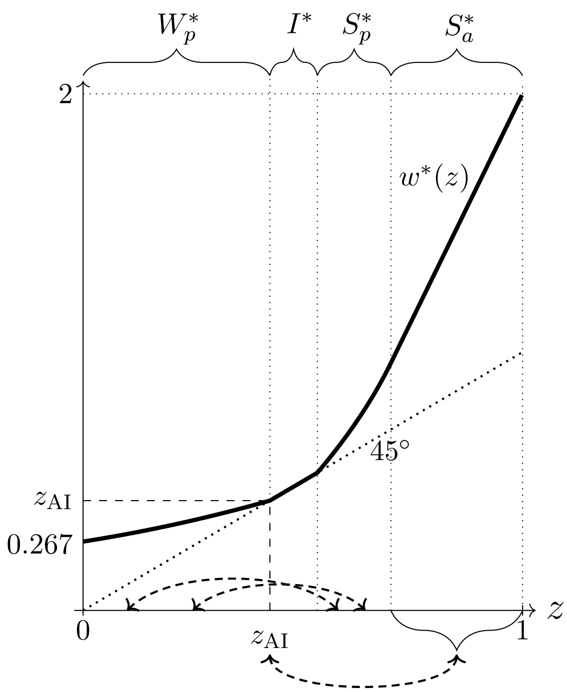


(b) Pre-AI Equilibrium when $h > h_0$

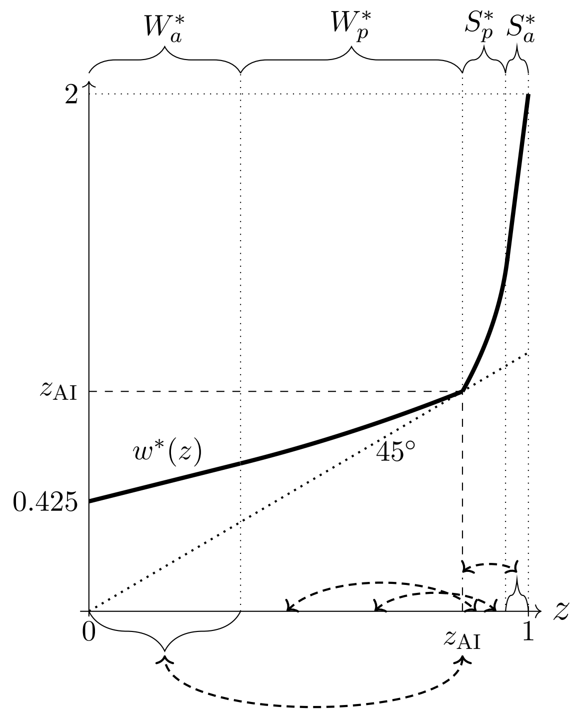
Figure 3: Illustration of the Pre-AI Equilibrium.

Notes. Distribution of knowledge: $G(z) = z$. Parameter values: For panel (a), $h = 1/2$ ($< h_0 = 3/4$), while for panel (b), $h = 0.8125$ ($> h_0 = 3/4$). The thick line depicts the equilibrium wage function. The dashed arrows illustrate the matching between workers and solvers.

Figure 4



(a) AI is used as a worker but not as a solver

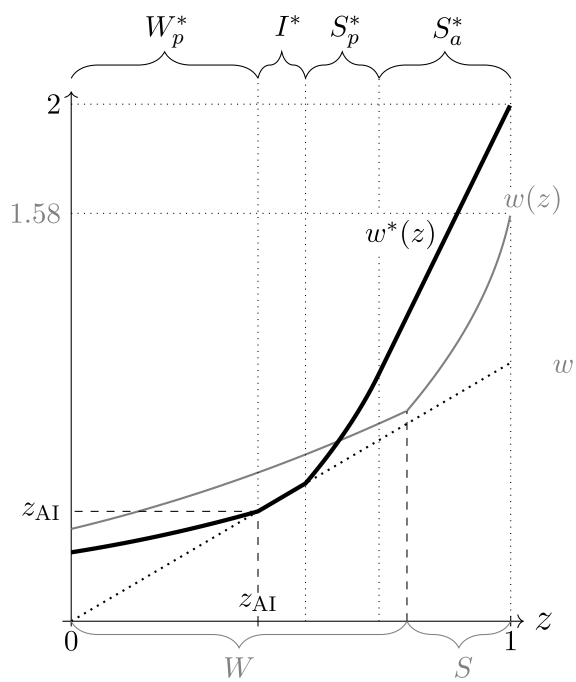


(b) AI is used as a worker and a solver

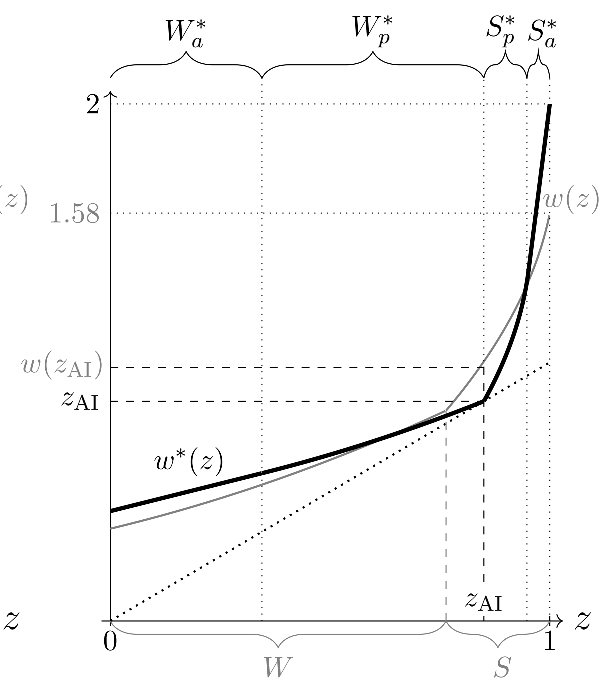
Figure 4: Illustration of the Post-AI Equilibrium for Two Different Values of z_{AI}

Notes. Distribution of knowledge: $G(z) = z$. Parameter values: Both panels have $h = 1/2$. For panel (a), $z_{\mathrm{AI}} = 0.425$, while for panel (b), $z_{\mathrm{AI}} = 0.85$. The thick line depicts the equilibrium wage function. The dashed arrows illustrate the matching between workers and solvers, both humans and AI. Human workers in W_p^* are endogenously matched with the human solvers in S_p^* according to m^*. All the humans in W_a^* are matched with an AI solver, while all humans in S_a^* are matched with AI workers.

Figure 5



(a) Basic AI (i.e., $z_{\text{AI}} \in \text{int}W$)

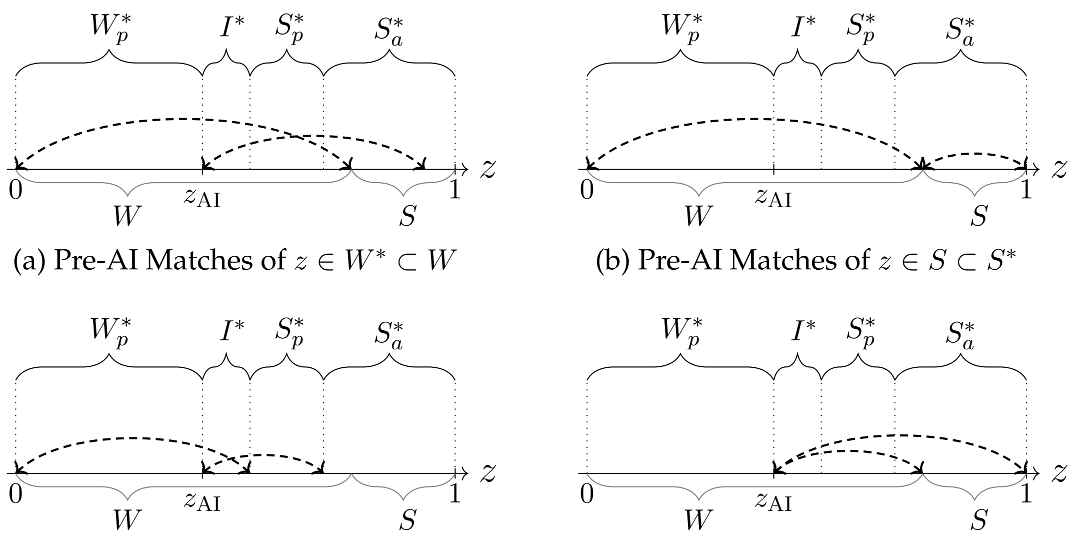


(b) Advanced AI (i.e., $z_{\text{AI}} \in \text{int}S$)

Figure 5: Comparison of the Pre- and Post-AI Equilibrium

Notes. Distribution of knowledge: $G(z) = z$. Parameter values: For both panels, $h = 1/2$ ($< h_0 = 3/4$). For panel (a), $z_{\text{AI}} = 0.425$, while for panel (b), $z_{\text{AI}} = 0.85$. In the post-AI equilibrium depicted in panel (a), AI is used as a worker and independent producer. In panel (b), AI is used in all three possible roles.

Figure 6



(a) Pre-AI Matches of $z \in W^* \subset W$

(b) Pre-AI Matches of $z \in S \subset S^*$

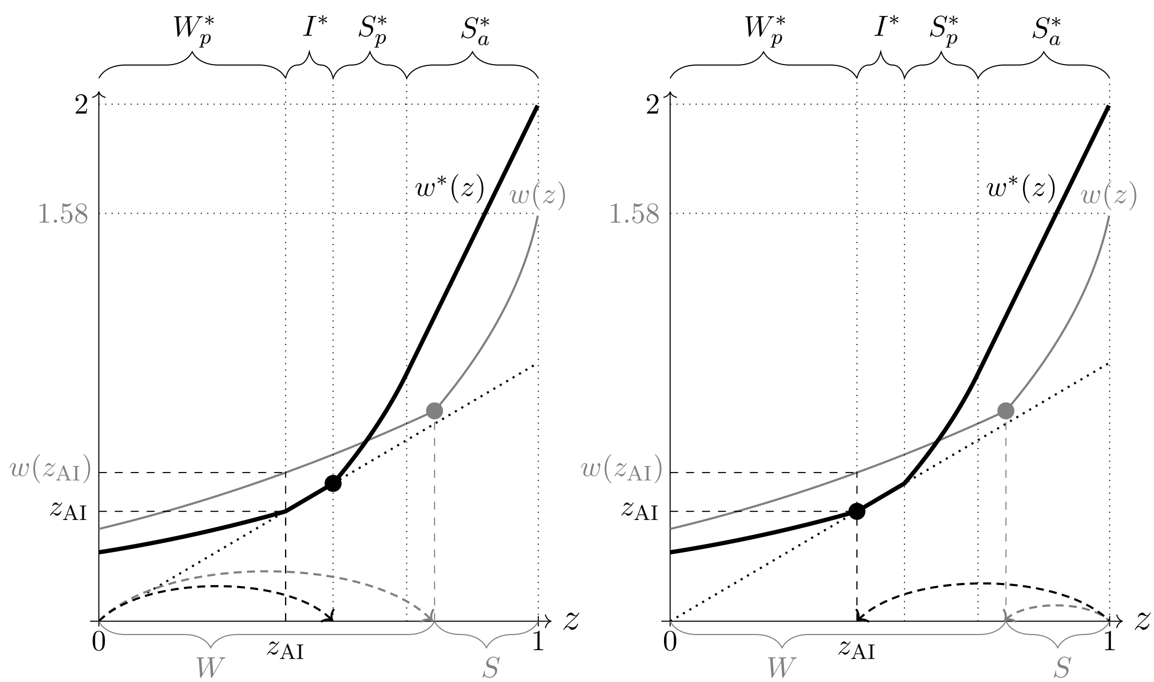
(c) Post-AI Matches of $z \in W^* \subset W$

(d) Post-AI Matches of $z \in S \subset S^*$

Figure 6: An Illustration of Proposition 4 - $z_{\text{AI}} \in \text{int}W$

Notes. Distribution of knowledge: $G(z) = z$. Parameter values: $h = 1/2$ and $z_{\text{AI}} = 0.425$. In the post-AI equilibrium shown in the figure, AI is used as a worker and independent producer. The dashed arrows illustrate the matching between workers and solvers.

Figure 7

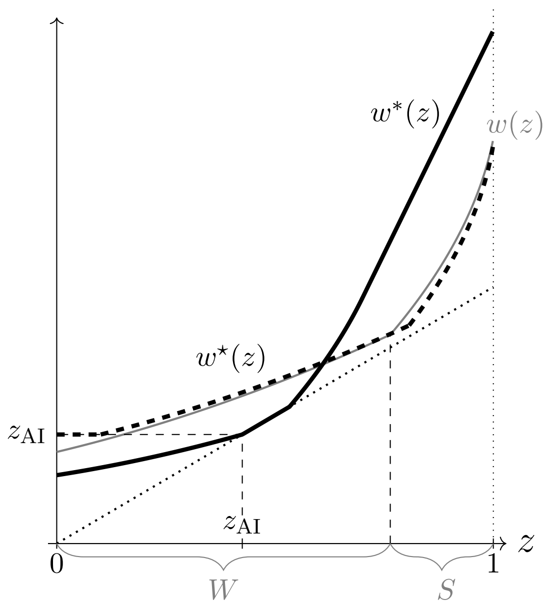


(a) The Least Knowledgeable Individuals (b) The Most Knowledgeable Individuals

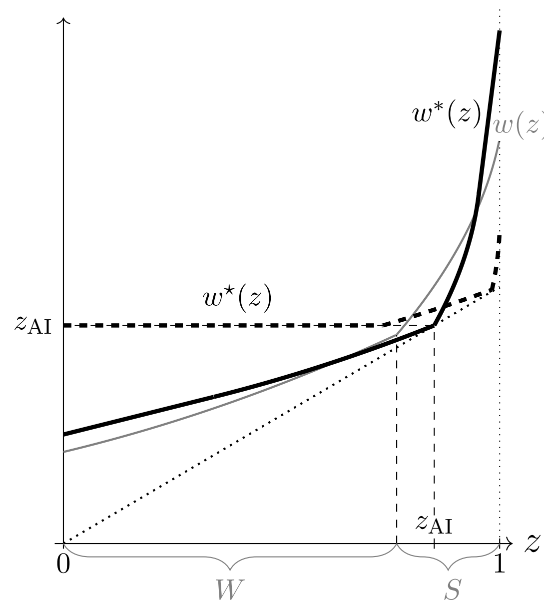
Figure 7: Decomposing the Effects of AI on Wages

Notes. Distribution of knowledge: $G(z) = z$. Parameter values: $h = 1/2$ ($< h_0 = 3/4$) and $z_{\mathrm{AI}} = 0.425$, so AI is used as a worker and independent producer. Panel (a) shows how AI affects the match of $z = 0$ and the wages of their match before and after AI, while panel (b) provides the same analysis for $z = 1$.

Figure 9



(a) Basic AI (i.e., $z_{\text{AI}} \in \text{int}W$)



(b) Advanced AI (i.e., $z_{\text{AI}} \in \text{int}S$)

Figure 9: Non-Autonomous AI vs. Autonomous AI vs. No AI

Notes. Distribution of knowledge: $G(z) = z$. Parameter values: $h = 1/2$. Moreover, for panel (a), $z_{\text{AI}} = 0.425$, while for panel (b), $z_{\text{AI}} = 0.85$. The thick gray line represents the pre-AI equilibrium wage function w. The thick black dashed line depicts the equilibrium wage function with non-autonomous AI, $w^\star$. The thick black line represents the equilibrium wage function with autonomous AI, w^*.

Online Appendix for:
Artificial Intelligence in the Knowledge Economy
(Not for Publication)

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Contents

Chapter 1

Limitations of Our Model and Avenues for Future Research

In this chapter, we discuss some limitations of our model and suggest avenues for future research.

Firms with at Most Two Layers.—To keep the analysis transparent and provide simple, testable predictions, we focus on the simplest hierarchical organizations—those with at most two layers. As discussed in the main text, this simple environment aligns well with emerging experimental and observational evidence on AI's impact on knowledge work. It is also sufficiently rich to demonstrate the reorganizations AI can induce and how these depend on AI's autonomy and problem-solving capabilities.

Allowing for more complex organizational structures could undoubtedly reveal additional subtleties in how AI reorganizes knowledge work. However, our goal is to strike a balance between simplicity and analytical richness. Introducing a more elaborate model risks obscuring the key mechanisms and insights we aim to emphasize. Furthermore, as we show in Chapter 4.6, many insights we uncover—such as the conditions under which the least and most knowledgeable individuals benefit from AI—naturally generalize to settings with an arbitrary number of layers. Nevertheless, a more comprehensive examination of AI's effects on more complex organizational structures is a promising area for future research.

Equal Access to AI.—In light of the rise of foundation models, we assume negligible customization costs and equal access to AI across firms. This serves as a useful first approximation since customizing foundation models typically costs several orders of magnitude less than developing AI systems from scratch (Appenzeller et al., 2023; Xia et al., 2024), with costs likely to decline further.[1] However, certain applications still require domain-specific AI models, as demonstrated by programs like Google's AlphaFold. Developing a multi-sector model of the knowledge economy—where some sec-

[1]Developing more effective and cost-efficient techniques for customizing foundation models is an especially active area in the computer science literature. Techniques such as RAG, LoRA, QLoRA, and PETF, or variations thereof, are being proposed at an accelerated pace. For an overview of the most commonly used methods, see Shah and Patel (2023).

tors rely on shared foundation models while others require specialized models—presents a promising avenue for future research.

Variation in AI's Capabilities Across Sectors.— AI systems often exhibit varying capabilities across industries. For example, Goldman Sachs (2023) estimates that Generative AI's automation potential in healthcare support is roughly half that in office and administrative support. A multi-sector framework that accounts for these differences would offer valuable insights into how AI might drive inter-industry reallocation.

The Industrial Organization of the AI Ecosystem.— For simplicity, we treat AI technology as a given and model the stock of compute as an endowment. This allows us to bypass the complexities of modeling the AI training technology and the industrial organization of the AI sector—such as foundation model developers, application developers, and compute providers. It also keeps the model static. While our model touches on some of these issues—see the discussion on the commoditization of foundation models in Section 3.2 of the main text—a detailed analysis of the AI sector and the drivers of AI and compute development are promising areas for future research.

Chapter 2

Proofs Omitted from the Main Text

2.1 Proof of Proposition 1

As noted in the main text, the pre-AI equilibrium was first described in general by Fuchs et al. (2015). Proposition 1 follows directly from Lemmas 2.1 and 2.2 and Corollary 2.1 below.

To start, note that the First Welfare Theorem holds in this setting (see Fuchs et al., 2015, Online Supplement, p. 1). Thus, the competitive equilibrium is efficient. The following lemma shows that in the pre-AI world, there is a unique efficient allocation:

Lemma 2.1. *In the absence of AI, there is a unique allocation that maximizes output:*

- *When* $h \geq h_0 \in (0,1)$, $W = [0, \underline{z}]$, $I = (\underline{z}, \bar{z})$, *and* $S = [\bar{z}, 1]$. *Moreover, workers and solvers are matched according to the strictly increasing function* $m(z; \bar{z})$ *given by* $\int_{\bar{z}}^{m(z;\bar{z})} dG(u) = \int_0^z h(1-u)dG(u)$. *Finally, the cutoffs* $0 < \underline{z} < \bar{z} < 1$ *satisfy:*

$$\frac{1}{h} - \bar{z} = \int_{\bar{z}}^{1} n\big(e(u;\bar{z})\big)du, \text{ and } m(\underline{z};\bar{z}) = 1, \text{ where } e(z;\bar{z}) = m^{-1}(z;\bar{z}) \tag{2.1}$$

- *When* $h < h_0$, $W = [0, \hat{z}]$, $I = \emptyset$, *and* $S = [\hat{z}, 1]$. *Moreover, workers and solvers are matched according to the strictly increasing function* $m(z; \hat{z})$ *given by* $\int_{\hat{z}}^{m(z;\hat{z})} dG(u) = \int_0^z h(1-u)dG(u)$. *Finally, the cutoff* $\hat{z} \in (0,1)$ *satisfies* $m(0;\hat{z}) = \hat{z}$, $m(\hat{z};\hat{z}) = 1$, *and:*

$$\frac{1}{h} - \hat{z} > \int_{\hat{z}}^{1} n(e(z;\hat{z}))dz, \text{ where } e(z;\hat{z}) = m^{-1}(z;\hat{z}) \tag{2.2}$$

Proof. For the proof see Fuchs et al. (2015, Lemma 2). □

Note that the efficient (and, therefore, equilibrium) allocation satisfies occupational stratification and strict positive assortative matching. Moreover, $W \neq \emptyset$ and $S \neq \emptyset$ but $I \neq \emptyset$ if and only if $h > h_0 \in (0,1)$. The next step is characterizing the wages that support the allocation of Lemma 2.1 as a competitive equilibrium:

Lemma 2.2. *In the absence of AI, the equilibrium wage function is given by:*

- *When $h \geq h_0$:*

$$w(z) = \begin{cases} m(z;\bar{z}) - w(m(z;\bar{z}))/n(z) & \text{if } z \in W \\ z & \text{if } z \in I \\ \bar{z} + \int_{\bar{z}}^{z} n\big(e(u;\bar{z})\big)du & \text{if } z \in S \end{cases}$$

- *When $h < h_0$:*

$$w(z) = \begin{cases} m(z;\hat{z}) - w(m(z;\hat{z}))/n(z) & \text{if } z \in W \\ \hat{z} + \frac{1}{1+n(\hat{z})}\left\{\frac{1}{h} - \hat{z} - \int_{\hat{z}}^{1} n\big(e(u;\hat{z})\big)du\right\} + \int_{\hat{z}}^{z} n\big(e(u;\hat{z})\big)du & \text{if } z \in S \end{cases}$$

Proof. See the Online Supplement of Fuchs et al. (2015, pp. 1–4). □

We end by showing that w is continuous, strictly increasing, and convex (strictly so when $z \in W \cup S$) and that $w(z) > z$ for all $z \notin \text{cl}I$.

Corollary 2.1. *Irrespective of whether $h \gtreqless h_0$, the equilibrium wage function $w(z)$ is continuous, strictly increasing, and (weakly) convex. Moreover, it satisfies:*

- *$w(z) > z$ for $z \in W$ (except possibly at $z = \sup W$), $w'(z) \in (0,1)$, and $w''(z) > 0$ for $z \in \text{int}W$.*
- *$w(z) = z$ for all $z \in I$.*
- *$w(z) > z$ for $z \in S$ (except possibly at $z = \inf S$), $w'(z) > 1$, and $w''(z) > 0$ for $z \in \text{int}S$.*

Proof. Consider first $h \geq h_0$. To show continuity, it is sufficient to verify that $w(z)$ is continuous at $z = \underline{z}$ and $z = \bar{z}$. Continuity at $z = \underline{z}$ follows from the fact that $\lim_{z\uparrow\underline{z}} w(z) = 1 - w(1)h(1-\underline{z}) = \underline{z} = \lim_{z\downarrow\underline{z}} w(z)$, where we are using the fact that $m(\underline{z};\bar{z}) = 1$ and that $w(1) = \int_{\bar{z}}^{1} n\big(e(u;\bar{z})\big)du + \bar{z} = \frac{1}{h}$ (due to condition (2.1)). Continuity at $z = \bar{z}$ follows because, by Lemma 2.2, $\lim_{z\downarrow\bar{z}} w(z) = w(\bar{z}) = \bar{z}$.

We now show the remaining properties of $w(z)$. Note that if $z \in \text{int}S$, then $w'(z) = n(e(z;\bar{z})) > 1$, which also implies that $w''(z) > 0$ because both $n(z)$ and $e(z;\bar{z})$ are strictly increasing in their arguments. Given that $\lim_{z\downarrow\bar{z}} w(z) = \bar{z}$, the previous results then imply that $w(z) > z$ for all $z \in (\bar{z}, 1]$. Similarly, if $z \in \text{int}W$, then $w'(z) = hw(m(z;\bar{z})) > 0$, which immediately implies that $w''(z) > 0$, since both $w(z)$ and $m(z;\bar{z})$ are strictly increasing in their arguments. Given that $\lim_{z\uparrow\underline{z}} w(z) = \underline{z}$, the previous results then imply that $w(z) > z$ for all $z \in [0, \underline{z})$. Finally, $w'(z) \in (0,1)$ follows from:

$$w'(z) = hw(m(z;\bar{z})) = h\left[\frac{m(z;\bar{z}) - w(z)}{h(1-z)}\right] < 1$$

where the second-to-last equality comes from the firms' zero-profit condition, and the last inequality holds because $m(z;\bar{z}) \leq 1$ and $w(z) > z$ when $z \in \text{int}W$.

Now consider $h \leq h_0$. To show continuity, it is sufficient to verify that $w(z)$ is continuous at $z = \hat{z}$. Given that $m(\hat{z};\hat{z}) = 1$, from Lemma 2.2, we have that:

$$\lim_{z\uparrow\hat{z}} w(z) = 1 - \tfrac{1}{n(\hat{z})}w(1) \quad \text{and} \quad \lim_{z\downarrow\hat{z}} w(z) = \tfrac{1}{1+n(\hat{z})}\left\{n(\hat{z}) - \int_{\hat{z}}^{1} n\big(e(u;\hat{z})\big)du\right\}$$

Note then that $w(1) = \lim_{z\downarrow\hat{z}} w(z) + \int_{\hat{z}}^{1} n\big(e(u;\hat{z})\big)du$. Combining the latter with the expression for $\lim_{z\uparrow\hat{z}} w(z)$ above and rearranging terms yields $\lim_{z\uparrow\hat{z}} w(z) = \lim_{z\downarrow\hat{z}} w(z)$. The proof that $\lim_{z\downarrow\hat{z}} w(z) >$

$\hat{z}$, in turn, can be found in Fuchs et al. (2015, Online Supplement, p. 4). Finally, the proofs for the remaining properties of $w(z)$ (i.e., their monotonicity and convexity, among others) follow the exact same logic as in the case when $h > h_0$. □

2.2 Proof of Proposition 2

In this chapter, we provide a complete characterization of the equilibrium with autonomous AI. As noted in the main text, we focus on $h < h_0$. The proof of Proposition 2 is a direct implication of Lemmas 2.3 and 2.4 below.

Lemma 2.3. *Any equilibrium with autonomous AI has the following features:*

- *Efficiency (i.e., it maximizes total output).*
- *Some compute must be allocated to independent production:* $\mu_i^* > 0$.
- *The price of compute is equal to AI's knowledge:* $r^* = z_{\text{AI}}$.
- *Occupational stratification:* $W^* \preceq I^* \preceq S^*$.
- *No worker is more knowledgeable than AI, and no solver is less knowledgeable than AI:* $W^* \preceq \{z_{\text{AI}}\} \preceq S^*$.
- *Positive assortative matching:* $m^* : W_p^* \to S_p^*$ *is strictly increasing and* $W_a^* \preceq W_p^*$ *and* $S_p^* \preceq S_a^*$.

Proof. • *The equilibrium is efficient.*— The following proof is adapted from Garicano and Rossi-Hansberg (2006, Proposition 6): Consider a competitive equilibrium with wage function w^* and rental rate of compute r^*. Suppose for contradiction that there is an alternative feasible allocation where some workers and/or compute are allocated to different firms and where output is strictly higher. This constitutes a contradiction of the definition of competitive equilibrium since, in equilibrium, firms maximize profits given w^* and r^*.

• *Some compute must be allocated to independent production.*— This result follows because compute is abundant relative to human time. Hence, there are not enough humans to interact with all AI agents (i.e., compute) inside two-layer organizations.

• *The price of compute is equal to AI's knowledge.*— This follows because the single-layer firms using AI must obtain zero profits.

• *Occupational stratification.*— Throughout the proof, let $\Pi(s, z)$ be the profit of a two-layer firm that hires a solver with knowledge s and workers with knowledge z. Moreover, we use that $w^*(z) \geq z$ for all $z \in [0, 1]$, as it cannot be strictly profitable to hire z as an independent producer.

We first show that $W^* \preceq I^*$. Suppose for contradiction that there exist $z_L < z_H$ such that $z_H \in W^*$ and $z_L \in I^*$. Since $z_L \in I^*$, then $w^*(z_L) = z_L$. Let $s > z_H$ be the knowledge of the solver matched with z_H (which can be human or AI). In equilibrium, $\Pi(s, z_H) = 0$, so $w^*(s) = n(z_H)(s - w^*(z_H))$. We claim that this implies that $\Pi(s, z_L) > 0$, yielding a contradiction. Indeed, $\Pi(s, z_L) = n(z_L)(s - z_L) - w^*(s) = n(z_L)(s - z_L) - n(z_H)(s - w^*(z_H)) \geq n(z_L)(s - z_L) - n(z_H)(s - z_H) > 0$, where the second-to-last inequality follows because $w^*(z_H) \geq z_H$ and the last inequality because $z_H > z_L$ and $n(x)(s-x)$ is strictly decreasing in x.

We now show that $I^* \preceq S^*$. Suppose for contradiction that there exist $s_L < s_H$ such that $s_H \in I^*$ and $s_L \in S^*$. Since $s_H \in I^*$, then $w^*(s_H) = s_H$. Let $z < s_L$ be the knowledge of the worker matched with s_L (which can be human or AI). In equilibrium, $\Pi(s_L, z) = 0$, so $n(z)w^*(z) = n(z)s_L - w^*(s_L)$. We claim that $\Pi(s_H, z) > 0$, yielding a contradiction. Indeed, $\Pi(s_H, z) = n(z)(s_H - w^*(z)) - s_H = n(z)(s_H - s_L) - s_H + w^*(s_L) \geq (n(z)-1)(s_H - s_L) > 0$, where the second-to-last inequality follows because $w^*(s_L) \geq s_L$ and the last inequality because $n(z) > 1$ and $s_H > s_L$.

Finally, we show that $W^* \preceq S^*$. Suppose for contradiction that there exist $z_H \in W^*$ and $s_L \in S^*$ such that $z_H > s_L$. Let $z_L < s_L$ be the knowledge of the worker matched with s_L (which can be human or AI). Similarly, let $s_H > z_H$ be the solver matched with z_H (which can be human or AI). Note that $z_L < s_L < z_H < s_H$.

We claim that $n(s_L)\Pi(z_H, z_L) + \Pi(s_H, s_L) > 0$, which implies that either $\Pi(z_H, z_L) > 0$ or $\Pi(s_H, s_L) > 0$, leading to the desired contradiction. Indeed, $n(s_L)\Pi(z_H, z_L) + \Pi(s_H, s_L) = n(s_L)n(z_L)(z_H - s_L) - (n(z_H) - n(s_L))(s_H - w^*(z_H)) \geq n(s_L)n(z_L)(z_H - s_L) - (n(z_H) - n(s_L))(s_H - z_H)$, where the first equality follows from using $\Pi(s_L, z_L) = 0$ and $\Pi(s_H, z_H) = 0$, and the last inequality from $w^*(z_H) \geq z_H$. Moreover,

$$n(s_L)n(z_L)(z_H - s_L) - (n(z_H) - n(s_L))(s_H - z_H)$$
$$\geq n(s_L)n(z_L)(z_H - s_L) - (n(z_H) - n(s_L))(1 - z_H) = \frac{(z_H - s_L)(1 - h(1 - z_L))}{h^2(1 - z_L)(1 - s_L)} > 0$$

where the first inequality follows because $n(s_L)n(z_L)(z_H - s_L) - (n(z_H) - n(s_L))(s_H - z_H)$ is strictly decreasing in s_H and $s_H \leq 1$. Hence, we conclude that $n(s_L)\Pi(z_H, z_L) + \Pi(s_H, s_L) > 0$.

• *No worker is more knowledgeable than AI, and no solver is less knowledgeable than AI.*— This result follows from occupational stratification, the fact that some compute must necessarily be used for independent production, and that AI agents are perfect substitutes for humans with equivalent knowledge.

• *Positive assortative matching.*— To prove this part, we use the following claim:

Claim 2.1. *In equilibrium, if a firm matches workers with knowledge z with a solver with knowledge s, then there is no firm that matches workers with knowledge $z' > z$ with a solver with knowledge $s' < s$.*

Proof. Let w^* be the equilibrium wage schedule, and consider a firm that, in equilibrium, hires workers with knowledge x. Since the firm maximizes its profits, it must select a solver with knowledge $\sigma(x)$ that satisfies $\sigma(x) \in \arg\max_\sigma \Pi(\sigma, x)$, where $\Pi(\sigma, x) = n(x)(\sigma - w^*(x)) - w^*(\sigma)$. Because $\Pi(\sigma, x)$ exhibits increasing differences, standard results in monotone comparative statics (see, e.g., Tadelis and Segal, 2005, Theorem 1) imply that $\sigma(x'') \geq \sigma(x')$ whenever $x'' > x'$. Consequently, if z is matched with $s = \sigma(z)$, then $s' = \sigma(z') \geq \sigma(z) = s$ for any $z' > z$. □

The results $W_a^* \preceq W_p^*$ and $S_p^* \preceq S_a^*$ follow immediately from Claim 2.1 in combination with $W^* \preceq \{z_{\text{AI}}\} \preceq S^*$. Hence, it only remains to show that $m^* : W_p^* \to S_p^*$ is strictly increasing. Claim 2.1 implies that m^* is weakly increasing. Suppose for contradiction it is not strictly increasing. This means there exist two different elements of W_p^*, z' and $z'' > z'$, such that $m^*(z') = m^*(z'')$. Occupational stratification and the fact that $W_a^* \preceq W_p^*$ imply that W_p^* is an interval, so all elements between z' and z'' are also in W_p^*. Hence, $\int_{z'}^{z''} h(1-z)dG(z) = \int_{m^*(z')}^{m^*(z'')} h(1-z)dG(z) = 0$ (as m^* must satisfy condition (1) of the main text), contradicting the fact that $\int_{z'}^{z''} h(1-z)dG(z) > 0$ (as $z' < z''$ and G has strict positive density). □

The next corollary is a direct implication of Lemma 2.3:

Corollary 2.2. *An equilibrium allocation with autonomous AI must take one of the following four potential configurations:*

- ***Type 1 configuration:***

$$W_a^* = \emptyset,\ W_p^* = [0, z_{\text{AI}}],\ I^* = (z_{\text{AI}}, \underline{z}_1^*),\ S_p^* = [\underline{z}_1^*, \bar{z}_1^*],\ S_a^* = [\bar{z}_1^*, 1],\ \textit{where } z_{\text{AI}} < \underline{z}_1^* \le \bar{z}_1^* \le 1$$
$$\textit{So } \mu_w^* = \int_{\bar{z}_1^*}^1 n(z_{\text{AI}})dG(z),\ \mu_s^* = 0,\ \mu_i^* = \mu - \mu_w^*$$

- ***Type 2 configuration:***

$$W_a^* = [0, \underline{z}_2^*],\ W_p^* = [\underline{z}_2^*, z_{\text{AI}}],\ I^* \subseteq \{z_{\text{AI}}\},\ S_p^* = [z_{\text{AI}}, \bar{z}_2^*],\ S_a^* = [\bar{z}_2^*, 1],\ \textit{where } 0 \le \underline{z}_2^* \le z_{\text{AI}} \le \bar{z}_2^* \le 1$$
$$\textit{So } \mu_w^* = \int_{\bar{z}_2^*}^1 n(z_{\text{AI}})dG(z),\ \mu_s^* = \int_0^{\underline{z}_2^*} n(z)^{-1}dG(z),\ \mu_i^* = \mu - \mu_w^* - \mu_s^*$$

- ***Type 3 configuration:***

$$W_a^* = [0, \underline{z}_3^*],\ W_p^* = [\underline{z}_3^*, \bar{z}_3^*],\ I^* = (\bar{z}_3^*, z_{\text{AI}}),\ S_p^* = [z_{\text{AI}}, 1],\ S_a^* = \emptyset,\ \textit{where } 0 < \underline{z}_3^* \le \bar{z}_3^* < z_{\text{AI}}$$
$$\textit{So } \mu_w^* = 0,\ \mu_s^* = \int_0^{\underline{z}_3^*} n(z)^{-1}dG(z),\ \mu_i^* = \mu - \mu_s^*$$

- ***Type 4 configuration:***

$$W_a^* = \emptyset,\ W_p^* = [0, \underline{z}_4^*],\ I^* = (\underline{z}_4^*, \bar{z}_4^*) \ni z_{\text{AI}},\ S_p^* = [\bar{z}_4^*, 1],\ S_a^* = \emptyset,\ \textit{where } 0 \le \underline{z}_4^* < \bar{z}_4^* \le 1$$
$$\textit{So } \mu_w^* = 0,\ \mu_s^* = 0,\ \mu_i^* = \mu$$

Proof. As mentioned above, the proof of this corollary is a direct implication of Lemma 2.3. Note that in a Type 2 configuration, I^* can either be $\{z_{\text{AI}}\}$ or $\emptyset$ because the human with knowledge z_{AI} is indifferent between any of the three roles. However, this is irrelevant for all practical purposes because I^* has measure zero in this case. □

Intuitively, in a Type 1 configuration, AI is used as a worker and independent producer. In a Type 2 configuration, AI is used in all three possible roles (i.e., as a worker, an independent producer, and a solver). In a Type 3 configuration, AI is used as a solver and independent producer, while in a Type 4 configuration, AI is used exclusively as an independent producer.

Now, recall that W and S are the sets of human workers and solvers in the pre-AI equilibrium. For $z_{\text{AI}} \in W$, define the function $m_w : [0, z_{\text{AI}}] \to [z_{\text{AI}}, 1]$ by $\int_{z_{\text{AI}}}^{m_w(z;z_{\text{AI}})} dG(u) = \int_0^z h(1-u)dG(u)$ and note that $z_{\text{AI}} \in W$ implies that $m_w(z_{\text{AI}}; z_{\text{AI}}) \le 1$. Let then $e_w(z; z_{\text{AI}}) \equiv m_w^{-1}(z; z_{\text{AI}})$ and define:

$$\Gamma_w(x) \equiv n(x)(m_w(x;x) - x) - x - \int_x^{m_w(x;x)} n(e_w(u;x))du$$

Similarly, for $z_{\text{AI}} \in S$, define the function $e_s : [z_{\text{AI}}, 1] \to [0, z_{\text{AI}}]$ by $\int_z^1 dG(u) = \int_{e_s(z;z_{\text{AI}})}^{z_{\text{AI}}} h(1-u)dG(u)$, note that $z_{\text{AI}} \in S$ implies that $e_s(z_{\text{AI}}; z_{\text{AI}}) \geq 0$, and define:

$$\Gamma_s(x) \equiv \tfrac{1}{h} - x - \textstyle\int_x^1 n(e_s(u;x))du$$

Consider the following partition of the knowledge space (note that $W \cup S = [0,1]$ when $h < h_0$): $\mathcal{R}_1 \equiv \{z \in W : \Gamma_w(z) < 0\}$, $\mathcal{R}_2 \equiv \{z \in W : \Gamma_w(z) \geq 0\} \cup \{z \in S : \Gamma_s(z) \geq 0\}$, and $\mathcal{R}_3 \equiv \{z \in S : \Gamma_s(z) < 0\}$. The following lemma—which is the main result of this appendix—characterizes in detail the post-AI equilibrium:

Lemma 2.4. *In the presence of autonomous AI, there is a unique competitive equilibrium. It is given as follows:*

- *If $z_{\text{AI}} \in \mathcal{R}_1$, then the equilibrium allocation is Type 1. The equilibrium cutoffs $\underline{z}_1^*$ and $\bar{z}_1^*$ satisfy:*

$$\bar{z}_1^* = m_1^*(z_{\text{AI}}; \underline{z}_1^*) \quad \textit{and} \quad n(z_{\text{AI}})(m_1^*(z_{\text{AI}}; \underline{z}_1^*) - z_{\text{AI}}) = \underline{z}_1^* + \int_{\underline{z}_1^*}^{m_1^*(z_{\text{AI}};\underline{z}_1^*)} n\big(e_1^*(z; \underline{z}_1^*)\big)dz$$

  where $m_1^* : [0, z_{\text{AI}}] \to [\underline{z}_1^*, \bar{z}_1^*]$ is given by $\int_{\underline{z}_1^*}^{m_1^*(z;\underline{z}_1^*)} dG(u) = \int_0^z h(1-u)dG(u)$ and $e_1^*(z;\cdot) = (m_1^*)^{-1}(z;\cdot)$

- *If $z_{\text{AI}} \in \mathcal{R}_2$, then the equilibrium allocation is Type 2. The equilibrium cutoffs $\underline{z}_2^*$ and $\bar{z}_2^*$ satisfy:*

$$\bar{z}_2^* = m_2^*(z_{\text{AI}}; \underline{z}_2^*) \quad \textit{and} \quad n(z_{\text{AI}})(m_2^*(z_{\text{AI}}; \underline{z}_2^*) - z_{\text{AI}}) = z_{\text{AI}} + \int_{z_{\text{AI}}}^{m_2^*(z_{\text{AI}};\underline{z}_2^*)} n\big(e_2^*(z; \underline{z}_2^*)\big)dz$$

  where $m_2^* : [\underline{z}_2^*, z_{\text{AI}}] \to [z_{\text{AI}}, \bar{z}_2^*]$ is given by $\int_{z_{\text{AI}}}^{m_2^*(z;\underline{z}_2^*)} dG(u) = \int_{\underline{z}_2^*}^z h(1-u)dG(u)$ and $e_2^*(z;\cdot) = (m_2^*)^{-1}(z;\cdot)$

- *If $z_{\text{AI}} \in \mathcal{R}_3$, then the equilibrium allocation is Type 3. The equilibrium cutoffs $\underline{z}_3^*$ and $\bar{z}_3^*$ satisfy:*

$$\bar{z}_3^* = e_3^*(1; \underline{z}_3^*) \quad \textit{and} \quad \tfrac{1}{h} = z_{\text{AI}} + \int_{z_{\text{AI}}}^1 n\big(e_3^*(z; \underline{z}_3^*)\big)dz$$

  where $m_3^* : [\underline{z}_3^*, \bar{z}_3^*] \to [z_{\text{AI}}, 1]$ given by $\int_{z_{\text{AI}}}^{m_3^*(z;\underline{z}_3^*)} dG(u) = \int_{\underline{z}_3^*}^z h(1-u)dG(u)$ and $e_3^*(z;\cdot) = (m_3^*)^{-1}(z;\cdot)$

The equilibrium matching function is given by $m^*(z) = m_j^*(z; \underline{z}_j^*)$ if $z_{\text{AI}} \in \mathcal{R}_j$, while the equilibrium wage w^* is continuous, strictly increasing, and (weakly) convex, and satisfies:

- *$w^*(z) = z_{\text{AI}}(1 - 1/n(z)) > z$ for all $z \in W_a^*$.*
- *$w^*(z) = m^*(z) - w^*(m^*(z))/n(z)$ for all $z \in W_p^*$.*
- *$w^*(z) = z$ for all $z \in I^*$.*
- *$w^*(z) = \inf S_p^* + \int_{\inf S_p^*}^z n(e^*(u))du$ for all $z \in S_p^*$.*
- *$w^*(z) = n(z_{\text{AI}})(z - z_{\text{AI}}) > z$ for all $z \in S_a^*$.*

For the interested reader, in Chapter 4.1.1 we also provide a version of Lemma 2.4 for the case in which human knowledge is uniformly distributed. In such a case, the equilibrium expressions (i.e., the wage function w^*, the equilibrium cutoffs, and the partition $\mathcal{R}_j$ for $j = 1, 2, 3$) can be obtained in (almost) closed form.

According to Lemma 2.3, AI is always used for independent production. Lemma 2.4 adds to this point by stating that if $z_{\text{AI}} \in W$, then AI is also used as a worker and possibly as a solver, while if $z_{\text{AI}} \in S$, then AI is also used as a solver and possibly as a worker. Indeed, if $z_{\text{AI}} \in W$, then the post-AI equilibrium is either Type 1 (in which AI is a worker but not a solver) or Type 2 (in which AI is both a worker and solver). Similarly, if $z_{\rm AI} \in S$, then the post-AI equilibrium is either Type 3 (in which AI is a solver but not a worker) or Type 2.

Before formally proving this lemma, we informally derive the equilibrium in one of the regions to provide insight into its construction:

Informal Construction of the Equilibrium of a Type 2 Equilibrium.— A Type 2 equilibrium partitions the human population as follows:

$$W_a^* = [0, \underline{z}_2^*],\ W_p^* = [\underline{z}_2^*, z_{\rm AI}],\ I^* = \emptyset,\ S_p^* = [z_{\rm AI}, \bar{z}_2^*],\ S_a^* = [\bar{z}_2^*, 1],\ \text{where } 0 \le \underline{z}_2^* \le z_{\rm AI} \le \bar{z}_2^* \le 1$$

As mentioned in the main text, given that the equilibrium price of compute is $r^* = z_{\rm AI}$, the zero-profit condition of a tA firm pins down the wage $w^*(z) = z_{\rm AI}(1 - 1/n(z))$ of a human worker with knowledge $z \in W_a^*$. Similarly, the zero-profit condition of a bA firm determines the wage $w^*(z) = n(z_{\rm AI})(z - z_{\rm AI})$ of a human solver with knowledge $z \in S_a^*$.

Now consider those firms that do not use AI, i.e., the nA firms. First, let $m_2^*(z)$ be the equilibrium matching function in this case. Due to Lemma 2.3, this function must be strictly increasing and satisfy $\int_{z_{\rm AI}}^{m_2^*(z)} dG(u) = \int_{\underline{z}_2^*}^{z} h(1-u)dG(u)$ for all $z \in [\underline{z}_2^*, z_{\rm AI}]$ (which is simply constraint (1) of the main text in this specific case). Moreover, given that $\sup W_p^* = z_{\rm AI}$ and $\sup S_p^* = \bar{z}_2^*$, it must also be that $m_2^*(z_{\rm AI}) = \bar{z}_2^*$.

Note that for any given $z \in W_p$, there exists a unique increasing function $m_2^*(z)$ that satisfies both constraints. It is given by the solution to the differential equation $m_2^{*\prime}(z) = h(1-z)g(z)/g(m_2^*(z))$ with border condition $m_2^*(z_{\rm AI}) = \bar{z}_2^*$ (which comes from differentiating both sides of the resource constraint). We denote such a unique function by $m_2^*(z; \underline{z}_2^*)$ (as it depends on $\underline{z}_2^*$ through its domain).

Consider the problem of an nA firm that recruited $n(z)$ workers of type $z \in W_p^*$ and is deciding which solver $z \in S_p^*$ to hire: $\max_{s \in S_p^*} \Pi_2^{n\rm A}(s, z) = n(z)[s - w(z)] - w(s)$. The corresponding first-order condition evaluated at $s = m_2^*(z; \underline{z}_2^*)$ implies that $w^{*\prime}(z) = n(e_2^*(z; \underline{z}_2^*))$ for any $z \in S_p^*$. Thus, $w^*(z) = C^* + \int_{z_{\rm AI}}^{z} n(e_2^*(u; \underline{z}_2^*))du$ for any $z \in S_p^*$. The wages of the workers hired by such firms are then determined by the zero-profit condition of nA firms: $w^*(z) = m_2^*(z; \underline{z}_2^*) - w^*(m_2^*(z; \underline{z}_2^*))/n(z)$ for any $z \in W_p^*$.

The final step is determining the constant C^*, the cutoff $\underline{z}_2^*$, and arguing that no firms have incentives to deviate. To do this, note that the least knowledgeable human solver has the same knowledge as AI, i.e., $\inf S_p^* = z_{\rm AI}$. Hence, her wage must be equal to the price of one unit of compute, so $C^* = z_{\rm AI}$. Moreover, the most knowledgeable solver hired by an nA firm has the same knowledge as the least knowledgeable solver of a bA firm. As a result, these two individuals must receive the same wage, i.e., $\lim_{z \uparrow \bar{z}_2^*} w^*(z) = \lim_{z \downarrow \bar{z}_2^*} w^*(z)$. Since $\bar{z}_2^* = m_2^*(z_{\rm AI}; \underline{z}_2^*)$, we obtain the following:

$$n(z_{\rm AI})(m_2^*(z_{\rm AI}; \underline{z}_2^*) - z_{\rm AI}) = z_{\rm AI} + \int_{z_{\rm AI}}^{m_2^*(z_{\rm AI}; \underline{z}_2^*)} n(e_2^*(z; \underline{z}_2^*))dz \tag{2.3}$$

which is the equilibrium condition in the statement of Lemma 2.4. It is then possible to prove that there is a unique cutoff $\underline{z}_2^*$ that satisfies this condition and that such a cutoff is contained in $[0, z_{\rm AI}]$

if and only if $z_{\text{AI}} \in \mathcal{R}_2$. This explains why this equilibrium can only arise in such a region of the parameter space. The fact that $C^* = z_{\text{AI}}$ and that $\underline{z}_2^*$ satisfies (2.3) then implies that $w^*(z)$ is also continuous at the juncture between W_p^* and S_p^*:

$$\begin{aligned}\lim_{z\downarrow \underline{z}_2^*} w^*(z) &= m_2^*(\underline{z}_2^*; \underline{z}_2^*) - \tfrac{w^*(m_2^*(\underline{z}_2^*;\underline{z}_2^*))}{n(\underline{z}_2^*)} = z_{\text{AI}}\left(1 - \tfrac{1}{n(\underline{z}_2^*)}\right) = \lim_{z\uparrow \underline{z}_2^*} w^*(z) \\ \lim_{z\uparrow z_{\text{AI}}} w^*(z) &= m_2^*(z_{\text{AI}}; \underline{z}_2^*) - \tfrac{w^*(m_2^*(z_{\text{AI}};\underline{z}_2^*))}{n(z_{\text{AI}})} = z_{\text{AI}} = \lim_{z\downarrow z_{\text{AI}}} w^*(z)\end{aligned}$$

This is sufficient sufficient to guarantee that $w^*(z)$ is continuous in all its domain, i.e., for all $z \in [0,1]$.

From here, arguing that no firm has incentives to deviate is straightforward. Indeed, note that the wage function is continuous, strictly increasing, and weakly convex. This implies that if a firm does not have incentives to deviate locally, then it does not have incentives to deviate globally either. That bA firms do not want to deviate locally follows because such deviation also leads to no profits (as all firms who hire a human solver in S_p^* earn zero profits). Similar reasoning also explains why tA and nA firms do not have incentives to deviate locally either. □

We now formally prove the lemma. We do this in two steps. In the first step, we show that the outcomes described in the lemma are indeed an equilibrium by verifying that (i) markets clear, and (ii) firms are maximizing their profits while obtaining zero profits. In the second step, we prove that the equilibrium is unique.

Proof of Step 1. We show this part only for $z_{\text{AI}} \in \mathcal{R}_1$, as the other two cases are analogous. We begin by verifying market clearing in the market for compute. By Corollary 2.2, it is immediate that $\mu_i^* + \mu_w^* + \mu_s^* = \mu$. Moreover, the total time required to consult on the problems left unsolved by AI is equal to the total time available of solvers in S_a^*, i.e., $h(1 - z_{\text{AI}})\mu_w^* = \int_{\bar{z}_1^*}^1 dG(z)$.

We now turn to verifying labor market clearing. First, it must be that the total time required to consult on the problems left unsolved by the human workers in the interval $[0, z] \subseteq W_p^*$ is equal to the total time available of human solvers in the interval $[\underline{z}_1^*, m_1^*(z; \underline{z}_1^*)] \subseteq S_p^*$. This resource constraint is satisfied as $m_1^*(z; \underline{z}_1^*)$ is given by $\int_{\underline{z}_1^*}^{m_1^*(z;\underline{z}_1^*)} dG(u) = \int_0^z h(1-u)dG(u)$ and $\bar{z}_1^* = m_1^*(z_{\text{AI}}; \underline{z}_1^*)$.

Second, it must be that the union of the sets $(W_p^*, W_a^*, I^*, S_p^*, S_a^*)$ is $[0,1]$, and the intersection of any two of these sets has measure zero. By Corollary 2.2, this occurs if and only if $z_{\text{AI}} < \underline{z}_1^* \leq \bar{z}_1^* \leq 1$. Verifying that $\underline{z}_1^* \leq \bar{z}_1^*$ is straightforward: It follows because $m_1^*(z; \underline{z}_1^*)$ is strictly increasing in z plus the fact that $\underline{z}_1^* = m_1^*(0; \underline{z}_1^*)$ and $\bar{z}_1^* = m_1^*(z_{\text{AI}}; \underline{z}_1^*)$.

Showing that $z_{\text{AI}} < \underline{z}_1^*$ requires more work. Note that $\underline{z}_1^*$ is given by the solution $\Gamma_1(\underline{z}_1^*; z_{\text{AI}}) = 0$, where $\Gamma_1(x; z_{\text{AI}}) \equiv n(z_{\text{AI}})(m_1^*(z_{\text{AI}}; x, z_{\text{AI}}) - z_{\text{AI}}) - x - \int_x^{m_1^*(z_{\text{AI}};x,z_{\text{AI}})} n\big(e_1^*(z; x, z_{\text{AI}})\big)dz$ (here we are making explicit that $m_1^*(\cdot)$ and $e_1^*(\cdot)$ also depend indirectly on z_{AI} through the boundary of the set $W_p^* = [0, z_{\text{AI}}]$ to avoid any type of confusion[1]). It is then not difficult to prove that $\Gamma_1(x; z_{\text{AI}})$ is strictly increasing in x, so $z_{\text{AI}} < \underline{z}_1^*$ if and only if $\Gamma_1(0; z_{\text{AI}}) < 0$. Furthermore, note that $m_1^*(z; 0, z_{\text{AI}})$ satisfies $\int_0^{m_1^*(z;0,z_{\text{AI}})} dG(u) = \int_0^z h(1-u)dG(u)$ for $z \in [0, z_{\text{AI}}]$, which implies that $m_1^*(z; 0, z_{\text{AI}}) = m_w(z; z_{\text{AI}})$ (and, therefore, that $e_1^*(z; 0, z_{\text{AI}}) = e_w(z; z_{\text{AI}})$). Hence, $\Gamma_1(0; z_{\text{AI}}) = \Gamma_w(z_{\text{AI}}) < 0$ as $z_{\text{AI}} \in \mathcal{R}_1$.

[1]In the statement of Lemma 2.4, we simply wrote $m_1^*(z; \underline{z}_1^*)$ instead of $m_1^*(z; \underline{z}_1^*, z_{\text{AI}})$ to avoid cluttering notation.

Finally, we show that $\bar{z}_1^* \leq 1$. To prove it, we show that $\underline{z}_1^* < \hat{z}$ and then use this result to conclude that $\bar{z}_1^* \leq 1$. As a first step, note that $m_1^*(z; \hat{z}, \hat{z})$ satisfies $\int_{\hat{z}}^{m_1^*(z;\hat{z},\hat{z})} dG(u) = \int_0^z h(1-u)dG(u)$ for $z \in [0, \hat{z}]$, so $m_1^*(z; \hat{z}, \hat{z}) = m(z; \hat{z})$ where $m(z; \hat{z})$ is matching function of the pre-AI equilibrium. Recall then that $\underline{z}_1^*$ is the unique solution $\Gamma_1(\underline{z}_1^*; z_{\rm AI}) = 0$, where $\Gamma_1(x; z_{\rm AI})$ is strictly increasing in x. It is not difficult to prove that $\Gamma_1(x; z_{\rm AI})$ is strictly decreasing in $z_{\rm AI}$ for any given x. We claim that $\Gamma_1(\hat{z}; z_{\rm AI}) > 0$, which immediately implies that $\underline{z}_1^* < \hat{z}$. Indeed, note that $\Gamma_1(\hat{z}; z_{\rm AI}) \geq \Gamma_1(\hat{z}; \hat{z}) = \frac{1}{h} - \hat{z} - \int_{\hat{z}}^1 n(e(z; \hat{z}))dz$, where the first inequality follows because $\Gamma_1(x; z_{\rm AI})$ is strictly decreasing in $z_{\rm AI}$ and $z_{\rm AI} \leq \hat{z}$ (as $z_{\rm AI} \in W$) and the last equality follows because $e_1^*(z; \hat{z}, \hat{z}) = e(z; \hat{z})$ for all $z \in [\hat{z}, 1]$. However, by Lemma 2.1, $\frac{1}{h} - \hat{z} - \int_{\hat{z}}^1 n(e(z; \hat{z}))dz > 0$ when $h < h_0$, so $\Gamma_1(\hat{z}; z_{\rm AI}) > 0$.

Having proven that $\underline{z}_1^* < \hat{z}$, we now show that $\bar{z}_1^* \leq 1$. By construction, $\bar{z}_1^*$ satisfies $\int_{\underline{z}_1^*}^{\bar{z}_1^*} dG(u) = \int_0^{z_{\rm AI}} h(1-u)dG(u)$. Note, moreover, that $z \in W$ implies that $\int_0^{z_{\rm AI}} h(1-u)dG(u) \leq \int_0^{\hat{z}} h(1-u)dG(u) = \int_{\hat{z}}^1 dG(u)$ (as $z_{\rm AI} \leq \hat{z}$ and $\int_0^{\hat{z}} h(1-u)dG(u) = \int_{\hat{z}}^1 dG(u)$). Hence, $\int_{\underline{z}_1^*}^{\bar{z}_1^*} dG(u) \leq \int_{\hat{z}}^1 dG(u)$. Given that $\underline{z}_1^* < \hat{z}$, it must be that $\bar{z}_1^* < 1$.

Having verified market clearing, we now show that, in the candidate equilibrium, firms maximize their profits while obtaining zero profits. Given how the wages are constructed, it is clear that firms are optimizing their profits locally while obtaining zero profits. Thus, we only need to consider global deviations. As discussed above, to discard such global deviations, it is sufficient to show that w^* is continuous, strictly increasing, and weakly convex. This is what we prove next.

To show continuity, it suffices to verify that w^* is continuous at the junctures of (i) S_p^* and S_a^*, (ii) I^* and S_p^*, and (iii) W_p^* and I^*. For (i), note that $\lim_{z \uparrow \bar{z}_1^*} w^*(z) = \bar{z}_1^* + \int_{\underline{z}_1^*}^{\bar{z}_1^*} n(e_1^*(z; \underline{z}_1^*))dz$ and $\lim_{z \downarrow \bar{z}_1^*} w^*(z) = n(z_{\rm AI})(\bar{z}_1^* - z_{\rm AI})$. Hence, from the conditions determining $\underline{z}_1^*$ and $\bar{z}_1^*$, we obtain that $\lim_{z \uparrow \bar{z}_1^*} w^*(z) = \lim_{z \downarrow \bar{z}_1^*} w^*(z)$. For (ii) note that by construction, $\lim_{z \uparrow \underline{z}_1^*} w^*(z) = \underline{z}_1^* = \lim_{z \downarrow \underline{z}_1^*} w^*(z)$. Finally, for (iii) note that $\lim_{z \uparrow z_{\rm AI}} w^*(z) = \bar{z}_1^* - w^*(\bar{z}_1^*)/n(z_{\rm AI}) = z_{\rm AI} = \lim_{z \downarrow z_{\rm AI}} w^*(z)$ given that $w^*(\bar{z}_1^*) = n(z_{\rm AI})(\bar{z}_1^* - z_{\rm AI})$.

With continuity at hand, proving that w^* is strictly increasing and weakly convex is straightforward: The logic is analogous to the proof of Corollary 2.1 (which shows that the pre-AI wage function satisfies these two properties). Consequently, in the case $z_{\rm AI} \in \mathcal{R}_1$, the outcome described in the statement is indeed a competitive equilibrium. □

Proof of Step 2. We show this part only for $z_{\rm AI} \in \mathcal{R}_1$, as the other two cases follow the same logic. In particular, we show that if $z_{\rm AI} \in \mathcal{R}_1$, then there cannot be any other type of equilibrium.

Suppose first for contradiction that there is a Type 3 equilibrium. By Corollary 2.2, we know that such an equilibrium features the following partition of the human population:

$$W_a^* = [0, \underline{z}_3^*],\ W_p^* = [\underline{z}_3^*, \bar{z}_3^*],\ I^* = (\bar{z}_3^*, z_{\rm AI}),\ S_p^* = [z_{\rm AI}, 1],\ S_a^* = \emptyset, \text{ where } 0 < \underline{z}_3^* \leq \bar{z}_3^* < z_{\rm AI}$$

To show that such an equilibrium cannot arise when $z_{\rm AI} \in \mathcal{R}_1$, we first establish that $z_{\rm AI} \in \mathcal{R}_1$ implies $z_{\rm AI} < \hat{z}$. To do this, it suffices to show that $z_{\rm AI} \notin \mathcal{R}_1$ (as $\mathcal{R}_1 \subseteq W = [0, \hat{z}]$), or, equivalently, that $\Gamma_w(\hat{z}) > 0$. Indeed, note that $m_w(z; \hat{z}) = m(z; \hat{z})$, where $m(z; \hat{z})$ is the matching function of the pre-AI equilibrium. This implies that $m_w(\hat{z}; \hat{z}) = 1$, so $\Gamma_w(\hat{z}) = 1/h - \hat{z} - \int_{\hat{z}}^1 n(e(z; \hat{z}))dz > 0$, where the last inequality follows from Lemma 2.1.

Given that $z_{\text{AI}} \in \mathcal{R}_1$ implies $z_{\text{AI}} < \hat{z}$, we then have the following:

$$\int_{\underline{z}_3^*}^{\bar{z}_3^*} h(1-u)dG(u) < \int_0^{z_{\text{AI}}} h(1-u)dG(u) < \int_0^{\hat{z}} h(1-u)dG(u) = \int_{\hat{z}}^1 dG(u) < \int_{z_{\text{AI}}}^1 dG(u)$$

which violates the resource constraint that the total time required to consult on the problems left unsolved by the human workers in the interval W_p^* equals the total time available of human solvers in the interval S_p^*, i.e., $\int_{\underline{z}_3^*}^{\bar{z}_3^*} h(1-u)dG(u) = \int_{z_{\text{AI}}}^1 dG(u)$. Hence, a Type 3 configuration cannot arise when $z_{\text{AI}} \in \mathcal{R}_1$.

Now suppose for contradiction that there is a Type 2 equilibrium. As explained above (see "Informal Construction of the Equilibrium"), for this to be an equilibrium, there must exist a cutoff $\underline{z}_2^* \in [0, z_{\text{AI}}]$ with the property that $\Gamma_2(\underline{z}_2^*; z_{\text{AI}}) = 0$, where $\Gamma_2(x; z_{\text{AI}}) \equiv n(z_{\text{AI}})(m_2^*(z_{\text{AI}}; x) - z_{\text{AI}}) - z_{\text{AI}} - \int_{z_{\text{AI}}}^{m_2^*(z_{\text{AI}};x)} n(e_2^*(z; x))dz$. It is not difficult to prove that $\Gamma_2(x; z_{\text{AI}})$ is strictly decreasing in x. Hence, there exists at most one $\underline{z}_2^*$ that satisfies $\Gamma_2(\underline{z}_2^*; z_{\text{AI}}) = 0$, and for $\underline{z}_2^* \geq 0$, it must be that $\Gamma_2(0; z_{\text{AI}}) \geq 0$, where $m_2^*(z; 0)$ is given by $\int_{z_{\text{AI}}}^{m_2^*(z;0)} dG(u) = \int_0^z h(1-u)dG(u)$ for $z \in [0, z_{\text{AI}}]$. However, from this last condition we have that $m_2^*(z; 0) = m_w(z; z_{\text{AI}})$ (as $m_2^*(z; 0)$ and $m_w(z; z_{\text{AI}})$ satisfy the same condition), so $\Gamma_2(0; z_{\text{AI}}) = \Gamma_w(z_{\text{AI}})$. Consequently, for $\underline{z}_2^* \geq 0$, we need that $\Gamma_w(z_{\text{AI}}) \geq 0$, which contradicts the fact that $z_{\text{AI}} \in \mathcal{R}_1$.

Finally, suppose for contradiction that there is a Type 4 equilibrium. By Corollary 2.2, we know that such an equilibrium features the following partition of the human population:

$$W_p^* = \emptyset,\ W_p^* = [0, \underline{z}_4^*],\ I^* = (\underline{z}_4^*, \bar{z}_4^*) \ni z_{\text{AI}},\ S_p^* = [\bar{z}_4^*, 1],\ S_a^* = \emptyset$$

Moreover, following similar reasoning as that developed above for a Type 2 equilibrium, for this to be an equilibrium, (i) it must be that $\bar{z}_4^* = m_4^*(0; \underline{z}_4^*)$, where $m_4^*(0; \underline{z}_4^*)$ satisfies $m_4^*(\underline{z}_4^*; \underline{z}_4^*) = 1$ and $\int_{m_4^*(0;\underline{z}_4^*)}^{m_4^*(z;\underline{z}_4^*)} dG(u) = \int_{\underline{z}_4^*}^z h(1-u)dG(u)$ for $z \in [0, \underline{z}_4^*]$, and (ii) there must exist a cutoff $\underline{z}_4^* < m_4^*(0; \underline{z}_4^*)$ such that $1/h - m_4^*(0; \underline{z}_4^*) - \int_{m_4^*(0;\underline{z}_4^*)}^1 n(e_4^*(z; \underline{z}_4^*))dz = 0$. By Lemma 2.1, we know that the unique solution to these equilibrium conditions is $\underline{z}_4^* = \underline{z}$ and $\bar{z}_4^* = m_4^*(0; \underline{z}_4^*) = \bar{z}$, where $\underline{z}$ and $\bar{z}$ are the equilibrium cutoffs of the pre-AI equilibrium when $h > h_0$. However, for $z_{\text{AI}} \in I^* = (\underline{z}, \bar{z})$, we also need that $\underline{z}_4^* = \underline{z} < \bar{z} = \bar{z}_4^*$. For this to be the case, it must be that $h > h_0$, which contradicts the assumption that $h < h_0$. □

We finish this Chapter by describing some properties of the partition $(\mathcal{R}_1, \mathcal{R}_2, \mathcal{R}_3)$ that will be useful later. In particular, the next lemma states that when z_{AI} is sufficiently low, the equilibrium is Type 1, so AI is used as a worker and as an independent producer, but not as a solver in that case. Similarly, when $z_{\text{AI}} = \hat{z}$ or $z_{\text{AI}} \to 1$ (where $\hat{z}$ is the knowledge cutoff to become a solver in the pre-AI equilibrium), the equilibrium is Type 2, so AI is used in all three possible roles.

Lemma 2.5. *(i) $z_{\text{AI}} \in \mathcal{R}_1$ if $z_{\text{AI}} \in [0, \epsilon)$ with $\epsilon \downarrow 0$, (ii) $z_{\text{AI}} = \hat{z} \in \mathcal{R}_2$ (where $\{\hat{z}\} = W \cap S$), and (iii) $z_{\text{AI}} \in \mathcal{R}_2$ if $z_{\text{AI}} \in [1-\epsilon, 1)$ with $\epsilon \downarrow 0$.*

Proof. First we show that $z_{\text{AI}} \in \mathcal{R}_1$ if $z_{\text{AI}} \in [0, \epsilon)$ with $\epsilon \downarrow 0$. Note that $\Gamma_w(0) = 0$ (as $m_w(z; 0) = 0$), and that $\Gamma_w'(0) = -1$, as:

$$\Gamma_w'(x) = \tfrac{1}{h} - 1 - \tfrac{1}{h(1-x)} + \tfrac{m_w(x;x)-x}{h(1-x)^2} + h\int_x^{m_w(x;x)} \tfrac{n(e_w(z;x))^3 g(x)}{g(e_w(z;x))} dz$$

Hence, if $z_{\text{AI}} \in [0,\epsilon)$ with $\epsilon \downarrow 0$, then $z_{\text{AI}} \in W$ and $\Gamma_w(z_{\text{AI}}) \le 0$, so $z_{\text{AI}} \in \mathcal{R}_1$.

Next, we prove that $\hat{z} \in \mathcal{R}_2$ by showing that $\Gamma_s(\hat{z}) = \Gamma_w(\hat{z}) > 0$. Note that $m_w(z;\hat{z}) = m(z;\hat{z})$, where $m(z;\hat{z})$ is the matching function of the pre-AI equilibrium. This implies that $m_w(\hat{z};\hat{z}) = 1$, so $\Gamma_w(\hat{z}) = 1/h - \hat{z} - \int_{\hat{z}}^1 n(e(z;\hat{z}))dz > 0$, where the last inequality follows from Lemma 2.1. Similarly, note that $e_s(z;\hat{z}) = e(z;\hat{z})$, where $e(z;\hat{z}) = m^{-1}(z;\hat{z})$. Thus, $\Gamma_s(\hat{z}) = 1/h - \hat{z} - \int_{\hat{z}}^1 n(e(z;\hat{z}))dz$, so $\Gamma_s(\hat{z}) = \Gamma_w(\hat{z}) > 0$.

Finally, we show that $z_{\text{AI}} \in \mathcal{R}_2$ if $z_{\text{AI}} \in [1-\epsilon, 1)$ with $\epsilon \downarrow 0$. First, note that if $z_{\text{AI}} \in [1-\epsilon,1)$ with $\epsilon \downarrow 0$, then $z_{\text{AI}} \in S$. Second, is not difficult to prove that $\Gamma_s(1) = 1/h - 1 > 0$ so, by continuity, $\Gamma_s(z_{\text{AI}}) > 0$ for all $z_{\text{AI}} \in [1-\epsilon,1)$ with $\epsilon \downarrow 0$. Hence, if $z_{\text{AI}} \in [1-\epsilon,1)$ with $\epsilon \downarrow 0$, then $z_{\text{AI}} \in S$ and $\Gamma_s(z_{\text{AI}}) > 0$, so $z_{\text{AI}} \in \mathcal{R}_2$. □

2.3 Proof of Proposition 3

• $z_{\text{AI}} \in \text{int}W$.— By Lemma 2.4, the AI equilibrium is either Type 1 or Type 2. If it is Type 2, then $\sup W^* = \inf S^* = z_{\text{AI}}$, so $W^* \subset W$ and $S^* \supset S$, since $z_{\text{AI}} < \hat{z}$ when $z_{\text{AI}} \in \text{int}W$. If it is Type 1, then $\sup W^* = z_{\text{AI}}$ and $\inf S^* = \underline{z}_1^*$. Note that $\sup W^* = z_{\text{AI}}$ implies that $W^* \subset W$, because $z_{\text{AI}} < \hat{z}$. Similarly, note that $\inf S^* = \underline{z}_1^*$ implies that $S^* \supset S$, given that $\underline{z}_1^* < \hat{z}$ (as shown in the proof of Lemma 2.4). □

• $z_{\text{AI}} \in \text{int}S$.— By Lemma 2.4, the AI equilibrium is either Type 2 or Type 3. Suppose first that it is Type 2. Then $\sup W^* = \inf S^* = z_{\text{AI}}$, so $W^* \supset W$ and $S^* \subset S$ given that $z_{\text{AI}} > \hat{z}$ when $z_{\text{AI}} \in \text{int}S$.

Now, suppose the equilibrium is Type 3. Then, $\inf S^* = z_{\text{AI}} > \hat{z}$, immediately implying that $S^* \subset S$. To prove that $W^* \supset W$, we show that $\sup W^* = \bar{z}_3^* > \hat{z}$. Indeed, recall that $\bar{z}_3^*$ is given by:[2]

$$\bar{z}_3^* = e_3^*(1; \underline{z}_3^*, z_{\text{AI}}) \quad \text{and} \quad \tfrac{1}{h} = z_{\text{AI}} + \int_{z_{\text{AI}}}^1 n\big(e_3^*(z;\underline{z}_3^*, z_{\text{AI}})\big)dz$$
$$\text{where } \int_{z_{\text{AI}}}^{m_3^*(z;\underline{z}_3^*,z_{\text{AI}})} dG(u) = \int_{\underline{z}_3^*}^{z} h(1-u)dG(u) \text{ for } z \in [\underline{z}_3^*, \bar{z}_3^*]$$

Using the fact that $m_3^*(\bar{z}_3^*; \underline{z}_3^*, z_{\text{AI}}) = 1$, the equilibrium conditions that determine $\underline{z}_3^*$ and $\bar{z}_3^*$ can be written as follows:[3]

$$\underline{z}_3^* = \tilde{e}_3^*(z_{\text{AI}}; \bar{z}_3^*, z_{\text{AI}}) \quad \text{and} \quad \tfrac{1}{h} = z_{\text{AI}} + \int_{z_{\text{AI}}}^1 n\big(\tilde{e}_3^*(z;\bar{z}_3^*, z_{\text{AI}})\big)dz$$
$$\text{where } G(z) = 1 - \int_{\tilde{e}_3^*(z;\bar{z}_3^*,z_{\text{AI}})}^{\bar{z}_3^*} h(1-u)dG(u) \text{ for } z \in [z_{\text{AI}}, 1]$$

Define $\Gamma_3(x; z_{\text{AI}}) \equiv \frac{1}{h} - z_{\text{AI}} - \int_{z_{\text{AI}}}^1 n\big(\tilde{e}_3^*(z;x,z_{\text{AI}})\big)dz$. It is not difficult to prove that $\Gamma_3(x;z_{\text{AI}})$ is strictly decreasing in x and strictly increasing in z_{AI}. Moreover, $\bar{z}_3^*$ is given by the unique solution

[2] To avoid any type of confusion, we are making explicit that $m_3^*(z;\underline{z}_3^*, z_{\text{AI}})$ and $e_3^*(1;\underline{z}_3^*, z_{\text{AI}})$ depend on both $\underline{z}_3^*$ and z_{AI} (in the statement of Lemma 2.4, we simply wrote $m_3^*(z;\underline{z}_3^*)$ instead of $m_3^*(z;\underline{z}_3^*, z_{\text{AI}})$ to avoid cluttering notation).

[3] Note that $\tilde{e}_3^*(z;\bar{z}_3^*, z_{\text{AI}})$ is the equilibrium employee function indexed by $\bar{z}_3^*$ instead of $\underline{z}_3^*$.

to $\Gamma_3(\bar{z}_3^*; z_{\text{AI}}) = 0$. Hence, to prove that $\bar{z}_3^* > \hat{z}$, it suffices to show that $\Gamma_3(\hat{z}; z_{\text{AI}}) > 0$. Indeed, since $z_{\text{AI}} > \hat{z}$, we have that $\Gamma_3(\hat{z}; z_{\text{AI}}) > \Gamma_3(\hat{z}; \hat{z}) = \frac{1}{h} - \hat{z} - \int_{z_{\text{AI}}}^{1} n\big(e(z; \hat{z})\big)dz > 0$, where the second-to-last inequality follows because $\tilde{e}_3^*(z; \hat{z}, \hat{z}) = e(z; \hat{z})$ for all $z \in [\hat{z}, 1]$, and the last inequality comes from the pre-AI equilibrium characterized in Lemma 2.1. □

2.4 Proof of Proposition 4

For the proof that follows, it is important that we make explicit that the pre-AI equilibrium matching and employee functions depend on $\hat{z}$ (i.e., the knowledge threshold that separates workers and solvers in the pre-AI equilibrium). For this reason, we write $m(z; \hat{z})$ and $e(z; \hat{z})$ instead of $m(z)$ and $e(z)$, respectively.

• $z_{\text{AI}} \in \text{int}W$.— First, we show that every $z \in W^*$ is matched with a less knowledgeable solver post-AI than pre-AI. Note that if $z \in W_a^*$, then such a worker is matched with AI in the post-AI equilibrium. This immediately gives the desired result, as $m(z; \hat{z}) \geq \hat{z} > z_{\text{AI}}$ (given that $z_{\text{AI}} \in \text{int}W$).

Proving that every $z \in W_p^*$ is also matched with a less knowledgeable solver is more involved. Since $z_{\text{AI}} \in \text{int}W$, the post-AI equilibrium is either Type 1 or Type 2. Suppose first that it is Type 1. Then, the matching functions pre- and post-AI are given by:

$$\int_{\underline{z}_1^*}^{m_1^*(z;\underline{z}_1^*)} dG(u) = \int_0^z h(1-u)dG(u) \text{ for } z \in W_p^* = [0, z_{\text{AI}}]$$
$$\int_{\hat{z}}^{m(z;\hat{z})} dG(u) = \int_0^z h(1-u)dG(u) \text{ for } z \in W = [0, \hat{z}]$$

Thus, if $z \in W_p^* \cap W = W_p^*$, then $\int_{\underline{z}_1^*}^{m_1^*(z;\underline{z}_1^*)} dG(u) = \int_{\hat{z}}^{m(z;\hat{z})} dG(u)$, so $m_1^*(z; \underline{z}_1^*) < m(z; \hat{z})$ as $\underline{z}_1^* < \hat{z}$.

Suppose instead that the post-AI equilibrium is Type 2. Then, the matching functions pre- and post-AI are given by:

$$\int_{z_{\text{AI}}}^{m_2^*(z;\underline{z}_2^*)} dG(u) = \int_{\underline{z}_2^*}^z h(1-u)dG(u) \text{ for } z \in W_p^* = [\underline{z}_2^*, z_{\text{AI}}]$$
$$\int_{\hat{z}}^{m(z;\hat{z})} dG(u) = \int_0^z h(1-u)dG(u) \text{ for } z \in W = [0, \hat{z}]$$

Consequently, if $z \in W_p^* \cap W = W_p^*$, then $\int_{\hat{z}}^{m(z;\hat{z})} dG(u) - \int_{z_{\text{AI}}}^{m_2^*(z;\underline{z}_2^*)} dG(u) = \int_0^{\underline{z}_2^*} h(1-u)dG(u) > 0$, which implies that $m_2^*(z; \underline{z}_2^*) < m(z; \hat{z})$ as $z_{\text{AI}} < \hat{z}$.

We now turn to solvers, i.e., those $z \in S \subset S^*$. We first claim that if $e(z; \hat{z}) = z_{\text{AI}}$, then $z \in S_a^* \cap S$, which immediately implies that if $e(z; \hat{z}) = z_{\text{AI}}$, then z has the same span of control pre- and post-AI. The proof is via the contrapositive. Suppose that $z \notin S_a^* \cap S$, so $z \in S_p^* \cap S$. It follows that $z_{\text{AI}} \geq e_j^*(z; \underline{z}_j^*) > e(z; \hat{z})$, where the first inequality follows because AI is the best worker, and the second inequality follows because $e_j^*(z'; \underline{z}_j^*) > e(z'; \hat{z})$ for all $z' \in S_p^* \cap S$ (this follows because, as have already established above, $m_j^*(z''; \underline{z}_j^*) < m(z''; \hat{z})$ for all $z'' \in W_p^* \cap W = W_p^*$ and $j = 1, 2$). In particular, $e(z; \hat{z}) \neq z_{\text{AI}}$.

The previous claim implies that if $e(z; \hat{z}) < z_{\text{AI}}$, then z's span of control increases with AI, while if $e(z; \hat{z}) > z_{\text{AI}}$, then z's span of control decreases with AI. Indeed, if $e(z; \hat{z}) < z_{\text{AI}}$, then either $z \in S_p^* \cap S$

or $z \in S_a^* \cap S$. In either case, z is assisting more knowledgeable workers post-AI than pre-AI: If $z \in S_p^*$, then we have already shown that $e_j^*(z; \underline{z}_j^*) > e(z; \hat{z})$, while if $z \in S_a^*$, then, post-AI, individuals with knowledge z are assisting AI, while pre-AI, they are assisting humans with knowledge $e(z; \hat{z}) < z_{\text{AI}}$. Similarly, if $e(z; \hat{z}) > z_{\text{AI}}$, then $z \in S_a^* \cap S$. The latter implies that individuals with knowledge z are assisting less knowledgeable workers post-AI than pre-AI, as post-AI, they are assisting the work of AI, while pre-AI, they are assisting humans with knowledge $e(z; \hat{z}) > z_{\text{AI}}$. $\square$

• $z_{\text{AI}} \in \text{int}S$.— First, we show that every $z \in S^* \subset S$ has a larger span of control post-AI than pre-AI. Note that if $z \in S_a^*$, then such a solver is matched with AI in the post-AI equilibrium. Hence, $e(z; \hat{z}) \leq \hat{z} < z_{\text{AI}}$, as $z_{\text{AI}} \in \text{int}S$.

Next, we show that the same holds for every $z \in S_p^*$. Since $z_{\text{AI}} \in \text{int}S$, the post-AI equilibrium is either Type 2 or Type 3. Suppose first that it is Type 2. Then, the employee functions pre- and post-AI are given by:

$$\int_z^{\bar{z}_2^*} dG(u) = \int_{e_2^*(z;\underline{z}_2^*)}^{z_{\text{AI}}} h(1-u)dG(u) \text{ for } z \in S_p^* = [z_{\text{AI}}, \bar{z}_2^*]$$
$$\int_z^{1} dG(u) = \int_{e(z;\hat{z})}^{\hat{z}} h(1-u)dG(u) \text{ for } z \in S = [\hat{z}, 1]$$

Consequently, if $z \in S_p^* \cap S = S_p^*$, then $0 < \int_{\bar{z}_2^*}^{1} dG(u) = \int_{e(z;\hat{z})}^{\hat{z}} h(1-u)dG(u) - \int_{e_2^*(z;\underline{z}_2^*)}^{z_{\text{AI}}} h(1-u)dG(u)$, which implies that $e_2^*(z; \underline{z}_2^*) > e(z; \hat{z})$ (since $z_{\text{AI}} > \hat{z}$).

Suppose instead that the post-AI equilibrium is Type 3. Then, the employee functions pre- and post-AI are given by:

$$\int_z^{1} dG(u) = \int_{e_3^*(z;\underline{z}_3^*)}^{\bar{z}_3^*} h(1-u)dG(u) \text{ for } z \in S_p^* = [z_{\text{AI}}, 1]$$
$$\int_z^{1} dG(u) = \int_{e(z;\hat{z})}^{\hat{z}} h(1-u)dG(u) \text{ for } z \in S = [\hat{z}, 1]$$

Consequently, for $z \in S_p^* \cap S = S_p^*$, we have that $\int_{e_3^*(z;\underline{z}_3^*)}^{\bar{z}_3^*} h(1-u)dG(u) = \int_{e(z;\hat{z})}^{\hat{z}} h(1-u)dG(u)$, so $e(z; \hat{z}) < e_3^*(z; \underline{z}_3^*)$ since $\hat{z} < \bar{z}_3^*$.

We now turn to workers, i.e., those with $z \in W \subset W^*$. We first claim that if $z = e(z_{\text{AI}}; \hat{z})$, then $z \in W_a^* \cap W$, which immediately implies that if $z = e(z_{\text{AI}}; \hat{z})$, then z is equally productive pre- and post-AI. The proof is via the contrapositive. Suppose that $z \notin W_a^* \cap W$, so $z \in W_p^* \cap W$. Hence, $z_{\text{AI}} \leq m_j^*(z; \underline{z}_j^*) < m(z; \hat{z})$, where the first inequality follows because AI is the worst solver, and the second inequality follows because $m_j^*(z'; \underline{z}_j^*) < m(z'; \hat{z})$ for all $z' \in W_p^* \cap W$ (because, as we already showed above, $e_j^*(z''; \underline{z}_j^*) > e(z''; \hat{z})$ for all $z'' \in S_p^* \cap S = S_p^*$ and $j = 2, 3$). In particular, $z \neq e(z_{\text{AI}}; \hat{z})$.

The previous claim implies that if $z < e(z_{\text{AI}}; \hat{z})$, then z is strictly more productive post-AI than pre-AI, while if $z > e(z_{\text{AI}}; \hat{z})$, then z is strictly less productive post-AI than pre-AI. Indeed, if $z < e(z_{\text{AI}}; \hat{z})$, then $z \in W_a^* \cap W$. Hence, individuals with knowledge z match with more knowledgeable solvers post-AI than pre-AI because post-AI, they are assisted by AI, while pre-AI, they are assisted by humans with knowledge $m(z; \hat{z}) < z_{\text{AI}}$. Similarly, if $z > e(z_{\text{AI}}; \hat{z})$, then $z \in W_a^* \cap W$ or $z \in W_p^* \cap W$. If $z \in W_a^* \cap W$, then individuals with knowledge z match with less knowledgeable solvers post-AI than pre-AI because post-AI, they are assisted by AI, while pre-AI, they are assisted by humans with knowledge $m(z;\hat{z}) > z_{\mathrm{AI}}$. Similarly, if $z \in W_p^* \cap W$, then individuals with knowledge z also match with less knowledgeable solvers post-AI than pre-AI because we already established that $m_j^*(z;\underline{z}_j^*) < m(z;\hat{z})$ for $j = 2, 3$. □

2.5 Proof of Proposition 5

2.5.1 A Preliminary Result

To prove this proposition, we begin with the following preliminary result:

Lemma 2.6. *Define $\Delta(z) \equiv w^*(z) - w(z)$. We have that $\Delta(z_{\mathrm{AI}}) < 0$ and:*

- *If $\Delta(z) > 0$ for some $z \in [0, z_{\mathrm{AI}}]$, then $\Delta(z') > 0$ for all $z' \in [0, z]$.*
- *If $\Delta(z) > 0$ for some $z \in [z_{\mathrm{AI}}, 1]$, then $\Delta(z') > 0$ for all $z' \in [z, 1]$.*

For ease of exposition, we divide the proof Lemma 2.6 into three smaller claims:

Claim 2.2. *(i) $\Delta(z_{\mathrm{AI}}) < 0$, (ii) $\Delta(1) > 0$, and (iii) $\Delta(0) > 0$ if $z_{\mathrm{AI}} \in S$.*

Proof. That $\Delta(z_{\mathrm{AI}}) = z_{\mathrm{AI}} - w(z_{\mathrm{AI}}) < 0$ follows directly from the fact that $w(z) > z$ for all $z \in [0, 1]$ when $h < h_0$ (see Proposition 1). Consider next $\Delta(1) = w^*(1) - w(1)$. By Lemma 2.4, we have that $w^*(1) = 1/h$, while by Lemma 2.1 and Corollary 2.1, we have that $w(1) < 1/h$. Hence, $\Delta(1) > 0$. Finally, consider $\Delta(0) = w^*(0) - w(0)$ and suppose that $z_{\mathrm{AI}} \in S$. From Lemma 2.4, we have that $w^*(0) = z_{\mathrm{AI}}(1 - h)$ as the human with zero knowledge is assisted by AI irrespective of whether $z_{\mathrm{AI}} \in \mathcal{R}_2$ or $z_{\mathrm{AI}} \in \mathcal{R}_3$. Moreover, from Lemma 2.1 and Corollary 2.1 we have that $w(0) = \hat{z} - hw(\hat{z})$. Thus, $\Delta(0) = z_{\mathrm{AI}}(1 - h) - (\hat{z} - hw(\hat{z})) > (1 - h)(z_{\mathrm{AI}} - \hat{z}) \geq 0$, where the first inequality follows because $w(\hat{z}) > \hat{z}$, and the second inequality follows because $z_{\mathrm{AI}} \geq \hat{z}$. □

Claim 2.3. *If $\Delta(z) > 0$ for some $z \in [z_{\mathrm{AI}}, 1)$, then $\Delta(z') > 0$ for all $z' \in [z, 1)$.*

Proof. Given that $\Delta(z_{\mathrm{AI}}) < 0$ (by Claim 2.2), it suffices to show that if $\Delta(z)$ crosses zero at some $z > z_{\mathrm{AI}}$, then it always does so from below. We first consider $z_{\mathrm{AI}} \in \mathrm{int}W$ and then $z_{\mathrm{AI}} \in \mathrm{int}S$.

• $z_{\mathrm{AI}} \in \mathrm{int}W$.— Lemma 2.4 implies that $\sup W^* = z_{\mathrm{AI}}$, while Proposition 3 that $W^* \subset W$ and $S^* \supset S$. Hence, if $z > z_{\mathrm{AI}}$, then z can only belong to either $I^* \cap W$, $S^* \cap W$, or $S^* \cap S$.

Now, irrespective of the presence of AI, the marginal return to knowledge is greater for solvers than for independent producers, and it is greater for independent producers than for workers. Hence, $\Delta'(z) = w^{*\prime}(z) - w'(z) \geq 0$ whenever z is in either $I^* \cap W$ or $S^* \cap W$. This implies that if $\Delta(z)$ crosses zero in either of these sets, then it necessarily does so from below.

Consider then $z \in S^* \cap S$. Since $S^* = S_p^* \cup S_a^*$, here we have two cases to consider: $z \in S_p^* \cap S$ and $z \in S_a^* \cap S$. If $z \in S_p^* \cap S$, then $\Delta'(z) = n(e_j^*(z;\underline{z}_j^*)) - n(e(z;\hat{z}))$, where $e_j^*(z;\underline{z}_j^*)$ is the employee function in a Type $j = 1, 2$ equilibrium and $e(z;\hat{z})$ employee function in the pre-AI equilibrium. By

Proposition 4, we know that $e_j^*(z;\underline{z}_j^*) > e(z;\hat{z})$, as every $z \in S_p^* \cap S$ supervises better workers post-AI than pre-AI. Hence, in this case, $\Delta'(z) > 0$ also. Consequently, if $\Delta(z)$ crosses zero when $z \in S_p^* \cap S$, then it necessarily crosses from below.

Finally, consider the possibility that $\Delta(z)$ crosses zero at $z \in S_a^* \cap S$. In this case, $\Delta'(z) = n(z_{\text{AI}}) - n(e(z;\hat{z})) \gtrless 0$, so $\Delta(z)$ is no longer monotone in z in this set. Note, however, that $\Delta''(z) = -n'(e(z;\hat{z}))e'(z;\hat{z}) < 0$, so $\Delta(z)$ is concave. Moreover, if $S_a^* \neq \emptyset$, then $1 \in S_a^*$. The fact that $\Delta(z)$ is concave and that $\Delta(1) > 0$ then immediately implies that if $\Delta(z)$ crosses zero in this set, then it can only cross once and from below (otherwise, $\Delta(1) \leq 0$ contradicting the fact $\Delta(1) > 0$).

• $z_{\text{AI}} \in \text{int}S$.— From Lemma 2.4, we know that $\inf S^* = z_{\text{AI}}$. Moreover, from Proposition 3, we have that $W^* \supseteq W$ and $S^* \subseteq S$. Consequently, if $z > z_{\text{AI}}$, then $z \in S^* \cap S$. Thus, we have two cases to consider: $z \in S_p^* \cap S$ and $z \in S_a^* \cap S$.

If $z \in S_p^* \cap S$, then $\Delta'(z) = n(e_j^*(z;\underline{z}_j^*)) - n(e(z;\hat{z}))$, while if $z \in S_a^* \cap S$, then $\Delta'(z) = n(z_{\text{AI}}) - n(e(z;\hat{z}))$, where $e_j^*(z;\underline{z}_j^*)$ is the employee function in a Type $j = 2,3$ equilibrium and $e(z;\hat{z})$ the employee function in the pre-AI equilibrium. In either case, $\Delta'(z) > 0$ since Proposition 4 states that $e_j^*(z;\underline{z}_j^*) > e(z;\hat{z})$ and $z_{\text{AI}} > e(z;\hat{z})$ when AI has the knowledge of a pre-AI solver. Consequently, $\Delta'(z) > 0$ for all $z \geq z_{\text{AI}}$, so if $\Delta(z)$ crosses zero at some $z > z_{\text{AI}}$, then it always crosses it from below. □

Claim 2.4. *If $\Delta(z) > 0$ for some $z \in [0, z_{\text{AI}}]$, then $\Delta(z') > 0$ for all $z' \in [0, z]$.*

Proof. Given that $\Delta(z_{\text{AI}}) < 0$, it suffices to show that if $\Delta(z)$ crosses zero at some $z < z_{\text{AI}}$, then it always crosses zero from above. We first consider $z_{\text{AI}} \in \text{int}W$, and then $z_{\text{AI}} \in \text{int}S$.

• $z_{\text{AI}} \in \text{int}W$.— Lemma 2.4 implies that $\sup W^* = z_{\text{AI}}$, while Proposition 3 that $W^* \subset W$ and $S^* \supset S$. Consequently, if $z < z_{\text{AI}}$, then $z \in W^* \cap W$, where $W^* = W_a^* \cup W_p^*$.

Now, if $z \in W_p^* \cap W$, then $\Delta'(z) = hw^*(m_j^*(z;\underline{z}_j^*)) - hw(m(z;\hat{z}))$, where $m_j^*(z;\underline{z}_j^*)$ is the matching function in a Type $j = 1,2$ equilibrium, and $m(z;\hat{z})$ the matching function of the pre-AI equilibrium. However, using the firms' zero-profit condition:

$$\Delta'(z) = hw^*(m_j^*(z;\underline{z}_j^*)) - hw(m(z;\hat{z})) = \frac{m_j^*(z;\underline{z}_j^*) - m(z;\hat{z}) - \Delta(z)}{1-z}$$

Consequently, if $\Delta(z) = 0$ at some z in this interval, say at $z = \zeta$, then $\Delta'(\zeta) = m_j^*(\zeta;\underline{z}_j^*) - m(\zeta;\hat{z}) \leq 0$, where the last inequality follows because every worker is assisted by a worse solver post-AI than pre-AI when $z_{\text{AI}} \in \text{int}W$ (Proposition 4).

On the other hand, if $z \in W_a^* \cap W$, then following the same reasoning as before, we have that:

$$\Delta'(z) = hw(z_{\text{AI}}) - hw(m(z;\hat{z})) = \frac{z_{\text{AI}} - m(z;\hat{z}) - \Delta(z)}{1-z}$$

Consequently, if $\Delta(z) = 0$ at some z in this interval, say at $z = \zeta$, then $\Delta'(\zeta) = z_{\text{AI}} - m(\zeta;\hat{z}) \leq 0$, where the inequality follows, again, from Proposition 4.

- $z_{\text{AI}} \in \text{int}S$.— In this case, Lemma 2.4 implies that $\inf S^* = z_{\text{AI}}$, while Proposition 3 that $W^* \supset W$ and $S^* \subset S$. Consequently, if $z < z_{\text{AI}}$, then z can only belong to either $I^* \cap S$, $W^* \cap S$, or $W^* \cap W$.

Now, $\Delta'(z) \leq 0$ whenever z is in either $I^* \cap S$ or $W^* \cap S$, given that the marginal return to knowledge is greater for solvers than for independent producers, and it is greater for independent producers than for workers. Hence, if $\Delta(z)$ crosses zero in either of these sets, then it necessarily does so from above.

Consider $z \in W^* \cap W$. We have two cases to consider: $z \in W_p^* \cap W$ and $z \in W_a^* \cap W$. If $z \in W_p^* \cap W$, then:

$$\Delta'(z) = hw^*(m_j^*(z; \underline{z}_j^*)) - hw(m(z; \hat{z})) = \frac{m_j^*(z; \underline{z}_j^*) - m(z; \hat{z}) - \Delta(z)}{1 - z}$$

where $m_j^*(z; \underline{z}_j^*)$ is the matching function in a Type $j = 2, 3$ equilibrium, and $m(z; \hat{z})$ the matching function of the pre-AI equilibrium. Consequently, if $\Delta(z) = 0$ at some z in this interval, say at $z = \zeta$, then $\Delta'(\zeta) = m_j^*(\zeta; \underline{z}_j^*) - m(\zeta; \hat{z}) \leq 0$, where the last inequality follows because every $z \in W^* \cap W$ is assisted by a worse solver post-AI than pre-AI when $z_{\text{AI}} \in \text{int}S$.

Finally, consider the possibility that $\Delta(z)$ crosses zero at $z \in W_a^* \cap W$. In this case, $\Delta'(z) = hz_{\text{AI}} - hw(m(z; \hat{z}))$, so $\Delta''(z) = -hw'(m(z; \hat{z}))f'(z; \hat{z}) < 0$, implying that $\Delta(z)$ is concave. Moreover, if $W_a^* \neq \emptyset$, then $0 \in W_a^*$, and we know that $\Delta(0) > 0$ in this case. Consequently, if $\Delta(z)$ crosses zero in this set, then it can only cross once and from above (otherwise, $\Delta(0) \leq 0$ contradicting the fact $\Delta(0) > 0$). □

2.5.2 Proof of Proposition 5

Note that Lemma 2.6 implies that B takes the form $[0, z_b)$ and T takes the form $(z_t, 1]$ for some $z_b \geq 0$ and $z_t \leq 1$. Hence, to verify the existence of winners below or above z_{AI}, it suffices to analyze AI's impact on the wages of the least and most knowledgeable individuals.

Part (i) ("There are winners at the bottom if AI is good enough").— To prove this part, we begin by constructing $\bar{z}_{\text{AI}}$ and then showing that the statement is true.

Let $\Delta(0; z_{\text{AI}}) \equiv w^*(0; z_{\text{AI}}) - w(0)$, and define $\bar{z}_{\text{AI}}$ as the solution to $\Delta(0; \bar{z}_{\text{AI}}) = 0$. We first show that $\bar{z}_{\text{AI}}$ exists, is unique, and that $\bar{z}_{\text{AI}} \in (0, \hat{z})$. We do this by showing that, as we move from $z_{\text{AI}} = 0$ to $z_{\text{AI}} = 1$, $\Delta(0; z_{\text{AI}})$ crosses zero once, and that this crossing point is at $z_{\text{AI}} < \hat{z}$. Indeed, if $z_{\text{AI}} \geq \hat{z}$, then Claim 2.2 states that $\Delta(0; z_{\text{AI}}) > 0$ (as $z_{\text{AI}} \in S$ in this case). Moreover, as shown in Lemma 2.5, $0 \in \mathcal{R}_1$, so when $z_{\text{AI}} = 0$, the equilibrium is always Type 1. The latter implies that $\Delta(0; 0) = \underline{z}_1^*(0)(1 - h) - w(0) = -w(0) < 0$,[4] where the second-to-last equality follows because $\underline{z}_1^*(0) = 0$, as can be easily be proven from the condition that determines $\underline{z}_1^*(z_{\text{AI}})$ (see the statement of Lemma 2.4).

When $z \in [0, \hat{z}] = W$, the equilibrium is either Type 1, in which case $w^*(0; z_{\text{AI}}) = \underline{z}_1^*(z_{\text{AI}})(1 - h)$, or Type 2, in which case $w^*(0; z_{\text{AI}}) = z_{\text{AI}}(1 - h)$. Using the equilibrium condition that determines

[4] To avoid any confusion, here we make explicit that $\underline{z}_1^*(z_{\text{AI}})$ depends on z_{AI} as seen from the condition that determines $\underline{z}_1^*(z_{\text{AI}})$ in the statement of Lemma 2.4.

$\underline{z}_1^*(z_{\mathrm{AI}})$, it is not difficult to prove that (i) $\underline{z}_1^*(z_{\mathrm{AI}})$ is strictly increasing in z_{AI}, and that (ii) $\underline{z}_1^*(z_{\mathrm{AI}}) = z_{\mathrm{AI}}$ whenever we switch from a Type 1 to a Type 2 equilibrium (and vice versa). Consequently, irrespective of the equilibrium type in this region, $w^*(0; z_{\mathrm{AI}})$ is continuous and strictly increasing in z_{AI}, which implies that $\Delta(0; z_{\mathrm{AI}})$ is also continuous and strictly increasing in z_{AI}. This result, combined with the fact that $\Delta(0; 0) < 0$ and $\Delta(0; \hat{z}) > 0$, immediately yields the desired result.

Having constructed $\bar{z}_{\mathrm{AI}}$, we prove that there exists $z < z_{\mathrm{AI}}$ such that $\Delta(z; z_{\mathrm{AI}}) > 0$ if and only if $z_{\mathrm{AI}} > \bar{z}_{\mathrm{AI}}$. First we show that if there exists $z < z_{\mathrm{AI}}$ such that $\Delta(z; z_{\mathrm{AI}}) > 0$, then $z_{\mathrm{AI}} > \bar{z}_{\mathrm{AI}}$. To do this, we prove the contrapositive statement: If $z_{\mathrm{AI}} \leq \bar{z}_{\mathrm{AI}}$, then there is no such z. Indeed, as shown above, $\Delta(0; z_{\mathrm{AI}}) \leq 0$ for all $z_{\mathrm{AI}} \leq \bar{z}_{\mathrm{AI}}$. Hence, Lemma 2.6 implies that $\Delta(z; z_{\mathrm{AI}}) \leq 0$ for all $z < z_{\mathrm{AI}}$.

Finally, we prove that if $z_{\mathrm{AI}} > \bar{z}_{\mathrm{AI}}$, then there exists $z < z_{\mathrm{AI}}$ such that $\Delta(z; z_{\mathrm{AI}}) > 0$. Indeed, as shown above, if $z_{\mathrm{AI}} > \bar{z}_{\mathrm{AI}}$, then $\Delta(0; z_{\mathrm{AI}}) > 0$. Consequently, Lemma 2.6 immediately implies that there exists $\zeta \in [0, z_{\mathrm{AI}})$ such that $\Delta(z; z_{\mathrm{AI}}) > 0$ for $z \in [0, \zeta)$ and $\Delta(z; z_{\mathrm{AI}}) < 0$ for $z \in (\zeta, z_{\mathrm{AI}}]$. □

Part (ii) ("There are always winners at the top").— This part follows directly from Lemma 2.6 and the fact that $\Delta(1) > 0$ (see Claim 2.2). □

2.6 Proof of Proposition 6

Proposition 6 is a direct implication of Lemmas 2.7– 2.10 below.

Lemma 2.7. *Any equilibrium with non-autonomous AI has the following features:*

- *Efficiency: Maximizes total output (which is equal to total labor income).*
- *Occupational Stratification: $W^\star \preceq I^\star \preceq S^\star = S_p^\star$.*
- *Positive assortative matching: $m^\star : W_p^\star \to S_p^\star$ is strictly increasing.*
- *$w^\star(z) = z_{\mathrm{AI}}$ for all $z \in W_a^\star$. Moreover, $W_a^\star \preceq \{z_{\mathrm{AI}}\}$.*
- *Only the least knowledgeable humans use AI: $W_a^\star \preceq W_p^\star$.*

Proof. The proof that the equilibrium (i) is efficient and (ii) satisfies both occupational stratification and positive assortative matching mirror those for the autonomous AI scenario. The fact that total output is equal to total labor income follows because total capital income, in this case, is zero, as $r^\star = 0$.

The result that $w^\star(z) = z_{\mathrm{AI}}$ for all $z \in W_a^\star$ follows from the zero profit condition of top-automated firms plus the fact that $r^\star = 0$. Showing that $W_a^\star \preceq \{z_{\mathrm{AI}}\}$ is also straightforward: no top-automated firm ever hires human workers who are more knowledgeable than its AI solver, as such workers can solve all problems that the AI solver can.[5]

[5]Strictly speaking, the equilibrium may exhibit top-automated firms that employ humans with knowledge exceeding that of the AI. Although these firms gain no direct benefit from using AI—since the human at the bottom layer can solve all problems the AI can—the zero compute cost allows them to rent compute without denting their profits. However, such top-automated firms are functionally identical to single-layer firms. For simplicity, we rule out this type of top-automated firms.

Finally, we show that $W_a^\star \preceq W_p^\star$. Suppose for contradiction that $z \in W_a^\star$ and $z' \in W_p^\star$ with $z' < z$. Note that no one can earn a wage lower than z_{AI} in equilibrium (otherwise, a top-automated firm could make strictly positive profits). Moreover, firm optimality plus firms' zero-profit condition implies that $w^\star$ is non-decreasing. Consequently, $w^\star(z') = z_{\text{AI}}$, as $z' < z$ and $z \in W_a^\star$. However, if so, then the non-automated firm employing z' could hire z instead for the same wage, strictly increasing its profits, as z is strictly more knowledgeable than z'. Contradiction. □

The next corollary is a direct implication of Lemma 2.7:

Corollary 2.3. *In the presence of non-autonomous AI, an equilibrium allocation must take one of the following three forms:*

- ***Pre-AI Configuration:***

$$W_a^\star = \emptyset,\ W_p^\star = [0, \hat{z}],\ I^\star = \emptyset,\ S_p^\star = [\hat{z}, 1],\ S_a^\star = \emptyset,\ \textit{where } \hat{z} \textit{ is given by } \int_0^{\hat{z}} h(1-z)dG(z) = \int_{\hat{z}}^1 dG(z)$$
$$\textit{So } \mu_w^\star = 0,\ \mu_s^\star = 0,\ \mu_i^\star = \mu$$

- ***Type 1a configuration:***

$$W_a^\star = [0, \underline{z}_1^\star],\ W_p^\star = [\underline{z}_1^\star, \bar{z}_1^\star],\ I^\star = \emptyset,\ S_p^\star = [\bar{z}_1^\star, 1],\ S_a^\star = \emptyset,\ \textit{where } 0 < \underline{z}_1^\star \leq \bar{z}_1^\star \leq 1$$
$$\textit{So } \mu_w^\star = 0,\ \mu_s^\star = \int_0^{\underline{z}_1^\star} h(1-z)dG(z),\ \mu_i^\star = \mu - \mu_s^\star$$

- ***Type 2a configuration:***

$$W_a^\star = [0, \underline{z}_2^\star],\ W_p^\star = [\underline{z}_2^\star, \hat{z}_2^\star],\ I^\star = [\hat{z}_2^\star, \bar{z}_2^\star],\ S_p^\star = [\bar{z}_2^\star, 1],\ S_a^\star = \emptyset,\ \textit{where } 0 \leq \underline{z}_2^\star \leq \hat{z}_2^\star \leq \bar{z}_2^\star \leq 1$$
$$\textit{So } \mu_w^\star = 0,\ \mu_s^\star = \int_0^{\underline{z}_2^\star} h(1-z)dG(z),\ \mu_i^\star = \mu - \mu_s^\star$$

Proof. By Lemma 2.7, we have that, in equilibriun, $W_a^\star \preceq W_p^\star \preceq I^\star \preceq S_p^\star$ and $S_a^\star = \emptyset$. Now, if AI is not used in equilibrium (i.e., $W_a^\star = \emptyset$), the equilibrium must coincide with the pre-AI equilibrium (this implies, in particular, that $I^\star = \emptyset$ necessarily in this case). Otherwise, there are only two alternative configurations, either $W_a^\star \preceq W_p^\star \preceq S_p^\star$ with $I^\star = \emptyset$, or $W_a^\star \preceq W_p^\star \preceq I^\star \preceq S_p^\star$ with all being non-empty. □

The following lemma—which is the main result of this chapter—characterizes in detail the post-AI equilibrium for a non-autonomous AI when compute is abundant relative to time:

Lemma 2.8. *In the presence of a non-autonomous AI, there is a unique competitive equilibrium. It is given as follows:*

- *When $z_{\text{AI}} \leq w(0)$, the equilibrium allocation follows a pre-AI configuration.*
- *When $z_{\text{AI}} > w(0)$, the equilibrium allocation is either Type 1a or Type 2a.*
  - *If the allocation is Type 1a, the equilibrium cutoffs $\underline{z}_1^\star$ and $\bar{z}_1^\star$ are the unique solution to:*

$$n(\underline{z}_1)(\bar{z}_1 - z_{\text{AI}}) = \bar{z}_1 + \left(\frac{1}{1+n(\bar{z}_1)}\right)\left[\frac{1}{h} - \bar{z}_1 - \int_{\bar{z}_1}^1 n(e_1^\star(z;\bar{z}_1))dz\right] \quad \textit{and} \quad m_1^\star(\underline{z}_1;\bar{z}_1) = \bar{z}_1$$

where $m_1^\star : [\underline{z}_1, \bar{z}_1] \to [\bar{z}_1, 1]$ satisfies $\int_z^{\bar{z}_1} h(1-u)dG(u) = \int_{m_1^\star(z;\bar{z}_1)}^1 dG(u)$ and $e_1^\star(z;\cdot) = (m_1^\star)^{-1}(z;\cdot)$

And $\bar{z}_1^\star$ must be such that $\frac{1}{h} > \bar{z}_1^\star + \int_{\bar{z}_1^\star}^1 n(e_1^\star(z;\bar{z}_1^\star))dz$. In this case, $m^\star(z) = m_1^\star(z;\bar{z}_1^\star)$.

– *If the allocation is Type 2a, the equilibrium cutoffs $\underline{z}_2^\star$, $\hat{z}_2^\star$, and $\bar{z}_2^\star$ are the unique solution to:*

$$\tfrac{1}{h} = \bar{z}_2 + \textstyle\int_{\bar{z}_2}^1 n(e_2^\star(z;\hat{z}_2))dz \quad \textit{and} \quad \bar{z}_2 = n(\underline{z}_2)(\bar{z}_2 - z_{\text{AI}}) \quad \textit{and} \quad m_2^\star(\underline{z}_2;\hat{z}_2) = \bar{z}_2$$

where $m_2^\star : [\underline{z}_2, \hat{z}_2] \to [\bar{z}_2, 1]$ is given by $\int_z^{\hat{z}_2} h(1-u)dG(u) = \int_{m_2^\star(z;\hat{z}_2)}^1 dG(u)$ and $e_2^\star(z;\cdot) = (m_2^\star)^{-1}(z;\cdot)$

The equilibrium matching function is then given by $m^\star(z) = m_2^\star(z;\hat{z}_2^\star)$.

Irrespective of whether the allocation is Type 1a or Type 2a, the equilibrium wage $w^\star$ is continuous, weakly increasing, and weakly convex, and is given by:

– $w^\star(z) = z_{\text{AI}}$ *for all* $z \in W_a^\star$.
– $w^\star(z) = m^\star(z) - w^\star(m^\star(z))/n(z)$ *for all* $z \in W_p^\star$.
– $w^\star(z) = z$ *for all* $z \in I^\star$.
– $w^\star(z) = C^\star + \int_{\inf S_p^\star}^z n(e^\star(u))du$ *for all* $z \in S_p^\star$*, where* $C^\star = \bar{z}_1^\star + \left(\frac{1}{1+n(\bar{z}_1^\star)}\right)\left(\frac{1}{h} - \bar{z}_1^\star - \int_{\bar{z}_1^\star}^1 n\big(e^\star(u)\big)du\right)$ *if the equilibrium is Type 1a, and* $C^\star = \bar{z}_2^\star$ *if it is Type 2a.*

To prove this lemma, we will make use of the following auxiliary result:

Claim 2.5. *For a fixed parameter $\kappa \in [0,1)$, consider an auxiliary economy identical to the 'true' economy, except that all individuals with $z \in [0,\kappa]$ are absent (so there is only a mass $1 - G(\kappa)$ of humans). The pre-AI equilibrium of such an economy is unique and given as follows. There exists $\tilde{h}_0(\kappa) \in [0,1]$ such that:*

- *If $h < \tilde{h}_0(\kappa)$, then $\tilde{W} = [\kappa, \hat{z}(\kappa)]$ and $\tilde{S} = [\hat{z}(\kappa), 1]$. Moreover, workers and solvers are matched according to the strictly increasing function $\tilde{m}_1(z;\hat{z}(\kappa))$ given by $\int_z^{\hat{z}(\kappa)} h(1-u)dG(u) = \int_{\tilde{m}_1(z;\hat{z}(\kappa))}^1 dG(z)$. Finally, the cutoff $\hat{z}(\kappa) \in (\kappa, 1)$ is the unique solution to $\int_\kappa^{\hat{z}(\kappa)} h(1-z)dG(z) = \int_{\hat{z}(\kappa)}^1 dG(z)$ and satisfies:*

$$\tfrac{1}{h} - \hat{z}(\kappa) > \textstyle\int_{\hat{z}(\kappa)}^1 n(\tilde{e}_1(z;\hat{z}(\kappa)))dz, \ \textit{where } \tilde{e}_1(z;\hat{z}(\kappa)) = \tilde{m}_1^{-1}(z;\hat{z}(\kappa))$$

- *If $h \geq \tilde{h}_0(\kappa)$, then $\tilde{W} = [\kappa, \underline{z}(\kappa)]$, $\tilde{I} = (\underline{z}(\kappa), \bar{z}(\kappa))$, and $\tilde{S} = [\bar{z}(\kappa), 1]$. Moreover, workers and solvers are matched according to the strictly increasing function $\tilde{m}_2(z;\underline{z}(\kappa))$, given by $\int_z^{\underline{z}(\kappa)} h(1-u)dG(u) = \int_{\tilde{m}_2(z;\underline{z}(\kappa))}^1 dG(u)$. Finally, the cutoffs $\kappa < \underline{z}(\kappa) < \bar{z}(\kappa) < 1$ are the unique solutions to:*

$$\tfrac{1}{h} - \bar{z}(\kappa) = \textstyle\int_{\bar{z}(\kappa)}^1 n\big(\tilde{e}_2(u;\underline{z}(\kappa))\big)du, \ \textit{and } \tilde{m}_2(\kappa;\underline{z}(\kappa)) = \bar{z}(\kappa), \ \textit{where } \tilde{e}_2(z;\underline{z}(\kappa)) = \tilde{m}_2^{-1}(z;\underline{z}(\kappa))$$

Let $\tilde{m}(z;\kappa) = \tilde{m}_1(z;\hat{z}(\kappa))$ if $h < \tilde{h}_0(\kappa)$ and $\tilde{m}(z;\kappa) = \tilde{m}_2(z;\underline{z}(\kappa))$, otherwise. Define $\tilde{e}(z;\kappa) = (\tilde{m})^{-1}(z;\kappa)$. Irrespective of $h \gtrless \tilde{h}_0(\kappa)$, the equilibrium wage $\tilde{w}$ is continuous, weakly increasing, and weakly convex in z, and is given by:

- $\tilde{w}(z;\kappa) = \tilde{m}(z;\kappa) - \tilde{w}(\tilde{m}(z;\kappa))/n(z)$ *for all* $z \in \tilde{W}$.
- $\tilde{w}(z;\kappa) = z$ *for all* $z \in \tilde{I}$.
- $\tilde{w}(z;\kappa) = \tilde{C}(\kappa) + \int_{\inf \tilde{S}}^z n(\tilde{e}(u;\kappa))du$ *for all* $z \in \tilde{S}$*, where:*

$$\tilde{C}(\kappa) = \begin{cases} \hat{z}(\kappa) + \left(\frac{1}{1+n(\hat{z}(\kappa))}\right)\left(\frac{1}{h} - \hat{z}(\kappa) - \int_{\hat{z}(\kappa)}^1 n\big(\tilde{e}(u;\kappa)\big)du\right) & \textit{if } h < \tilde{h}_0(\kappa) \\ \bar{z}(\kappa) & \textit{if } h \geq \tilde{h}_0(\kappa) \end{cases}$$

Finally, the wage $\tilde{w}(\kappa;\kappa)$ is continuous and strictly increasing in κ, and $\tilde{w}(0;0) = w(0)$, where $w(0)$ is the equilibrium wages of the least knowledgeable individual of the pre-AI economy of the 'true' economy.

Proof. The characterization of the equilibrium (i.e., the equilibrium occupations and wages) is virtually identical to the characterization of the pre-AI equilibrium of the 'true' economy (which has $\kappa = 0$) of Chapter 2.1. Furthermore, the fact that $\tilde{w}(0;0) = w(0)$ follows because the auxiliary economy with $\kappa = 0$ is identical to the 'true' economy, and the equilibrium is unique in both cases. Hence, here we only prove that the wage $\tilde{w}(\kappa;\kappa)$ is continuous and strictly increasing in κ.

First, note that the equilibrium is continuous in (h,κ) and therefore so is $\tilde{w}(\kappa;\kappa)$. This follows because equilibrium occupations and wages are continuous in (h,κ) when $h < \tilde{h}_0(\kappa)$ and when $h > \tilde{h}_0(\kappa)$, and because equilibrium wages and occupations coincide as we approach $\tilde{h}_0(\kappa)$ from above or below. The latter is due to the fact that for a given κ, $\tilde{h}_0(\kappa)$ is the unique $h \in (0,1)$ that satisfies $\frac{1}{h} - \hat{z}(\kappa;h) = \int_{\hat{z}(\kappa;h)}^{1} n(\tilde{e}(z;\hat{z}(\kappa;h),h),h)dz$ (as shown by Fuchs et al., 2015).[6]

Now, because $\tilde{w}(\kappa;\kappa)$ is continuous in κ, to show that $\tilde{w}(\kappa;\kappa)$ is strictly increasing in κ it suffices to show that it strictly increasing when $h < \tilde{h}_0(\kappa)$ and when $h > \tilde{h}_0(\kappa)$. When $h < \tilde{h}_0(\kappa)$, we have that:

$$\tilde{w}(\kappa;\kappa) = \hat{z}(\kappa)[1 - h(1-\kappa)] - h(1-\kappa)\left(\frac{1}{1+n(\hat{z}(\kappa))}\right)\left(\frac{1}{h} - \hat{z}(\kappa) - \int_{\hat{z}(\kappa)}^{1} n(\tilde{e}(z;\hat{z}(\kappa)))dz\right)$$

Using the facts that (i) $\hat{z}(\kappa)$ is strictly increasing in κ, (ii) $(\partial/\partial\hat{z})\tilde{e}(z;\hat{z}(\kappa)) > 0$, and (iii) $\frac{1}{h} - \hat{z}(\kappa) > \int_{\hat{z}(\kappa)}^{1} n(\tilde{e}(z;\hat{z}(\kappa)))dz$, it is not difficult to prove that $(d/d\kappa)\tilde{w}(\kappa;\kappa) > 0$. When $h > \tilde{h}_0(\kappa)$, in turn, we have that $\tilde{w}(\kappa;\kappa) = \bar{z}(\kappa)[1 - h(1-\kappa)]$. This is also strictly increasing in κ, as $\bar{z}(\kappa)$ is strictly increasing in κ. □

To understand the usefulness of Claim 2.5, note that the equilibrium wages and occupations of the humans who do not work with a non-autonomous AI—those in $W_p^\star$ and $S_p^\star$—must coincide with the pre-AI equilibrium of the auxiliary κ-economy when $\kappa = \inf W_p^\star$. Of course, the threshold $\kappa = \inf W_p^\star$ is endogenous. However, we know that $w^\star(z) = z_{\text{AI}}$ for all $z \in W_a^\star$. Hence, the endogenous cutoff $\kappa = \inf W_p^\star$ is determined either by $\tilde{w}(\kappa;\kappa) = z_{\text{AI}}$ (if the κ that solves this condition is weakly positive), or it is equal to $\kappa = 0$ (if the κ that solves $\tilde{w}(\kappa;\kappa) = z_{\text{AI}}$ is strictly negative).

We can now proceed to prove Lemma 2.8. We do this in two steps. First, we argue that when $z_{\text{AI}} \leq w(0)$, there is a unique equilibrium with the pre-AI configuration. Then, we show that when $z_{\text{AI}} > w(0)$, there is a unique equilibrium, and this equilibrium follows either a Type 1a or a Type 2a configuration. For each case, we obtain the conditions that determine the equilibrium cutoffs and wages.

Step 1. First, it is clear that if $z_{\text{AI}} \leq w(0)$, then the pre-AI equilibrium is also an equilibrium with a non-autonomous AI. Indeed, if at the prevailing pre-AI wages a top-automated firm hired a human with knowledge z, the most that firm would earn in profits is $n(z)[\max\{z, z_{\text{AI}}\} - w(z)] \leq 0$, where the inequality follows because $w(z) > z$ and $z_{\text{AI}} \leq w(0) < w(z)$ (as $w(z)$ is strictly increasing in

[6] Here we are making explicit that $\hat{z}(\kappa;h)$, $\tilde{e}(z;\hat{z},h)$, and $n(z;h) = [h \times (1-z)]^{-1}$ all depend on h.

z). Thus, no firm has incentives to deviate from the pre-AI equilibrium, even in the presence of non-autonomous AI.

To show that the equilibrium is unique, suppose for contradiction that there exists an equilibrium where AI is used. This implies that there exists $\kappa > 0$ such that $\tilde{w}(\kappa; \kappa) = z_{\text{AI}}$, as explained above. However, by Claim 2.5, we know that $\tilde{w}(0; 0) = w(0)$ and that $\tilde{w}(\kappa; \kappa)$ is strictly increasing in κ. Hence, when $z_{\text{AI}} \leq w(0)$, there does not exist a $\kappa > 0$ with the desired properties. Contradiction. □

Step 2. First, note that if $z_{\text{AI}} > w(0)$, then the pre-AI equilibrium cannot be an equilibrium with a non-autonomous AI. At the pre-AI wages, a top-automated firm can hire a human with knowledge $z = 0$ and earn strictly positive profits. Thus, in this case, the equilibrium can only follow a Type 1a or a Type 2a configuration.

To show equilibrium existence and uniqueness, we simply note that $\tilde{w}(0; 0) = w(0)$ and that $\tilde{w}(\kappa; \kappa)$ is strictly increasing in κ. Hence, there exists a unique $\kappa > 0$ such that $\tilde{w}(\kappa; \kappa) = z_{\text{AI}}$. This also yields uniqueness since we know that for a given $\kappa > 0$, the pre-AI equilibrium of the auxiliary κ-economy is unique (by Claim 2.5).

Finally, we obtain the equilibrium cutoffs for each of the two possible configurations. Suppose first that the post-AI equilibrium follows a Type 1a configuration. In this case, $\underline{z}_1^\star = \kappa$ and $\bar{z}_1^\star = \hat{z}(\kappa = \underline{z}_1^\star)$, which immediately implies that $e_1^\star(z; \bar{z}_1^\star) = \tilde{e}_1(z; \hat{z}(\kappa))$. Hence, $\underline{z}_1^\star$ and $\bar{z}_1^\star$ must satisfy the following two conditions:

$$
\begin{aligned}
&\tilde{w}(\kappa; \kappa) = z_{\text{AI}} \iff n(\underline{z}_1^\star)(\bar{z}_1^\star - z_{\text{AI}}) = \bar{z}_1^\star + \left(\tfrac{1}{1+n(\bar{z}_1^\star)}\right)\left[\tfrac{1}{h} - \bar{z}_1^\star - \textstyle\int_{\bar{z}_1^\star}^1 n(e_1^\star(z; \bar{z}_1^\star))dz)\right] \\
&\textstyle\int_\kappa^{\hat{z}(\kappa)} h(1-z)dG(z) = \int_{\hat{z}(\kappa)}^1 dG(z) \iff m_1^\star(\underline{z}_1^\star; \bar{z}_1^\star) = \bar{z}_1^\star
\end{aligned}
$$

These are exactly the conditions in the statement of Lemma 2.8. Moreover, note that, by construction, $\frac{1}{h} > \bar{z}_1^\star + \int_{\bar{z}_1^\star}^1 n(e_1^\star(z; \bar{z}_1^\star))dz$, since the equilibrium of the auxiliary κ-economy satisfies $\frac{1}{h} - \hat{z}(\kappa) > \int_{\hat{z}(\kappa)}^1 n(\tilde{e}_1(z; \hat{z}(\kappa)))dz$. Finally, a direct application of Claim 2.5 implies that the equilibrium wages are indeed the ones described in the statement of the lemma.

Now, suppose that the post-AI equilibrium follows a Type 2a configuration. In this case, $\underline{z}_2^\star = \kappa$, $\hat{z}_2^\star = \underline{z}(\kappa = \underline{z}_2^\star)$, and $\bar{z}_2^\star = \bar{z}(\kappa = \underline{z}_2^\star)$, which immediately implies that $e_2^\star(z; \hat{z}_2^\star) = \tilde{e}_2(u; \underline{z}(\kappa))$. Hence, $\underline{z}_2^\star$, $\hat{z}_2^\star$, and $\bar{z}_2^\star$ must satisfy the following three equilibrium conditions:

$$
\begin{aligned}
&\tilde{w}(\kappa; \kappa) = z_{\text{AI}} \iff \bar{z}_2^\star = n(\underline{z}_2^\star)(\bar{z}_2^\star - z_{\text{AI}}) \\
&\tfrac{1}{h} - \bar{z}(\kappa) = \textstyle\int_{\bar{z}(\kappa)}^1 n\big(\tilde{e}_2(u; \underline{z}(\kappa))\big)du \iff \tfrac{1}{h} = \bar{z}_2^\star + \int_{\bar{z}_2^\star}^1 n(e_2^\star(z; \hat{z}_2^\star))dz \\
&\tilde{m}_2(\kappa; \underline{z}(\kappa)) = \bar{z}(\kappa) \iff m_2^\star(\underline{z}_2^\star; \hat{z}_2^\star) = \bar{z}_2^\star
\end{aligned}
$$

These are exactly the conditions in the statement of Lemma 2.8. Finally, a direct application of Claim 2.5 implies that the equilibrium wages are indeed the ones described in the statement of the lemma.
□

The following lemma (i) shows that there is always an individual that loses from the introduction of a non-autonomous AI relative to the pre-AI equilibrium, and (ii) characterizes the effects of a non-autonomous AI on the wages of the least and most knowledgeable individuals relative to the pre-AI and the autonomous AI case.

Lemma 2.9.

1. *$\exists\, z \in (0,1]$ such $w^\star(z) \le w(z)$ (with strict inequality if $z_{\text{AI}} > w(0)$).*
2. *$\exists\, \epsilon > 0$ such that $w^\star(z) \ge \max\{w(z), w^*(z)\}$ for all $z \in [0,\epsilon)$ (with strict inequality if $z_{\text{AI}} > w(0)$)*
3. *$\exists\, \epsilon > 0$ such that $w^\star(z) \le w^*(z)$ for all $z \in (1-\epsilon, 1]$, with strict inequality if $z \neq 1$.*

Proof. 1. $\exists\, z \in (0,1]$ such $w^\star(z) \le w(z)$ (with strict inequality if $z_{\text{AI}} > w(0)$).— If $z_{\text{AI}} \le w(0)$, then $w^\star(z) = w(z)$, so the statement holds trivially. Hence, it only remains to show that when $z_{\text{AI}} > w(0)$, there exists a $z \in (0,1]$ such $w^\star(z) < w(z)$.

Let κ be the supremum of $W_a^\star$. If $w^\star(\kappa) < w(\kappa)$, we are done as κ loses from AI. So suppose otherwise without loss, i.e., $w^\star(\kappa) \ge w(\kappa)$. The fact that, pre-AI, $m^\star(\kappa)$ could have matched with κ paying her no more than post-AI implies that $w^\star(m^\star(\kappa)) \le w(m^\star(\kappa))$, with strict inequality unless $w^\star(\kappa) = w(\kappa)$. Hence, it only remains to consider the case $w^\star(\kappa) = w(\kappa)$ and $w^\star(m^\star(\kappa)) = w(m^\star(\kappa))$. Without loss of generality, we can restrict attention to a Type 1a configuration (because in a Type 2a configuration there are clearly losers, namely, the newly created independent producers). Clearly, $\overline{z}_1^\star > \hat{z}$, so $m^\star(\kappa) < m(\kappa)$ given that:

$$\int_{m(\kappa)}^1 dG(u) = h\int_\kappa^{\hat{z}} (1-u)dG(u) < h\int_\kappa^{\overline{z}_1^\star}(1-u)dG(u) = \int_{m^\star(\kappa)}^1 dG(u)$$

Recall that in equilibrium, for any worker z we have that $w'(z) = \frac{m(z)-w(z)}{1-z}$. Hence, $m^\star(\kappa) < m(\kappa)$ together with $w^\star(\kappa) = w(\kappa)$ imply that $w^{\star\prime}(\kappa) < w^{*\prime}(\kappa)$. It follows that there exists $z' > \kappa$ with $w^\star(z') < w(z')$.

2. $\exists\, \epsilon > 0$ such that $w^\star(z) \ge \max\{w(z), w^*(z)\}$ for all $z \in [0,\epsilon)$ (with strict inequality if $z_{\text{AI}} > w(0)$).— First, note that if $z_{\text{AI}} \le w(0)$, then $w^\star(z) = w(z)$ for all z. Now consider $z_{\text{AI}} > w(0)$. Because $w^\star(0) = z_{\text{AI}} > w(0)$, the individual with $z = 0$ is always strictly better off relative to the pre-AI equilibrium. Thus, by continuity, there exists $\epsilon' > 0$ such that $w^\star(z) \ge w(z)$ for all $z \in [0,\epsilon')$, with strict inequality if $z_{\text{AI}} > w(0)$.

To show the remaining part, note that in the equilibrium with autonomy $w^*(z_{\text{AI}}) = z_{\text{AI}}$ and $w^*(z)$ is strictly increasing in z. This immediately implies that $w^*(0) < w^*(z_{\text{AI}}) = z_{\text{AI}} \le \max\{z_{\text{AI}}, w(0)\} = w^\star(0)$. Consequently, by continuity, there exists $\epsilon'' > 0$ such that $w^\star(z) > w^*(z)$ for all $z \in [0,\epsilon'')$. Taking $\epsilon = \min\{\epsilon', \epsilon''\}$ we obtain the desired result.

3. $\exists\, \epsilon > 0$ such that $w^\star(z) \le w^*(z)$ for all $z \in (1-\epsilon, 1]$, with strict inequality if $z \neq 1$.— In the equilibrium with autonomous AI, we always have that $w^*(1) = 1/h$. In contrast, with a non-autonomous AI, $w^\star(1) = w(0) < 1/h$ in a pre-AI configuration, $w^\star(1) < 1/h$ in a Type 1a configuration, and $w^\star(1) = 1/h$ in a Type 2a configuration.

Thus, if the equilibrium with a non-autonomous AI follows a pre-AI configuration or is Type 1a, we are done: $w^\star(1) < 1/h = w^*(1)$, so by continuity there exists $\epsilon > 0$ such that $w^\star(z) < w^*(z)$

for all $z \in (1-\epsilon, 1]$. When the equilibrium with a non-autonomous AI is Type 2a, we have that $w^{\star}(1) = 1/h = w^{*}(1)$, so to prove that there exists $\epsilon > 0$ such that $w^{\star}(z) < w^{*}(z)$ for all $z \in (1-\epsilon, 1)$, we will show that $w^{*\prime}(1) < w^{\star\prime}(1)$, implying that $w^{*}(z) - w^{\star}(z)$ converges to zero from above when $z \to 1$.

Indeed, we have that $w^{*\prime}(1) = n(e^{*}(1))$ and that $w^{\star\prime}(1) = n(e^{\star}(1))$. Hence, $w^{*\prime}(1) < w^{\star\prime}(1)$ if and only if $e^{*}(1) < e^{\star}(1)$. Note then that $e^{*}(1) \leq z_{\mathrm{AI}}$,[7] while $e^{\star}(1) = \hat{z}_2^{\star}$ in a Type 2a configuration. Hence, if we show that $\hat{z}_2^{\star} > z_{\mathrm{AI}}$, we are done. To show this, note that $w^{\star}(z) = z_{\mathrm{AI}}$ for all $z \in [0, \underline{z}_2^{\star}]$ and that $w^{\star}(\hat{z}_2^{\star}) = \hat{z}_2^{\star}$. Since $w^{\star}(z)$ is strictly increasing (and continuous) in $z \in W_p^{\star}$, it must be that $w^{\star}(\hat{z}_2^{\star}) = \hat{z}_2^{\star} > w^{\star}(\underline{z}_2^{\star}) = z_{\mathrm{AI}}$. □

We finish this chapter with the following lemma comparing overall output with autonomous and non-autonomous AI:

Lemma 2.10. *Overall output is strictly higher with autonomous than non-autonomous AI.*

Proof. The competitive equilibrium is efficient both in the case of autonomous and in the case of non-autonomous AI (i.e., both maximize output subject to AI's technological constraints and the available resources). The only difference between autonomous and non-autonomous AI is that the latter has an additional constraint that AI cannot do production work. Moreover, given that compute is abundant relative to time, this constraint is strictly binding because, in the autonomous case, some compute is used for independent production. In contrast, non-autonomous AI necessarily leaves resources (namely, compute) idle. It follows that output is strictly smaller when this constraint is there than when it is not. □

[7] This follows because $e^{*}(1) = z_{\mathrm{AI}}$ in a Type 1 and Type 2 equilibrium, and $e^{*}(1) = \bar{z}_3^{*} < z_{\mathrm{AI}}$ in a Type 3 equilibrium.

Chapter 3

The Pre-AI Economy: Further Results

3.1 Pre-AI Equilibrium when Knowledge is Uniformly Distributed

When knowledge is uniformly distributed, the pre-AI equilibrium can almost be obtained in closed form. We present the following results without proof, as they are straightforward applications of the results found in Chapter 2.1.

Lemma 3.1. *In the absence of AI, there is a unique competitive equilibrium. It is given as follows:*

- *When $h > h_0 = 3/4$, then $W = [0, \underline{z}]$, $I = (\underline{z}, \bar{z})$, and $S = [\bar{z}, 1]$, where:*

$$\underline{z} = 1 - \frac{1}{h} + \frac{\sqrt{h^2 - 4h + 3}}{h} \qquad \text{and} \qquad \bar{z} = \frac{2}{h} - 1 - \frac{\sqrt{h^2 - 4h + 3}}{h}$$

  The equilibrium matching function is $m(z; \bar{z}) = \bar{z} + hz - hz^2/2$, while the wage function is given by:

$$w(z) = \begin{cases} z + \frac{h(\underline{z}-z)^2}{2} & \text{if } z \in W \\ z & \text{if } z \in I \\ 1 + \bar{z} - \sqrt{(1+\bar{z})^2 - \frac{2(z+\bar{z})}{h} + \frac{1}{h^2}} & \text{if } z \in S \end{cases}$$

- *When $h \leq h_0 = 3/4$, then $W = [0, \hat{z}]$, $I = \emptyset$, and $S = [\hat{z}, 1]$, where $\hat{z} = (1 + h - \sqrt{1+h^2})/h$. The equilibrium matching function is $m(z; \hat{z}) = \hat{z} + hz - hz^2/2$, while the wage function is given by:*

$$w(z) = \begin{cases} \hat{z} - \left(\frac{h\hat{z}(2+h\hat{z})}{2(1+h(1-\hat{z}))}\right)(1-z) + \frac{hz^2}{2} & \text{if } z \in W \\ \frac{h+2\hat{z}}{1+h(1-\hat{z})} - \sqrt{1 - \frac{2(z-\hat{z})}{h}} & \text{if } z \in S \end{cases}$$

3.2 Developing vs. Advanced Economies

In the main text, we claimed that the knowledge cutoff to become a solver in the pre-AI equilibrium is higher in advanced than in developing economies if the former have better communication technologies and/or a more knowledgeable population than the latter. In this appendix, we formalize this claim.

In particular, the following lemma characterizes how the communication cost h, and the distribution of knowledge $G(z)$ affect the knowledge cutoff to become a solver. For simplicity, we focus on the case $h < h_0$:

Lemma 3.2.

(i) Let $\hat{z}(h)$ be the knowledge of the best worker/worst solver of the pre-AI equilibrium as a function of h. Then $\hat{z}(h)$ is strictly decreasing in $h \in (0, h_0)$.

(ii) Let $\hat{z}^k(h)$ for $h \in (0, h_0^k)$ be the knowledge of the best worker/worst solver of the pre-AI equilibrium under the knowledge distribution $G^k(z)$. If $G^A \succcurlyeq_{\text{FOSD}} G^B$, then $\hat{z}^A(h) \geq \hat{z}^B(h)$ for any given $h \in (0, h_0^A) \cap (0, h_0^B)$.

Proof. The proof that $\hat{z}(h)$ is strictly decreasing in h is in Fuchs et al. (2015, Lemma 2). Hence, here we only provide the proof for (ii).

We want to show that if $G^A \succcurlyeq_{\text{FOSD}} G^B$, then $\hat{z}^A(h) \geq \hat{z}^B(h)$ for any given $h \in (0, h_0^A) \cap (0, h_0^B)$. To simplify notation, we omit the dependence of the equilibrium variables on h. Recall that $\hat{z}^k$ is given by $m^k(\hat{z}^k; \hat{z}^k) = 1$, where $m^k(z; \hat{z}^k)$ is the equilibrium matching function in the pre-AI equilibrium. This implies that $\hat{z}^k$ must satisfy $1 = G(\hat{z}^k) + \int_0^{\hat{z}^k} h(1-u)dG^k(u)$. Integrating by parts the right-hand side of this last expression and rearranging terms yields:

$$1 = \underbrace{G^k(\hat{z}^k)[1 + h(1 - \hat{z}^k)] + h \int_0^{\hat{z}^k} G^k(z)dz}_{\equiv \alpha^k(\hat{z}^k)}$$

Note that $\alpha^k(x)$ is strictly increasing in x, so $\hat{z}^k$ is the unique solution to $\alpha^k(x) = 1$. We claim that $G^A \succcurlyeq_{\text{FOSD}} G^B$ implies that $\alpha^A(x) \leq \alpha^B(x)$ for any given x, which immediately implies that $\hat{z}^A \geq \hat{z}^B$. Indeed, given that $G^B(z) \geq G^A(z)$ (as $G^A \succcurlyeq_{\text{FOSD}} G^B$):

$$\alpha^B(x) - \alpha^A(x) = [1 + h(1-x)][G^B(x) - G^A(x)] + h \int_0^x [G^B(u) - G^A(u)]du \geq 0$$

□

Chapter 4

Autonomous AI: Further Results

4.1 More on the Baseline Model

4.1.1 Post-AI Equilibrium when Knowledge is Uniformly Distributed

When knowledge is uniformly distributed, the equilibrium with autonomous AI can be obtained in almost closed form. As in the main text, we focus on $h < h_0$. We present the following statements without proof, as they are straightforward applications of the results found in Chapter 2.2.

Recall from Chapter 2.2 that $\mathcal{R}_1 \equiv \{z \in W : \Gamma_w(z) < 0\}$, $\mathcal{R}_2 \equiv \{z \in W : \Gamma_w(z) \geq 0\} \cup \{z \in S : \Gamma_s(z) \geq 0\}$, and $\mathcal{R}_3 \equiv \{z \in S : \Gamma_s(z) < 0\}$, where:

$$\begin{aligned}
\Gamma_w(x) &\equiv n(x)(m_w(x;x) - x) - x - \textstyle\int_x^{m_w(x;x)} n(e_w(u;x))du \\
\Gamma_s(x) &\equiv \tfrac{1}{h} - x - \textstyle\int_x^1 n(e_s(u;x))du
\end{aligned}$$

with $\int_{z_{\text{AI}}}^{m_w(z;z_{\text{AI}})} dG(u) = \int_0^z h(1-u)dG(u)$ for $z \in [0, z_{\text{AI}}]$, and $\int_z^1 dG(u) = \int_{e_s(z;z_{\text{AI}})}^{z_{\text{AI}}} h(1-u)dG(u)$ for $z \in [z_{\text{AI}}, 1]$. When $G(z) = z$, we have:

$$\begin{aligned}
m_w(z; z_{\text{AI}}) &= z_{\text{AI}} + hz - \tfrac{hz^2}{2} \text{ for } z \in [0, z_{\text{AI}}] \\
e_s(z; z_{\text{AI}}) &= 1 - \sqrt{(1-z_{\text{AI}})^2 + \tfrac{2(1-z)}{h}} \text{ for } z \in [z_{\text{AI}}, 1] \\
\Gamma_w(x) &= \tfrac{x(2-x)}{2(1-x)} - 2x \\
\Gamma_s(x) &= \tfrac{1}{h} + 1 - 2x - \sqrt{\tfrac{(1-x)(2+h-hx)}{h}}
\end{aligned}$$

We are now ready to present the post-AI equilibrium when knowledge is uniformly distributed:

Lemma 4.1. *When $G(z) = z$ and AI is autonomous, there is a unique competitive equilibrium. It is given as follows:*

- *If $z_{\text{AI}} \in \mathcal{R}_1$, then:*

$$W_a^* = \emptyset,\ W_p^* = [0, z_{\text{AI}}],\ I^* = (z_{\text{AI}}, \underline{z}_1^*),\ S_p^* = [\underline{z}_1^*, m_1^*(z_{\text{AI}}; \underline{z}_1^*)],\ S_a^* = [m_1^*(z_{\text{AI}}; \underline{z}_1^*), 1]$$

where $\underline{z}_1^* = \frac{z_{\mathrm{AI}}(2-hz_{\mathrm{AI}})}{2[1-h(1-z_{\mathrm{AI}})]}$ *and* $m_1^*(z;\underline{z}_1^*) = \underline{z}_1^* + hz - \frac{hz^2}{2}$ *for* $z \in [0, z_{\mathrm{AI}}]$. *The equilibrium wage function is:*

$$w^*(z) = \begin{cases} \underline{z}_1^* - h\underline{z}_1^*(1-z) + \frac{hz^2}{2} & \text{if } z \in W_p^* \\ z & \text{if } z \in I^* \\ 1 + \underline{z}_1^* - \sqrt{1 - \frac{2(z-\underline{z}_1^*)}{h}} & \text{if } z \in S_p^* \\ n(z_{\mathrm{AI}})(z - z_{\mathrm{AI}}) & \text{if } z \in S_a^* \end{cases}$$

- *If* $z_{\mathrm{AI}} \in \mathcal{R}_2$, *then:*

$$W_a^* = [0, \underline{z}_2^*],\ W_p^* = [\underline{z}_2^*, z_{\mathrm{AI}}],\ I^* = \emptyset,\ S_p^* = [z_{\mathrm{AI}}, m_2^*(z_{\mathrm{AI}}; \underline{z}_2^*)],\ S_a^* = [m_2^*(z_{\mathrm{AI}}; \underline{z}_2^*), 1]$$

where $\underline{z}_2^* = z_{\mathrm{AI}} - \sqrt{2z_{\mathrm{AI}}(1-z_{\mathrm{AI}})}$ *and* $m_2^*(z;\underline{z}_2^*) = z_{\mathrm{AI}} + h(z - z_{\mathrm{AI}}) - \frac{h(z^2 - z_{\mathrm{AI}}^2)}{2}$ *for* $[\underline{z}_2^*, z_{\mathrm{AI}}]$. *The equilibrium wage function is:*

$$w^*(z) = \begin{cases} z_{\mathrm{AI}}\left(1 - \frac{z_{\mathrm{AI}}}{n(z)}\right) & \text{if } z \in W_a^* \\ z_{\mathrm{AI}} - h\underline{z}_2^*\left(1 - \frac{\underline{z}_2^*}{2}\right) - h(z_{\mathrm{AI}} - \underline{z}_2^*)(1-z) + \frac{hz^2}{2} & \text{if } z \in W_p^* \\ 1 + z_{\mathrm{AI}} - \underline{z}_2^* - \sqrt{(1-\underline{z}_2^*)^2 - \frac{2(z - z_{\mathrm{AI}})}{h}} & \text{if } z \in S_p^* \\ n(z_{\mathrm{AI}})(z - z_{\mathrm{AI}}) & \text{if } z \in S_a^* \end{cases}$$

- *If* $z_{\mathrm{AI}} \in \mathcal{R}_3$, *then:*

$$W_a^* = [0, \underline{z}_3^*],\ W_p^* = [\underline{z}_3^*, \bar{z}_3^*],\ I^* = (\bar{z}_3^*, z_{\mathrm{AI}}],\ S_p^* = [z_{\mathrm{AI}}, 1],\ S_a^* = \emptyset$$

where $\underline{z}_3^* = 1 - \frac{1-z_{\mathrm{AI}}}{1-hz_{\mathrm{AI}}} - \frac{1-hz_{\mathrm{AI}}}{2h}$, $\bar{z}_3^* = 1 + \frac{1-z_{\mathrm{AI}}}{1-hz_{\mathrm{AI}}} + \frac{1-hz_{\mathrm{AI}}}{2h}$, *and* $m_3^*(z;\underline{z}_3^*) = z_{\mathrm{AI}} + h(z - \underline{z}_3^*) - \frac{h(z^2 - \underline{z}_3^{*2})}{2}$ *for* $[\underline{z}_3^*, \bar{z}_3^*]$. *The equilibrium wage function is:*

$$w^*(z) = \begin{cases} z_{\mathrm{AI}}\left(1 - \frac{z_{\mathrm{AI}}}{n(z)}\right) & \text{if } z \in W_a^* \\ 1 - h\bar{z}_3^*\left(1 - \frac{\bar{z}_3^*}{2}\right) - (1-z)(1 - h\bar{z}_3^*) + \frac{hz^2}{2} & \text{if } z \in W_p^* \\ z & \text{if } z \in I^* \\ 1 - \bar{z}_3^* + \frac{1}{h} - \sqrt{(1-\bar{z}_3^*)^2 + \frac{2(1-z)}{h}} & \text{if } z \in S_p^* \end{cases}$$

4.1.2 The Effects of AI on the Distribution of Firm Size, Productivity, and Span of Control

This chapter analyzes the effects of autonomous AI on the distribution of firm size, productivity, and span of control. We focus on two-layer organizations and begin with the following preliminary result:

Corollary 4.1. *Autonomous AI necessarily increases the number (i.e., the measure) of two-layer firms.*

Proof. Since all two-layer organizations hire a single solver, to prove this corollary, it is sufficient to show that AI increases the overall number of solvers in the economy. When $z_{\mathrm{AI}} \in \mathrm{int}W$, this follows because $S^* \supset S$ (see Proposition 3 of the main text). When $z_{\mathrm{AI}} \in \mathrm{int}S$, more humans become workers after AI's introduction (i.e., $W^* \supset W$, see Proposition 3 of the main text). Hence, the overall number of solvers—human plus AI—must increase as each worker requires the same amount of help post-AI as that required pre-AI. □

We take *firm size* as its output, and *firm productivity* as its output divided by the units of time or compute used in production (this is a measure of "average" productivity). Hence, a firm's productivity is equal to the knowledge of its solver. We define a firm's *span of control* as the amount of resources under its solver's supervision. Hence, the span of control of a firm that hires human workers with knowledge z is $n(z)$, while that of a firm that uses AI for production is $n(z_{\text{AI}})$. Note that span of control is a measure of decentralization because a solver can supervise more resources only if its workers solve more problems on their own.

Proposition 4.1.

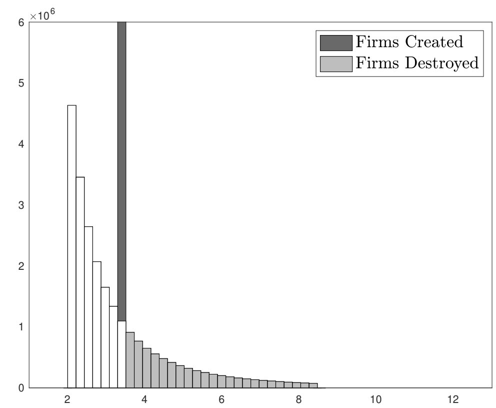
- *If $z_{\text{AI}} \in \text{int}W$, then autonomous AI decreases the maximum span of control, the minimum firm productivity, and both the minimum and maximum firm size.*
- *If $z_{\text{AI}} \in \text{int}S$, then autonomous AI increases the maximum span of control, the minimum firm productivity, and both the minimum and maximum firm size.*

Proof. Consider first span of control. The most decentralized firm of the pre-AI economy has a span of control $n(\sup W)$, while the most decentralized firm of the post-AI world has a span of control of $n(\sup W^*)$. That AI decreases the maximum span of control when $z_{\text{AI}} \in \text{int}W$, and increases it when $z_{\text{AI}} \in \text{int}S$, follows from occupational displacement, i.e. Proposition 3 of the main text, and the fact that $n(z) = [h \times (1-z)]^{-1}$ is strictly increasing in z.

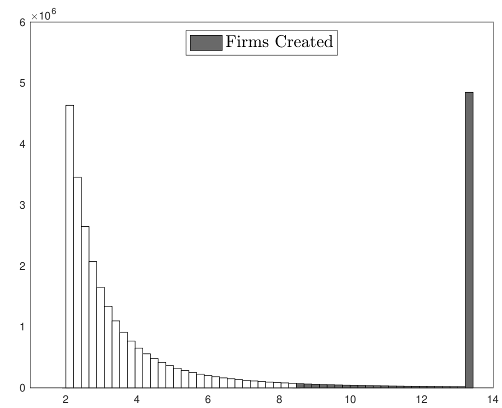
Consider next productivity. The least productive firm of the pre-AI economy has productivity $\inf S$, while the least productive firm of the post-AI world has productivity $\inf S^*$. That AI decreases the minimum firm productivity when $z_{\text{AI}} \in \text{int}W$, and increases it when $z_{\text{AI}} \in \text{int}S$, follows directly from occupational displacement (i.e. Proposition 3 of the main text).

Finally, consider size. By positive assortative matching, the size of the smallest firm is $n(0)$ times the minimum firm productivity, while the size of the largest firm is 1 times the maximum span of control. Consequently, when $z_{\text{AI}} \in \text{int}W$, AI reduces the minimum and maximum firm size, as it decreases the maximum span of control and the minimum firm productivity. Likewise, when $z_{\text{AI}} \in \text{int}S$, AI increases the minimum and maximum firm size, as it increases the maximum span of control and the minimum firm productivity. □

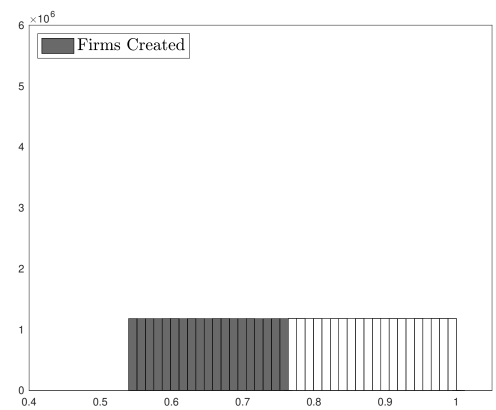
Proposition 4.1 is illustrated in Figure 4.1, which depicts the pre- and post-AI distributions of firm decentralization, productivity, and size when $z_{\text{AI}} = 0.425$ and $z_{\text{AI}} = 0.85$. The dark-shaded and light-shaded bars correspond to the firms created and destroyed by AI's introduction, respectively. Hence, the pre-AI distribution is equal to the white bars plus the light-shaded bars, while the post-AI distribution is equal to the white bars plus the dark-shaded bars.



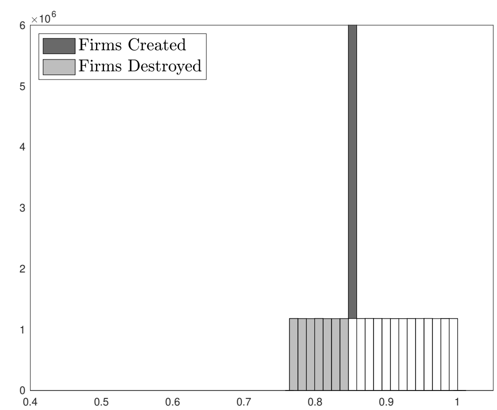
(a) Span of Control - $z_{\mathrm{AI}} \in \mathrm{int}W$



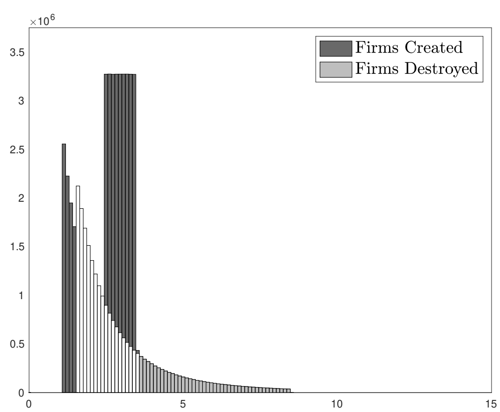
(b) Span of Control - $z_{\mathrm{AI}} \in \mathrm{int}S$



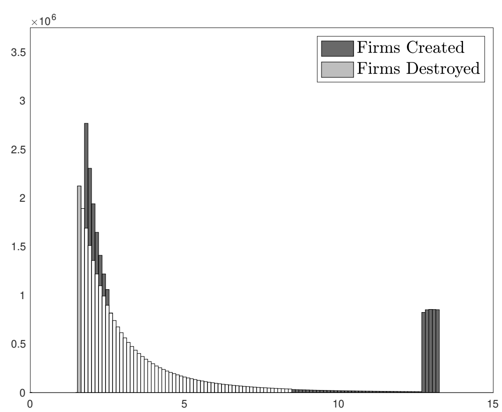
(c) Productivity - $z_{\mathrm{AI}} \in \mathrm{int}W$



(d) Productivity - $z_{\mathrm{AI}} \in \mathrm{int}S$



(e) Size - $z_{\mathrm{AI}} \in \mathrm{int}W$



(f) Size - $z_{\mathrm{AI}} \in \mathrm{int}S$

Figure 4.1: The Effects of AI on the Distribution of Firms' Span of Control, Productivity, and Size

Notes. This histogram is based on a human population of $N = 100 \times 10^6$ individuals. Distribution of knowledge: $G(z) = z$. For both panels, $h = 1/2$. Moreover, for panel (a), $z_{\mathrm{AI}} = 0.425$, while for panel (b), $z_{\mathrm{AI}} = 0.85$. Pre-AI, there are 23.6×10^6 firms. Post-AI, the number of firms increases to 46.7×10^6 in panel (a) and to 29.3×10^6 in panel (b). See Ide and Talamàs (2024) for the exact expressions of the pre- and post-AI distributions of firm size, productivity, and span of control.

As the figure shows, when $z_{\rm AI} \in {\rm int}W$, AI destroys the largest and most decentralized firms and leads to the creation of firms that are smaller and less productive than the smallest and least productive pre-AI firms. Similarly, when $z_{\rm AI} \in {\rm int}S$, AI destroys the smallest and least productive firms and leads to the creation of firms that are larger and more decentralized than the largest and most decentralized pre-AI firms.

We end this chapter by showing that when AI has the knowledge of a pre-AI solver, and this knowledge is sufficiently high, autonomous AI also leads to the creation of superstar firms that have "scale without mass." This term was coined by Brynjolfsson et al. (2008) and Autor et al. (2020) to refer to those firms that attain large market shares with relatively few employees.

Formally, we say that AI creates superstar firms with scale but no mass if the following two conditions are satisfied: (i) the largest post-AI firms are larger than the largest pre-AI firms, and (ii) the largest post-AI firms are bottom-automated (that is, they use a single human solver to supervise production work by AI).

Corollary 4.2. *There exists $\zeta \in {\rm int}S$ such that for all $z_{\rm AI} \geq \zeta$, autonomous AI leads to the creation of superstar firms with scale but no mass.*

Proof. First, note that when $z_{\rm AI} \in {\rm int}S$, the largest post-AI firms are larger than the largest pre-AI firms. Second, we know by Lemma 2.5 of Chapter 2.2 that $z_{\rm AI} \in \mathcal{R}_2$ if $z_{\rm AI} \in [1-\epsilon, 1) \subset S$ with $\epsilon \downarrow 0$. Thus, by Lemma 2.4 of Chapter 2.2, firms use AI in all three possible roles when $z_{\rm AI}$ is sufficiently close to 1. Since no worker is better than AI and the equilibrium exhibits positive assortative matching (Lemma 2.3 of Chapter 2.2), we then have that AI is the best worker in the economy and is therefore supervised by the most knowledgeable humans. Hence, when $z_{\rm AI}$ is sufficiently close to 1, the largest firms post-AI are bottom-automated firms.

Combining these two results, we have that when $z_{\rm AI}$ is sufficiently close to 1, AI leads to the creation of superstar firms with scale but no mass. □

Intuitively, AI increases the maximum firm size only when $z_{\rm AI} \in {\rm int}S$. In such a case, we know that AI is necessarily used as a solver and possibly also as a worker (Proposition 2 of the main text). The condition that $z_{\rm AI} \geq \zeta$ is necessary and sufficient for AI to be used in both roles, in which case the largest post-AI firms are bA firms (as AI is the best worker in the economy). These superstar firms with scale but no mass can, in fact, be seen in panel (f) of Figure 4.1.

4.1.3 The Knife-Edge Case $z_{\rm AI} \in W \cap S$

In this chapter, we describe the effects of AI when $z_{\rm AI} \in W \cap S = \{\hat{z}\}$.

Occupational Displacement

Proposition 4.2. *When $z_{\rm AI} \in W \cap S = \{\hat{z}\}$, autonomous AI does not lead to any human displacement between routine knowledge work and specialized problem solving, i.e., $W^* = W$ and $S^* = S$. However, it* *leads to the creation of bA and tA firms, i.e.,* $W_a^* \neq \emptyset$ *and* $S_a^* \neq \emptyset$.

Proof. By Lemma 2.5 of Chapter 2.2, we know that $z_{\mathrm{AI}} = \hat{z} \in \mathcal{R}_2$, so Lemma 2.4 of the same chapter implies that the equilibrium is necessarily Type 2. Hence, in this case, we have that $\sup W^* = \inf S^* = z_{\mathrm{AI}} = \hat{z}$, so there is no occupational displacement.

Showing that $W_a^* \neq \emptyset$ and $S_a^* \neq \emptyset$ requires more work. First, it is not difficult to prove that $\underline{z}_2^* > 0$ if and only if $\bar{z}_2^* < 1$. Hence, to prove that $W_a^* \neq \emptyset$ and $S_a^* \neq \emptyset$ it suffices to show that $\underline{z}_2^* > 0$. To do the latter, suppose for contradiction that $\underline{z}_2^* = 0$ ($\underline{z}_2^* < 0$ immediately contradicts that we are in a Type 2 equilibrium). Then the equilibrium matching function is given by $\int_{\hat{z}}^{m_2^*(z;0)} dG(u) = \int_0^z h(1-u)dG(u)$ for $z \in [\hat{z}, 1]$, so $m_2^*(z; 0) = m(z; \hat{z})$ for all $z \in [\hat{z}, 1]$, which implies that $n(z_{\mathrm{AI}})(m_2^*(z_{\mathrm{AI}}; 0) - z_{\mathrm{AI}}) = 1/h$. However, since $\frac{1}{h} > \hat{z} + \int_{\hat{z}}^1 n(e(z; \hat{z}))dz$ (by Lemma 2.1 of Chapter 2.1), we have that:

$$n(z_{\mathrm{AI}})(m_2^*(z_{\mathrm{AI}}; 0) - z_{\mathrm{AI}}) = \tfrac{1}{h} > \hat{z} + \int_{\hat{z}}^1 n\big(e(z; \hat{z})\big)dz = z_{\mathrm{AI}} + \int_{z_{\mathrm{AI}}}^{m_2^*(z_{\mathrm{AI}};0)} n\big(e_2^*(z; 0)\big)dz$$

so the equilibrium condition for $\underline{z}_2^* = 0$ is not satisfied (see the statement of Lemma 2.4 of Chapter 2.2). Contradiction. □

Distribution of Firm Size, Productivity, and Span of Control

Corollary 4.3. *When* $z_{\mathrm{AI}} \in W \cap S = \{\hat{z}\}$, *autonomous AI also increases the measure of two-layer firms.*

Proof. Since all two-layer organizations hire a single solver, to prove this corollary, it suffices to show that AI increases the overall number of solvers in the economy. Proving the latter is easy. By Proposition 4.2, we know that there is the same amount of human workers pre- and post-AI. Moreover, because $S_a^* \neq \emptyset$, there is a strictly positive amount of compute doing assisted production work. Both observations together imply that there are more overall workers (human or compute) post-AI than pre-AI. Hence, the overall number of solvers—human plus AI—must also be greater post-AI than pre-AI, as each worker requires the same amount of help post-AI as pre-AI. □

Proposition 4.3. *When* $z_{\mathrm{AI}} \in W \cap S = \{\hat{z}\}$, *autonomous AI does not affect the maximum span of control, the minimum firm productivity, or the minimum or maximum firm size.*

Proof. Recall that the most decentralized firm of the pre-AI economy has a span of control $n(\sup W)$, while the most decentralized firm of the post-AI world has a span of control of $n(\sup W^*)$. Hence, AI does not affect the maximum span of control because, by Proposition 4.2, we know that $W = W^*$.

Similarly, recall that the least productive firm of the pre-AI economy has productivity $\inf S$, while the least productive firm of the post-AI world has productivity $\inf S^*$. Hence, AI does not affect the minimum firm productivity because, by Proposition 4.2, we know that $S = S^*$.

Finally, consider size. As noted in the main text, the size of the smallest firm is $n(0)$ times the minimum firm productivity, while the size of the largest firm is 1 times the maximum span of control. Hence, AI does not affect the minimum or maximum firm size since, in this case, it does not affect the minimum firm productivity or the maximum span of control. □

Note, finally, that because AI does not increase the maximum firm size when $z_{\text{AI}} \in W \cap S = \{\hat{z}\}$, then in this case, AI cannot lead to the emergence of "superstar firms with scale but no mass," as defined in Chapter 4.1.2

The Effects of AI on Workers and Solvers who are Not Occupationally Displaced

Proposition 4.4. *If $z_{\text{AI}} \in W \cap S = \{\hat{z}\}$, then:*

- *The productivity of $z \in W^* = W$ decreases with autonomous AI (strictly so for all $z \neq 0$).*
- *The span of control of $z \in S^* = S$ increases with autonomous AI (strictly so for all $z \neq 1$).*

Proof. We first show that each $z \in W^* = W$ is assisted by a less knowledgeable solver post-AI than pre-AI (strictly so for all $z \neq 0$). By Lemma 2.5 of Chapter 2.2, we know that $z_{\text{AI}} = \hat{z} \in \mathcal{R}_2$, so Lemma 2.4 of the same chapter implies that the equilibrium is necessarily Type 2. Moreover, as shown in the proof of Proposition 4.2, in this case, we have that $\underline{z}_2^* > 0$ and $\bar{z}_2^* < 1$. Consequently, if $z \in W_a^*$, then $m(z; \hat{z}) \geq \hat{z} = z_{\text{AI}}$, where the first inequality is strict when $z > 0$. If $z \in W_p^*$ instead, then the matching functions pre- and post-AI are given by:

$$\int_{z_{\text{AI}}}^{m_2^*(z;\underline{z}_2^*)} dG(u) = \int_{\underline{z}_2^*}^{z} h(1-u)dG(u) \text{ for } z \in W_p^* = [\underline{z}_2^*, z_{\text{AI}}]$$
$$\int_{\hat{z}}^{m(z;\hat{z})} dG(u) = \int_0^z h(1-u)dG(u) \text{ for } z \in W = [0, \hat{z}]$$

when evaluating $z_{\text{AI}} = \hat{z}$. Consequently, for $z \in W^* = W$, $\int_{m_2^*(z;\underline{z}_2^*)}^{m(z;\hat{z})} dG(u) = \int_0^{\underline{z}_2^*} h(1-u)dG(u) > 0$, which implies that $m_2^*(z; \underline{z}_2^*) < m(z; \hat{z})$ given that $\underline{z}_2^* > 0$.

We now show that each $z \in S^* = S$ improves her match post-AI compared to pre-AI (strictly so for all $z \neq 1$). Indeed, if $z \in S_a^*$, then $e(z; \hat{z}) \leq \hat{z} = z_{\text{AI}}$, where the first inequality is strict when $z < 1$. If $z \in S_p^*$ instead, then the employee functions pre- and post-AI are given by:

$$\int_z^{\bar{z}_2^*} dG(u) = \int_{e_2^*(z;\underline{z}_2^*)}^{z_{\text{AI}}} h(1-u)dG(u) \text{ for } z \in S_p^* = [z_{\text{AI}}, \bar{z}_2^*]$$
$$\int_z^1 dG(u) = \int_{e(z;\hat{z})}^{\hat{z}} h(1-u)dG(u) \text{ for } z \in S = [\hat{z}, 1]$$

when evaluating $z_{\text{AI}} = \hat{z}$. Consequently, for $z \in S^* = S$, $\int_{\bar{z}_2^*}^1 dG(u) = \int_{e(z;\hat{z})}^{e_2^*(z;\underline{z}_2^*)} h(1-u)dG(u) > 0$, which implies that $e_2^*(z; \underline{z}_2^*) > e(z; \hat{z})$ since $\bar{z}_2^* < 1$. □

4.1.4 Superintelligent AI: $z_{\text{AI}} = 1$

We now characterize and discuss the effects of a superintelligent AI, i.e., $z_{\text{AI}} = 1$.

Proposition 4.5. *In the presence of a superintelligent autonomous AI, there is a unique equilibrium. The equilibrium allocations are:*

$$W_a^* = [0, 1],\ W_p^* = I^* = S_p^* = S_a^* = \emptyset$$
$$\mu_w^* = 0,\ \mu_s^* = \int_0^1 n(z)^{-1} dG(z),\ \mu_i^* = \mu - \mu_s^*$$

The equilibrium prices are $r^* = 1$ and $w^*(z) = 1 - h(1-z)$ for any $z \in [0, 1]$.

Proof. Given that compute is abundant relative to human time, some compute must be allocated to independent production. The zero-profit condition of single-layer automated firms then implies that $r^* = 1$. Moreover, because the equilibrium still exhibits occupational stratification, the unique candidate for the equilibrium has the following allocations:

$$W_a^* = [0,1],\ W_p^* = I^* = S_p^* = S_a^* = \emptyset$$
$$\mu_w^* = 0,\ \mu_s^* = \int_0^1 n(z)^{-1} dG(z),\ \mu_i^* = \mu - \mu_s^*$$

That is, all humans are employed as workers in top-automated firms. The zero-profit conditions of these firms then pin down the candidate wage schedule, i.e., $w^*(z) = 1 - h(1-z)$ for any $z \in [0,1]$. From here, verifying that this is indeed an equilibrium, i.e., that no firms have incentives to deviate and all market clearing conditions are satisfied, is straightforward. □

The effects of a superintelligent AI are depicted in Figure 4.2. As the figure shows—and Proposition 4.5 formalizes—in this case, all humans are hired as workers in top-automated firms. As a result, there are some similarities as well as some differences compared to the baseline setting studied in the main text.

Let us begin with the similarities. It is easy to see that in the case of a superintelligent AI, the results regarding (i) occupational displacement and (ii) the distribution of firm size, productivity, and span of control are the same as the ones in the baseline model when AI has the knowledge of a pre-AI

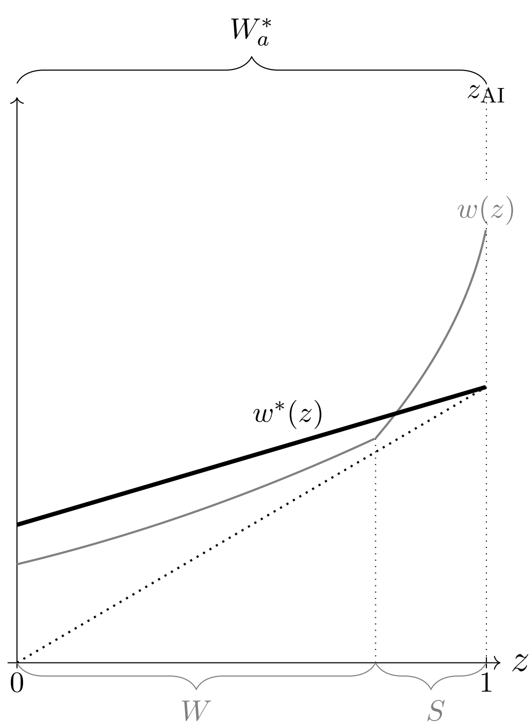


Figure 4.2: The Effects of a Superintelligent AI

Notes. Distribution of knowledge: $G(z) = z$. Parameter values: $h = 1/2$ and $z_{\text{AI}} = 1$. The thick gray line depicts the pre-AI equilibrium wage function w. The thick black line depicts the post-AI equilibrium wage function w^*.

solver. Indeed, humans are still displaced from specialized problem solving into routine production work, which leads to the destruction of the least productive firms and the creation of larger, more decentralized firms. The only difference is that when $z_{\text{AI}} = 1$, AI no longer leads to the creation of superstar firms with scale but no mass, as there are no bA firms.

Regarding the productivity of the workers who remain workers (note that there are no solvers who remain solvers), the results in the case of $z_{\text{AI}} = 1$ are also similar to those of the baseline. Indeed, Proposition 3 of Section 5.1 of the main text continues to hold. The only difference is that now all workers who are not occupationally displaced see a gain in productivity because $z \leq e(z_{\text{AI}} = 1) = \hat{z}$ for all $z \in W \subset W^*$

Thus, the main differences between the case of a superintelligent AI and the case studied in the baseline emerge regarding the distribution of labor income. In particular, as stated in Proposition 5 of Section 5.2 the main text, when $z_{\text{AI}} \in [0, 1)$, the most knowledgeable humans necessarily win from AI's introduction, even if $z_{\text{AI}} \to 1$. In contrast, when $z_{\text{AI}} = 1$, then the most knowledgeable humans necessarily lose from the introduction of the technology, as illustrated in Figure 4.2.

Intuitively, this difference arises because when $z_{\text{AI}} < 1$, the humans with $z \in (z_{\text{AI}}, 1]$ can still use AI to increase the extent to which they leverage their knowledge. In contrast, when $z_{\text{AI}} = 1$, that is no longer possible as the AI can solve the same amount of problems as the most knowledgeable humans.

4.2 Negligible Amount of Compute: $\mu \to 0$

We now characterize the equilibrium with an autonomous AI in the case where $\mu \to 0$ and compare it to the case where compute is abundant relative to time. This comparison allows us to disentangle in a clean way the role that abundant compute is playing in our results. As in the main text, we focus on the case $h < h_0$.

4.2.1 Overview

We begin with an overview of the equilibrium when μ is arbitrarily small and a comparison to the case where μ is abundant relative to time.

Proposition 4.6. *Regarding autonomous AI:*

- *If $z_{\text{AI}} \in \text{int}W$, then there exists $\bar{\mu}_w > 0$ such that if $\mu \in (0, \bar{\mu}_w)$, then the unique equilibrium involves AI being used exclusively as a worker. In particular, there exist cutoffs $\hat{z}^* < \underline{z}_s^* < \bar{z}_s^* < 1$ such that:*

$$W_a^* = \emptyset,\ W_p^* = [0, \hat{z}^*] \ni z_{\text{AI}},\ I^* = \emptyset,\ S_p^* = [\hat{z}^*, \underline{z}_s^*] \cup [\bar{z}_s^*, 1],\ S_a^* = [\underline{z}_s^*, \bar{z}_s^*]$$

where $\hat{z}^* < \hat{z}$ and satisfies $\int_0^{\hat{z}^*} h(1-z)dG(z) + h(1-z_{\text{AI}})\mu = \int_{\hat{z}^*}^1 dG(z)$.

- *If $z_{\text{AI}} \in \text{int}S$, then there exists $\bar{\mu}_s > 0$ such that if $\mu \in (0, \bar{\mu}_s)$, then the unique equilibrium involves AI being used exclusively as a solver. In particular, there exist cutoffs $0 < \underline{z}_w^* < \bar{z}_w^* < \hat{z}^*$ such that:*

$$W_a^* = [\underline{z}_w^*, \bar{z}_w^*],\ W_p^* = [0, \underline{z}_w^*] \cup [\bar{z}_w^*, \hat{z}^*],\ I^* = \emptyset,\ S_p^* = [\hat{z}^*, 1] \ni z_{\text{AI}},\ S_a^* = \emptyset$$

where $\hat{z}^* > \hat{z}$ and satisfies $\int_0^{\hat{z}^*} h(1-z)dG(z) = \mu + \int_{\hat{z}^*}^1 dG(z)$.

- *If $z_{\text{AI}} = \hat{z}$, then there exists $\bar{\mu}_k > 0$ such that if $\mu \in (0, \bar{\mu}_k)$, then the unique equilibrium involves AI simultaneously being used as a worker and as a solver (but not used as an independent producer). In particular, there exist cutoffs $0 < \bar{z}_w^* < \hat{z}^* < \underline{z}_s^* < 1$ such that:*

$$W_a^* = [0, \bar{z}_w^*],\ W_p^* = [\bar{z}_w^*, \hat{z}^*],\ I^* = \emptyset,\ S_p^* = [\hat{z}^*, \underline{z}_s^*],\ S_a^* = [\underline{z}_s^*, 1]$$

where $\hat{z}^*$ is equal to the pre-AI cutoff and, therefore, equal to z_{AI}, i.e., $\hat{z}^* = \hat{z} = z_{\text{AI}}$.

Proof. See Chapter 4.2.2 below. □

Proposition 4.6 highlights two key differences between small and large compute. First, when μ is small, AI is used exclusively as a worker when it has the knowledge of a pre-AI worker, while it is used exclusively as a solver when it has the knowledge of a pre-AI solver. Second, when AI is used exclusively as a worker, it is not the best worker in the economy. Similarly, when AI is used exclusively as a solver, it is not the worst solver of the economy.

Intuitively, when compute is large relative to human time, using all of the economy's compute requires allocating some of it to independent production, irrespective of AI's knowledge (as mentioned in Section 4 of the main text). Occupational stratification then leads to AI being the best worker and/or the worst solver in that case. These forces are absent when μ is small, as compute can be completely absorbed in production in two-layer firms (if $z_{\text{AI}} \in \text{int}W$) or completely absorbed in the supervision of humans (if $z_{\text{AI}} \in \text{int}S$).

As in Section 5 of the main text, we now compare the pre- and post-AI equilibrium. We start by noting that Proposition 3 of the main text (concerning occupational displacement) remains the same. This follows because—as Proposition 4.6 shows—when $z_{\text{AI}} \in \text{int}W$ then $\hat{z} < \hat{z}^*$, while $z_{\text{AI}} \in \text{int}S$, then $\hat{z} > \hat{z}^*$. This implies that humans are again displaced from routine production work into specialized problem solving when AI has the knowledge of a pre-AI worker, while they are displaced in the opposite direction when AI has the knowledge of a pre-AI solver.

The comparison is more subtle in the case of changes in the quality of matches (Proposition 4 of the main text). We now provide its analog for the case of small compute (recall that $e(z; \hat{z})$ is the equilibrium employee function of the pre-AI equilibrium):

Proposition 4.7. *When AI is autonomous and $\mu > 0$, but arbitrarily small:*

- *If $z_{\text{AI}} \in \text{int}W$, then:*
  - *The productivity of $z \in W \subset W^*$ strictly decreases with autonomous AI if $z < z_{\text{AI}}$, and strictly increases with autonomous AI if $z > z_{\text{AI}}$.*

  - *The span of control of $z \in S \subset S^*$ strictly increases with autonomous AI if $e(z;\hat{z}) < z_{\text{AI}}$, and strictly decreases with autonomous AI if $e(z;\hat{z}) > z_{\text{AI}}$.*

- *If $z_{\text{AI}} \in \text{int}S$, then:*
  - *The productivity of $z \in W \subset W^*$ strictly increases with autonomous AI if $z < e(z_{\text{AI}};\hat{z})$, and strictly increases with autonomous AI if $z > e(z_{\text{AI}};\hat{z})$.*
  - *The span of control of $z \in S \subset S^*$ strictly decreases with autonomous AI if $z < z_{\text{AI}}$, and strictly increases with autonomous AI if $z > z_{\text{AI}}$.*

- *If $z_{\text{AI}} = \hat{z}$, then:*
  - *The productivity of $z \in W \subset W^*$ decreases with autonomous AI (strictly so for all $z \neq 0$).*
  - *The span of control of $z \in S \subset S^*$ increases with autonomous AI (strictly so for all $z \neq 1$).*

Proof. See Chapter 4.2.3 below. □

Proposition 4.7 is similar to Proposition 4 of the main text, except in the following two aspects. First, while all workers are worse matched when compute is abundant if $z_{\text{AI}} \in \text{int}W$, there is a set of workers—those with knowledge above z_{AI}—who are better matched post-AI than pre-AI when compute is small. Second, while the span of control of all solvers increases when compute is abundant and $z_{\text{AI}} \in \text{int}S$, there is a set of solvers—those with knowledge below z_{AI}—whose span of control decreases when compute is small.

Intuitively, these differences arise because, when compute is abundant, AI is the best worker and/or the worst solver. Thus, in this case, there are no human workers with knowledge above z_{AI} nor human solvers with knowledge below z_{AI}. In that sense, Proposition 4.7 is true both with small and large compute, but Proposition 4 of the main text exploits the fact that compute is abundant to deliver stronger predictions.

Finally, we turn to the consequences of AI for labor income. In contrast to the case in which compute is abundant, when μ is small, the winners from AI are not necessarily the ones at the extremes of the knowledge distribution. For example, Figure 4.3 depicts the function $\Delta(z) \equiv w^*(z) - w(z)$ in a case with relatively low compute and $z_{\text{AI}} = 0$. In the example provided in the figure, the humans in the middle of knowledge distribution benefit from AI, while those at the extreme of the distribution are worse off from its introduction.

Intuitively, in this case, the reduction of wages at knowledge 0 increases the wages of the least knowledgeable solvers post-AI, which, in turn, increases the wages of the most knowledgeable workers post-AI. This makes the solvers of the latter—the most knowledgeable humans—strictly worse off since they can now appropriate a smaller share of the output produced. This intuition highlights the key role that abundant compute plays in our analysis of the distributional consequences of AI: By guaranteeing that the best workers and the worst solvers post-AI do not gain from AI, it prevents situations like the one depicted in Figure 4.3, where both extremes of the knowledge distribution lose from AI.

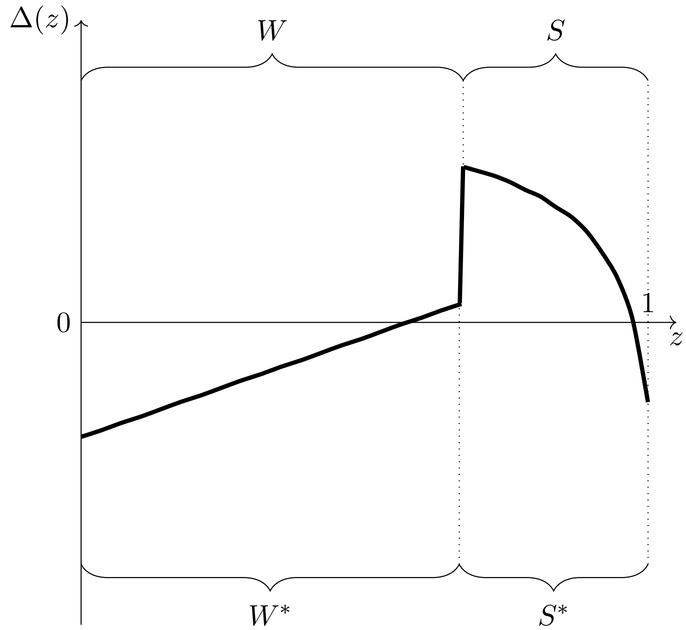


Figure 4.3: An Illustration of $\Delta(z)$ for low compute μ.

Notes. Distribution of knowledge: $G(z) = z$. Parameter values: $z_{\text{AI}} = 0.01$, $h = 0.73$, $\mu = 0.01$. Moreover, $\Delta(0) = -1.89 \times 10^{-3}$, $\Delta(1) = -1.26 \times 10^{-3}$, $\max_z \Delta(z) = 2.58 \times 10^{-3}$, $\min_z \Delta(z) = -1.89 \times 10^{-3}$.

4.2.2 Proof of Proposition 4.6

We begin with some preliminary observations. First, recall that we are assuming that $h < h_0$. Second, as in the pre-AI equilibrium and the AI equilibrium with abundant compute, it is not difficult to prove that in this setting, the equilibrium continues to exhibit positive assortative matching and occupational stratification. Third, we have the following result, which applies regardless of the value of z_{AI}:

Claim 4.1. *When μ is sufficiently small, there is no independent production in equilibrium with autonomous AI.*

Proof. Suppose that agent $z \in [0, 1]$ is hired as an independent producer in equilibrium. This implies that $w^*(z; \mu) = z$, where we are making explicit that the post-AI equilibrium depends on μ. Note then that $w^*(z; \mu)$ must be continuous in μ, and that $\lim_{\mu \to 0} w^*(z; \mu) > z$, since $h < h_0$. Therefore, by continuity, it must be that $w^*(z; \mu) > z$ for a positive but sufficient small μ, contradicting the premise that z is an independent producer. The exact same argument with $z = z_{\text{AI}}$ can be used to show that no positive mass of compute can be allocated to independent production in equilibrium. $\square$

• *AI has the Knowledge of a Pre-AI Worker.*— We begin by characterizing the equilibrium when $z_{\text{AI}} \in \text{int}W$.

Claim 4.2. *When μ is sufficiently small and $z_{\text{AI}} \in \text{int}W$, autonomous AI is not used as a solver in equilibrium.*

Proof. The proof is by contradiction. If AI is used as a solver in equilibrium, occupational stratification implies that $W^* = [0, z_{\text{AI}}]$ and $S^* = [z_{\text{AI}}, 1]$ (that $W^* = [0, z_{\text{AI}}]$ and not $W^* \subset [0, z_{\text{AI}}]$ is due to the fact that there is no independent production in equilibrium). This implies that a necessary condition for market clearing is $\int_0^{z_{\text{AI}}} h(1-u)dG(u) + h(1-z_{\text{AI}})\mu_w = \int_{z_{\text{AI}}}^1 dG(u) + \mu_s$. Given that $\mu_w \to 0$ and $\mu_s \to 0$ as $\mu \to 0$, then it must be that:

$$\lim_{\mu\to 0}\left(\int_0^{z_{\text{AI}}} h(1-u)dG(u) - \int_{z_{\text{AI}}}^1 dG(u)\right) = 0$$

However, given that $z_{\text{AI}} < \hat{z}$ then for any $\mu > 0$ we have that $\int_0^{z_{\text{AI}}} h(1-u)dG(u) < \int_{z_{\text{AI}}}^1 dG(u)$, so $\lim_{\mu\to 0}\left(\int_0^{z_{\text{AI}}} h(1-u)dG(u) - \int_{z_{\text{AI}}}^1 dG(u)\right) < 0$, contradiction. □

Lemma 4.2. *When μ is sufficiently small and $z_{\text{AI}} \in \text{int}W$, the unique equilibrium with autonomous AI is as follows. The equilibrium allocation is:*

$$W_a^* = \emptyset,\; W_p^* = [0, \hat{z}^*] \ni z_{\text{AI}},\; I^* = \emptyset,\; S_p^* = [\hat{z}^*, \underline{z}_s^*] \cup [\bar{z}_s^*, 1],\; S_a^* = [\underline{z}_s^*, \bar{z}_s^*]$$
$$\mu = \mu_w^* = \int_{\underline{z}_s^*}^{\bar{z}_s^*} n(z_{\text{AI}})dG(u)$$

where $\hat{z}^*$ satisfies $\int_0^{\hat{z}^*} h(1-z)dG(z) + h(1-z_{\text{AI}})\mu = \int_{\hat{z}^*}^1 dG(z)$, while $\underline{z}_s^* = m_-^*(z_{\text{AI}}; \hat{z}^*)$ and $\bar{z}_s^* = m_+^*(z_{\text{AI}}; \hat{z}^*)$, where $m_-^* : [0, z_{\text{AI}}] \to [\hat{z}^*, \underline{z}_s^*]$ and $m_+^* : [z_{\text{AI}}, \hat{z}^*] \to [\bar{z}_s^*, 1]$ are given by:

$$\text{(4.1)}\qquad \begin{aligned} &\int_{\hat{z}^*}^{m_-^*(z;\hat{z}^*)} dG(u) = \int_0^z h(1-u)dG(u), \text{ for } z \in [0, z_{\text{AI}}] \\ &\int_{m_+^*(z;\hat{z}^*)}^1 dG(u) = \int_z^{\hat{z}^*} h(1-u)dG(u), \text{ for } z \in [z_{\text{AI}}, \hat{z}^*] \end{aligned}$$

Moreover, the equilibrium wage schedule is:

$$\text{(4.2)}\qquad w^*(z) = \begin{cases} m_-^*(z;\hat{z}^*) - w^*(m_-^*(z;\hat{z}^*))/n(z) & \text{if } z \in [0, z_{\text{AI}}] \\ m_+^*(z;\hat{z}^*) - w^*(m_+^*(z;\hat{z}^*))/n(z) & \text{if } z \in [z_{\text{AI}}, \hat{z}^*] \\ C^* + \int_{\hat{z}^*}^z n(e_-^*(u;\hat{z}^*))du & \text{if } z \in [\hat{z}^*, \underline{z}_s^*] \\ n(z_{\text{AI}})(z - r^*) & \text{if } z \in [\underline{z}_s^*, \bar{z}_s^*] \\ n(z_{\text{AI}})(\bar{z}_s^* - r^*) + \int_{\bar{z}_s^*}^z n(e_+^*(u;\hat{z}^*))du & \text{if } z \in [\bar{z}_s^*, 1] \end{cases}$$

where (C^*, r^*) are given by the following system of linear equations:

$$n(\hat{z}^*)(1 - C^*) = n(z_{\text{AI}})(\bar{z}_s^* - r^*) + \int_{\bar{z}_s^*}^1 n(e_+^*(u;\hat{z}^*))du$$
$$C^* + \int_{\hat{z}^*}^{\underline{z}_s^*} n(e_-^*(u;\hat{z}^*))du = n(z_{\text{AI}})(\underline{z}_s^* - r^*)$$

Proof. Claims 4.1 and 4.2 imply that compute is used exclusively for production in two-layer organizations when μ is sufficiently small. Occupational stratification and positive assortative matching then imply that the equilibrium allocations must necessarily take the following form: There exist cutoffs $z_{\text{AI}} \le \hat{z}^* \le \underline{z}_s^* < \bar{z}_s^* < 1$ such that:

$$W_a^* = \emptyset,\; W_p^* = [0, \hat{z}^*] \ni z_{\text{AI}},\; I^* = \emptyset,\; S_p^* = [\hat{z}^*, \underline{z}_s^*] \cup [\bar{z}_s^*, 1],\; S_a^* = [\underline{z}_s^*, \bar{z}_s^*]$$
$$\mu = \mu_w^* = \int_{\underline{z}_s^*}^{\bar{z}_s^*} n(z_{\text{AI}})dG(u)$$

Following similar reasoning as in Chapter 2.2 (see "Informal Construction of the Equilibrium"), the matching functions $m^*_-(z;\hat{z}^*)$ and $m^*_+(z;\hat{z}^*)$ must satisfy (4.1), which implies that $\underline{z}^*_s = m^*_-(z_{\text{AI}};\hat{z}^*)$ and $\bar{z}^*_s = m^*_+(z_{\text{AI}};\hat{z}^*)$. Combining these expressions with the fact that $\mu = \int_{\underline{z}^*_s}^{\bar{z}^*_s} n(z_{\text{AI}})dG(u)$, we obtain the equilibrium condition for $\hat{z}^*$: $\int_0^{\hat{z}^*} h(1-z)dG(z) + h(1-z_{\text{AI}})\mu = \int_{\hat{z}^*}^1 dG(z)$. Notice from this last condition that $\hat{z}^* < \hat{z}$ and that $\hat{z}^* \to \hat{z}$ as $\mu \to 0$. This implies that, for any $z_{\text{AI}} < \hat{z}$, we can always find a sufficiently small μ such that $z_{\text{AI}} < \hat{z}^*$.

Having determined the equilibrium allocation, we now turn to wages. Following a similar reasoning as in Chapter 2.2 (see "Informal Construction of the Equilibrium"), the equilibrium wages must be given as in (4.2). Moreover, the equilibrium wage function has to be continuous, so $\lim_{z\uparrow\hat{z}^*} w^*(z) = \lim_{z\downarrow\hat{z}^*} w^*(z)$ and $\lim_{z\uparrow\underline{z}^*_s} w^*(z) = \lim_{z\downarrow\underline{z}^*_s} w^*(z)$. These two conditions give the system of linear equations that determines (C^*, r^*). It is straightforward to prove that this system has a unique solution.

The final step is verifying that the proposed allocations and prices are indeed an equilibrium. To do this, we need to argue that no firm has incentives to deviate. This is relatively straightforward: first, by construction, no firm has incentives to deviate locally. Moreover, also by construction, the wage function is continuous in z. Now, following a similar logic as the proof of Corollary 2.1 of Chapter 2.1, it is easy to prove that $w^*(z)$ is strictly increasing and weakly convex. This implies that if a firm does not have incentives to deviate locally, then it does not have incentives to deviate globally either. Thus, given the wage function and the equilibrium allocation, no firms have incentives to deviate. □

• *AI has the Knowledge of a Pre-AI solver*.—We now consider the case when $z_{\text{AI}} \in \text{int}S$.

Claim 4.3. *When μ is sufficiently small and $z_{\text{AI}} \in \text{int}S$, autonomous AI is exclusively used as a solver in equilibrium.*

Proof. The proof is by contradiction. If AI is used as a worker in equilibrium, occupational stratification implies that $W^* = [0, z_{\text{AI}}]$ and $S^* = [z_{\text{AI}}, 1]$ (because there is no independent production in equilibrium). This implies that a necessary condition for market clearing is $\int_0^{z_{\text{AI}}} h(1-u)dG(u) + h(1-z_{\text{AI}})\mu_w = \int_{z_{\text{AI}}}^1 dG(u) + \mu_s$. Given that $\mu_w \to 0$ and $\mu_s \to 0$ as $\mu \to 0$, it must be that:

$$\lim_{\mu\to 0}\left(\int_0^{z_{\text{AI}}} h(1-u)dG(u) - \int_{z_{\text{AI}}}^1 dG(u)\right) = 0$$

However, given that $z_{\text{AI}} > \hat{z}$, then for any $\mu > 0$ we have that $\int_0^{z_{\text{AI}}} h(1-u)dG(u) > \int_{z_{\text{AI}}}^1 dG(u)$, so $\lim_{\mu\to 0}\left(\int_0^{z_{\text{AI}}} h(1-u)dG(u) - \int_{z_{\text{AI}}}^1 dG(u)\right) > 0$, contradiction. □

Lemma 4.3. *When μ is sufficiently small and $z_{\text{AI}} \in \text{int}S$, the unique equilibrium with autonomous AI is as follows. The equilibrium allocation involves:*

$$W^*_a = [\underline{z}^*_w, \bar{z}^*_w],\ W^*_p = [0, \underline{z}^*_w] \cup [\bar{z}^*_w, \hat{z}^*],\ I^* = \emptyset,\ S^*_p = [\hat{z}^*, 1] \ni z_{\text{AI}},\ S^*_a = \emptyset$$
$$\mu = \mu^*_s = \int_{\underline{z}^*_w}^{\bar{z}^*_w} h(1-z)dG(z)$$

where $\hat{z}^*$ satisfies $\int_0^{\hat{z}^*} h(1-z)dG(z) = \mu + \int_{\hat{z}^*}^1 dG(z)$, while $\underline{z}_w^* = e_-^*(z_{\rm AI}; \hat{z}^*)$ and $\bar{z}_w^* = e_+^*(z_{\rm AI}; \hat{z}^*)$, where $e_-^* : [\hat{z}^*, z_{\rm AI}] \to [0, \underline{z}_w^*]$ and $e_+^* : [z_{\rm AI}, 1] \to [\bar{z}_w^*, \hat{z}^*]$ are given by:

$$(4.3) \qquad \begin{aligned} &\int_{\hat{z}^*}^z dG(u) = \int_0^{e_-^*(z;\hat{z})} h(1-u)dG(u), \text{ for } z \in [\hat{z}^*, z_{\rm AI}] \\ &\int_z^1 dG(u) = \int_{e_+^*(z;\hat{z}^*)}^{\hat{z}^*} h(1-u)dG(u), \text{ for } z \in [z_{\rm AI}, 1] \end{aligned}$$

Moreover, the equilibrium wage schedule is:

$$(4.4) \qquad w^*(z) = \begin{cases} m_-^*(z;\hat{z}^*) - w^*(m_-^*(z;\hat{z}^*))/n(z) & \text{if } z \in [0, \underline{z}_w^*] \\ z_{\rm AI} - r^*/n(z) & \text{if } z \in [\underline{z}_w^*, \bar{z}_w^*] \\ m_+^*(z;\hat{z}^*) - w^*(m_+^*(z;\hat{z}^*))/n(z) & \text{if } z \in [\bar{z}_w^*, \hat{z}^*] \\ C^* + \int_{\hat{z}^*}^z n(e_-^*(u;\hat{z}^*))du & \text{if } z \in [\hat{z}^*, z_{\rm AI}] \\ r^* + \int_{z_{\rm AI}}^z n(e_+^*(u;\hat{z}^*))du & \text{if } z \in [z_{\rm AI}, 1] \end{cases}$$

where (C^*, r^*) satisfy:

$$(1-C^*)n(\hat{z}^*) = r^* + \int_{z_{\rm AI}}^1 n(e_+^*(z;\hat{z}^*))dz \quad \text{and} \quad r^* = C^* + \int_{\hat{z}^*}^{z_{\rm AI}} n(e_-^*(z;\hat{z}^*))dz$$

Proof. According to Claim 4.3, compute is only used to supervise humans in equilibrium. Occupational stratification and positive assortative matching then imply that the equilibrium allocations must necessarily take the following form: There exist cutoffs $0 < \underline{z}_w^* < \bar{z}_w^* \le \hat{z}^* \le z_{\rm AI} \le 1$ such that:

$$W_a^* = [\underline{z}_w^*, \bar{z}_w^*],\ W_p^* = [0, \underline{z}_w^*] \cup [\bar{z}_w^*, \hat{z}^*],\ I^* = \emptyset,\ S_p^* = [\hat{z}^*, 1] \ni z_{\rm AI},\ S_a^* = \emptyset$$
$$\mu = \mu_s^* = \int_{\underline{z}_w^*}^{\bar{z}_w^*} h(1-z)dG(z)$$

Following similar reasoning as in Chapter 2.2 (see "Informal Construction of the Equilibrium"), the employee functions $e_-^*(z;\hat{z}^*)$ and $e_+^*(z;\hat{z}^*)$ then satisfy (4.3), which implies that $\underline{z}_w^* = e_-^*(z_{\rm AI};\hat{z}^*)$ and $\bar{z}_w^* = e_+^*(z_{\rm AI};\hat{z}^*)$. Combining these expressions with the fact that $\mu = \int_{\underline{z}_w^*}^{\bar{z}_w^*} h(1-z)dG(z)$, we obtain the equilibrium condition for $\hat{z}^*$: $\int_0^{\hat{z}^*} h(1-z)dG(z) = \int_{\hat{z}^*}^1 dG(z) + \mu$. From this last condition, we have that $\hat{z}^* > \hat{z}$ and that $\hat{z}^* \to \hat{z}$ as $\mu \to 0$. This implies that for any $z_{\rm AI} > \hat{z}$, we can always find a sufficiently small μ such that $z_{\rm AI} > \hat{z}^*$.

Having determined the equilibrium allocation, we now turn to wages. Following a similar reasoning as in Chapter 2.2 (see "Informal Construction of the Equilibrium"), the equilibrium wages must then be given as in (4.4). Moreover, the equilibrium wage function has to be continuous, so $\lim_{z\uparrow\hat{z}^*} w^*(z) = \lim_{z\downarrow\hat{z}^*} w^*(z)$ and $\lim_{z\uparrow\underline{z}_s^*} w^*(z) = \lim_{z\downarrow\underline{z}_s^*} w^*(z)$. These two conditions give the system of linear equations that determines (C^*, r^*). It is straightforward to prove that this system has a unique solution.

The final step is verifying that the proposed allocations and prices are indeed an equilibrium. To do this, we need to argue that no firm has incentives to deviate. This is relatively straightforward: first, by construction, no firm has incentives to deviate locally. Moreover, also by construction, the wage function is continuous in z. Now, following a similar logic as the proof of Corollary 2.1 of Chapter 2.1, it is easy to prove that $w^*(z)$ is strictly increasing and weakly convex. This implies that if a firm does not have incentives to deviate locally, then it does not have incentives to deviate globally either. Thus, given the wage function and the equilibrium allocation, no firms have incentives to deviate. □

• *AI has the Knife-Edge Knowledge of a Pre-AI Worker and Pre-AI solver.*— Finally, we consider the case where $z_{\text{AI}} = \hat{z}$.

Claim 4.4. *When μ is sufficiently small and $z_{\text{AI}} = \hat{z}$, autonomous AI must be used as a worker and a solver in equilibrium.*

Proof. As shown in the proof of Lemma 4.2, if AI is used exclusively as a worker, then $z_{\text{AI}} < \hat{z}^*$, where $\int_0^{\hat{z}^*} h(1-z)dG(z) + h(1-z_{\text{AI}})\mu = \int_{\hat{z}^*}^1 dG(z)$. Hence, $z_{\text{AI}} < \hat{z}$ (as $\hat{z}^* \to \hat{z}$ when $\mu \to 0$), which contradicts the premise that $z_{\text{AI}} = \hat{z}$. Similarly, as shown in the proof of Lemma 4.3, if AI is used exclusively as a solver, then $z_{\text{AI}} > \hat{z}^*$, where $\int_0^{\hat{z}^*} h(1-z)dG(z) = \int_{\hat{z}^*}^1 dG(z) + \mu$. Hence, $z_{\text{AI}} > \hat{z}$ (as $\hat{z}^* \to \hat{z}$ when $\mu \to 0$), which contradicts the premise that $z_{\text{AI}} = \hat{z}$. □

Lemma 4.4. *When μ is sufficiently small and $z_{\text{AI}} = \hat{z}$, the unique equilibrium with autonomous AI is as follows. The equilibrium allocation is:*

$$W_a^* = [0, \bar{z}_w^*],\ W_p^* = [\bar{z}_w^*, \hat{z}],\ I^* = \emptyset,\ S_p^* = [\hat{z}, \underline{z}_s^*],\ S_a^* = [\underline{z}_s^*, 1]$$
$$\mu_s^* = \int_0^{\bar{z}_w^*} h(1-z)dG(z),\ h(1-z_{\text{AI}})\mu_w^* = \int_{\underline{z}_s^*}^1 dG(u),\ \mu_s^* = h(1-z_{\text{AI}})\mu_w^*,\ \mu = \mu_w^* + \mu_s^*$$

where $\hat{z} = z_{\text{AI}}$ is the pre-AI equilibrium cutoff. Moreover, the equilibrium wage schedule is:

$$w^*(z) = \begin{cases} z_{\text{AI}} - r^*/n(z) & \text{if } z \in [0, \bar{z}_w^*] \\ m^*(z;\hat{z}) - w^*(m^*(z;\hat{z}))/n(z) & \text{if } z \in [\bar{z}_w^*, \hat{z}] \\ r^* + \int_{\hat{z}}^z n(e^*(u;\hat{z}))du & \text{if } z \in [\hat{z}, \underline{z}_s^*] \\ n(\hat{z})(z - r^*) & \text{if } z \in [\underline{z}_s^*, 1] \end{cases} \tag{4.5}$$

where r^* satisfies $r^* + \int_{\hat{z}}^{\underline{z}_s^*} n(e^*(u;\hat{z}))du = n(\hat{z})(\underline{z}_s^* - r^*)$, and $m^* : [\bar{z}_w^*, \hat{z}] \to [\hat{z}, \underline{z}_s^*]$ is given by:

$$\int_{\hat{z}}^{m^*(z;\hat{z})} dG(u) = \int_{\bar{z}_w^*}^z h(1-u)dG(u), \text{ for } z \in [\bar{z}_w^*, \hat{z}] \text{ with } m^*(\hat{z};\hat{z}) = \underline{z}_s^* \tag{4.6}$$

Proof. According to Claim 4.4, AI is used as a worker and as a solver in equilibrium. Occupational stratification and positive assortative matching then imply that the equilibrium allocations must necessarily take the following form: There exist cutoffs $0 < \bar{z}_w^* \le z_{\text{AI}} = \hat{z} \le \underline{z}_s^* < 1$ such that:

$$W_a^* = [0, \bar{z}_w^*],\ W_p^* = [\bar{z}_w^*, \hat{z}],\ I^* = \emptyset,\ S_p^* = [\hat{z}, \underline{z}_s^*],\ S_a^* = [\underline{z}_s^*, 1]$$
$$\mu_s^* = \int_0^{\bar{z}_w^*} h(1-z)dG(z),\ h(1-z_{\text{AI}})\mu_w^* = \int_{\underline{z}_s^*}^1 dG(u),\ \mu = \mu_w^* + \mu_s^*$$

Following similar reasoning as in Chapter 2.2 (see "Informal Construction of the Equilibrium"), the employee function $e^*(u;\hat{z})$ must satisfy (4.6), which implies that $m^*(\hat{z};\hat{z}) = \underline{z}_s^*$. Consequently, we have that $\int_{\hat{z}}^{\underline{z}_s^*} dG(z) = \int_{\bar{z}_w^*}^{\hat{z}} h(1-z)dG(z)$. Combining this last expression with the fact that $\mu_s^* = \int_0^{\bar{z}_w^*} h(1-z)dG(z)$ and $h(1-z_{\text{AI}})\mu_w^* = \int_{\underline{z}_s^*}^1 dG(u)$ we obtain that:

$$\int_{\hat{z}}^1 dG(u) + \mu_s^* = \int_0^{\hat{z}} h(1-z)dG(z) + h(1-z_{\text{AI}})\mu_w^*$$

Given that $\int_{\hat{z}}^1 dG(u) = \int_0^{\hat{z}} h(1-z)dG(z)$, this implies that $\mu_s^* = h(1-z_{\text{AI}})\mu_w^*$.

Having determined the equilibrium allocation, we now turn to wages. Following a similar reasoning as in Chapter 2.2 (see "Informal Construction of the Equilibrium"), the equilibrium wages must then be given as in (4.5). Moreover, a necessary condition for this to be an equilibrium is for the wage function to be continuous. The latter requires that $\lim_{z\uparrow \underline{z}_s^*} w^*(z) = \lim_{z\downarrow \underline{z}_s^*} w^*(z)$. This gives a single equation for r^*: $r^* + \int_{\hat{z}}^{\underline{z}_s^*} n(e^*(u;\hat{z}))du = n(\hat{z})(\underline{z}_s^* - r^*)$.

The final step is verifying that the proposed allocations and prices are indeed an equilibrium. To do this, we need to argue that no firm has incentives to deviate. This is relatively straightforward: first, by construction, no firm has incentives to deviate locally. Moreover, also by construction, the wage function is continuous in z. Now, following a similar logic as the proof of Corollary 2.1 of Chapter 2.1, it is easy to prove that $w^*(z)$ is strictly increasing and weakly convex. This implies that if a firm does not have incentives to deviate locally, it does not have incentives to deviate globally either. Thus, given the wage function and the equilibrium allocation, no firms have incentives to deviate. □

4.2.3 Proof of Proposition 4.7

As noted in the main text, a worker's productivity increases if and only if her solver match improves. Similarly, a given solver's span of control increases if and only if her workers' knowledge increases.

• $z_{\text{AI}} \in \text{int}W$.— Consider first $z \in W^* \subset W$. We want to show that if $z < z_{\text{AI}}$, then individuals with knowledge z are strictly less productive post-AI than pre-AI, while if $z > z_{\text{AI}}$, then individuals with knowledge z are strictly more productive post-AI than pre-AI. Recall that the pre-AI matching function for any $z \in W^* \subset W$ satisfies $\int_{\hat{z}}^{m(z;\hat{z})} dG(u) = \int_0^z h(1-u)dG(u)$ for $z \in [0,\hat{z}]$, or, equivalently, $\int_{m(z;\hat{z})}^1 dG(u) = \int_z^{\hat{z}} h(1-u)dG(u)$ for $z \in [0,\hat{z}]$. Post-AI matching, in turn, can be written as:

$$\begin{aligned} \int_{\hat{z}^*}^{m_-^*(z;\hat{z}^*)} dG(u) &= \int_0^z h(1-u)dG(u), \text{ for } z \in [0, z_{\text{AI}}] \\ \int_{m_+^*(z;\hat{z}^*)}^1 dG(u) &= \int_z^{\hat{z}^*} h(1-u)dG(u), \text{ for } z \in [z_{\text{AI}}, \hat{z}^*] \end{aligned}$$

Hence, if $z < z_{\text{AI}}$, the pre- and post-AI matching conditions can be combined to obtain $\int_{\hat{z}}^{m(z;\hat{z})} dG(u) = \int_{\hat{z}^*}^{m_-^*(z;\hat{z}^*)} dG(u)$, which implies that $m_-^*(z;\hat{z}^*) < m(z;\hat{z})$ as $\hat{z} > \hat{z}^*$. In contrast, if $z > z_{\text{AI}}$, the pre- and post-AI matching conditions can be combined to obtain $\int_{m(z;\hat{z})}^{m_+^*(z;\hat{z}^*)} dG(u) = \int_{\hat{z}^*}^{\hat{z}} h(1-u)dG(u)$, which implies that $m_+^*(z;\hat{z}^*) > m(z;\hat{z})$ as $\hat{z} > \hat{z}^*$.

Now consider $z \in S \subset S^*$. We want to show that if $e(z;\hat{z}) < z_{\text{AI}}$, then z's span of control increases with AI, while if $e(z;\hat{z}) > z_{\text{AI}}$, then z's span of control decreases with AI.

We first claim that if $e(z;\hat{z}) = z_{\text{AI}}$, then $z \in S_a^* \cap S$. The proof is via the contrapositive. Suppose that $z \notin S_a^* \cap S$, so $z \in S_p^* \cap S$, where $S_p^* = [\hat{z}^*, \underline{z}_s^*] \cup [\bar{z}_s^*, 1]$. Note then that if $z \in [\hat{z}^*, \underline{z}_s^*] \cap S$, then $e(z;\hat{z}) < e_-^*(z;\hat{z}^*) \leq z_{\text{AI}}$, where the first inequality follows because $e(z';\hat{z}) < e_-^*(z';\hat{z}^*)$ for all $z' \in [\hat{z}^*, \underline{z}_s^*] \cap S$ if $m(z'';\hat{z}) > m_-^*(z'';\hat{z}^*)$ for all $z'' \in [0, z_{\text{AI}}]$ (which we already showed is true). Hence, $e(z;\hat{z}) \neq z_{\text{AI}}$ in this case. Suppose instead that $z \in [\bar{z}_s^*, 1] \cap S$, then $e(z;\hat{z}) > e_+^*(z;\hat{z}^*) \geq z_{\text{AI}}$, where the first inequality follows because $e(z';\hat{z}) > e_+^*(z';\hat{z}^*)$ for all $z' \in [\bar{z}_s^*, 1] \cap S$ if $m(z'';\hat{z}) < m_+^*(z'';\hat{z}^*)$ for all $z'' \in [z_{\text{AI}}, \hat{z}^*]$ (which we already showed is true). Hence, $e(z;\hat{z}) \neq z_{\text{AI}}$ in this case also.

The above implies that if $e(z;\hat{z}) < z_{\text{AI}}$, then either $z \in [\hat{z}^*, \underline{z}_s^*] \cap S$ or $z \in S_a^* \cap S$. In the former case, we already know that $e(z;\hat{z}) < e_-^*(z;\hat{z}^*)$, i.e., individuals with knowledge z assist more knowledgeable workers post-AI than pre-AI. In the latter case, the knowledge of the workers under the supervision of solvers with knowledge z also increases because post-AI, these solvers assist AI, which has knowledge z_{AI}, while pre-AI, they assist humans with knowledge $e(z;\hat{z})$. Similarly, if $e(z;\hat{z}) > z_{\text{AI}}$, then either $z \in [\bar{z}_s^*, 1] \cap S$ or $z \in S_a^* \cap S$. In the former case, we already know that $e(z;\hat{z}) > e_+^*(z;\hat{z}^*)$, i.e., individuals with knowledge z assist less knowledgeable workers post-AI than pre-AI. In the latter case, the knowledge of the workers under the supervision of solvers with knowledge z also decreases because post-AI, these solvers assist AI, which has knowledge z_{AI}, while pre-AI, they assist humans with knowledge $e(z;\hat{z})$. □

• $z_{\text{AI}} \in \text{int}S$.— Consider first $z \in S^* \subset S$. We want to show that if $z < z_{\text{AI}}$, then individuals with knowledge z assist a strictly smaller team of workers post-AI than pre-AI, while if $z > z_{\text{AI}}$, then individuals with knowledge z assist a strictly larger team of workers post-AI than pre-AI. Recall that the pre-AI employee function for any $z \in S^* \subset S$ satisfies $\int_{\hat{z}}^{z} dG(u) = \int_0^{e(z;\hat{z})} h(1-u)dG(u)$ for $z \in [\hat{z}, 1]$, or, equivalently, $\int_z^1 dG(u) = \int_{e(z;\hat{z})}^{\hat{z}} h(1-u)dG(u)$ for $z \in [0, \hat{z}]$. Post-AI matching, in turn, can be written as:

$$\int_{\hat{z}^*}^{z} dG(u) = \int_0^{e_-^*(z;\hat{z})} h(1-u)dG(u), \text{ for } z \in [\hat{z}^*, z_{\text{AI}}]$$
$$\int_z^1 dG(u) = \int_{e_+^*(z;\hat{z}^*)}^{\hat{z}^*} h(1-u)dG(u), \text{ for } z \in [z_{\text{AI}}, 1]$$

Hence, if $z < z_{\text{AI}}$, the pre- and post-AI employee functions can be combined to obtain $\int_{\hat{z}}^{\hat{z}^*} dG(u) = \int_{e_-^*(z;\hat{z}^*)}^{e(z;\hat{z})} dG(u)$, which implies that $e_-^*(z;\hat{z}^*) < e(z;\hat{z})$ as $\hat{z} < \hat{z}^*$. In contrast, if $z > z_{\text{AI}}$, the pre- and post-AI matching conditions can be combined to obtain $\int_{\hat{z}^*}^{e_+^*(z;\hat{z}^*)} h(1-u)dG(u) = \int_{\hat{z}}^{e(z;\hat{z})} h(1-u)dG(u)$, which implies that $e_+^*(z;\hat{z}^*) > e(z;\hat{z})$ as $\hat{z} < \hat{z}^*$.

Consider now $z \in W \subset W^*$. We want to show that if $z < e(z_{\text{AI}};\hat{z})$, then z is strictly more productive post-AI than pre-AI, while if $z > e(z_{\text{AI}};\hat{z})$, z is strictly less productive post-AI than pre-AI.

We first claim that if $z = e(z_{\text{AI}};\hat{z})$, then $z \in W_a^* \cap W$. The proof is via the contrapositive. Suppose that $z \notin W_a^* \cap W$, so $z \in W_p^* \cap W$, where $W_p^* = [0, \underline{z}_w^*] \cup [\bar{z}_w^*, \hat{z}^*]$. Note then that if $z \in [0, \underline{z}_w^*] \cap W$, then $m(z;\hat{z}) < m_-^*(z;\hat{z}^*) \leq z_{\text{AI}}$, where the first inequality follows because $m(z';\hat{z}) < m_-^*(z';\hat{z}^*)$ for all $z' \in [0, \underline{z}_w^*] \cap W$ if $e_-^*(z'';\hat{z}^*) < e(z'';\hat{z})$ for all $z'' \in [\hat{z}^*, z_{\text{AI}}]$ (which we already showed is true). This implies that $m(z;\hat{z}) < z_{\text{AI}}$, or, equivalently, $z < e(z_{\text{AI}};\hat{z})$. Suppose instead that $z \in [\bar{z}_w^*, \hat{z}^*] \cap W$. Then $m(z;\hat{z}) > m_+^*(z;\hat{z}^*) \geq z_{\text{AI}}$, where the first inequality follows because $m(z';\hat{z}) > m_+^*(z';\hat{z}^*)$ for all

$z' \in [\bar{z}_w^*, \hat{z}^*] \cap W$ if $e_+^*(z''; \hat{z}^*) > e(z''; \hat{z})$ in $[z_{\text{AI}}, 1]$ (which we already showed is true). Consequently, $m(z; \hat{z}) > z_{\text{AI}}$, or, equivalently, $z > e(z_{\text{AI}}; \hat{z})$. Thus, in either case, $z \neq e(z_{\text{AI}}; \hat{z})$.

The previous claim implies that if $z < e(z_{\text{AI}}; \hat{z})$, then either $z \in [0, \underline{z}_w^*] \cap W$ or $z \in W_a^* \cap W$. In the former case, we already showed that $m(z; \hat{z}) < m_-^*(z; \hat{z}^*)$, so individuals with knowledge z are strictly more productive post-AI than pre-AI. In the latter case, the individuals with knowledge z are also strictly more productive post-AI than pre-AI because post-AI, they are assisted by AI—which has knowledge z_{AI}—while pre-AI, they are assisted by humans with knowledge $m(z; \hat{z}) < z_{\text{AI}}$. Similarly, if $z > e(z_{\text{AI}}; \hat{z})$, then either $z \in [\bar{z}_w^*, \hat{z}^*] \cap W$ or $z \in W_a^* \cap W$. In the former case, we already established that $m(z; \hat{z}) > m_+^*(z; \hat{z}^*)$, so the individuals with knowledge z are strictly less productive post-AI than pre-AI. In the latter case, the individuals with knowledge z are also less productive post-AI than pre-AI because post-AI, they are being assisted by AI, while pre-AI, they are assisted by humans with knowledge $m(z; \hat{z}) > z_{\text{AI}}$. □

• $z_{\text{AI}} = \hat{z}$.— We first show that individuals with knowledge $z \in W^* = W$ are assisted by a less knowledgeable solver post-AI than pre-AI (strictly so for all $z \neq 0$). Indeed, if $z \in W_a^*$, then $m(z; \hat{z}) \geq \hat{z} = z_{\text{AI}}$, where the first inequality is strict when $z > 0$. If $z \in W_p^*$ instead, then the matching functions pre- and post-AI are given by:

$$\begin{aligned} \textstyle\int_{\hat{z}}^{m(z;\hat{z})} dG(u) &= \textstyle\int_0^z h(1-u)dG(u), \text{ for } z \in [0, \hat{z}] \\ \textstyle\int_{\hat{z}}^{m^*(z;\hat{z})} dG(u) &= \textstyle\int_{\bar{z}_w^*}^z h(1-u)dG(u), \text{ for } z \in [\bar{z}_w^*, \hat{z}] \end{aligned}$$

Since $\bar{z}_w^* > 0$, this immediately implies that $m^*(z; \hat{z}) < m(z; \hat{z})$ for all $z \in W_p^*$.

We now show that individuals with knowledge $z \in S^* = S$ assist more knowledgeable workers post-AI than pre-AI (strictly so for all $z \neq 1$). Indeed, if $z \in S_a^*$, then $e(z; \hat{z}) \leq \hat{z} = z_{\text{AI}}$, where the first inequality is strict when $z < 1$. If $z \in S_p^*$ instead, then the employee functions pre- and post-AI are given by:

$$\begin{aligned} \textstyle\int_z^1 dG(u) &= \textstyle\int_{e(z;\hat{z})}^{\hat{z}} h(1-u)dG(u), \text{ for } z \in [\hat{z}, 1] \\ \textstyle\int_z^{\underline{z}_s^*} dG(u) &= \textstyle\int_{e^*(z;\hat{z})}^{\hat{z}} h(1-u)dG(u), \text{ for } z \in [\hat{z}, \underline{z}_s^*] \end{aligned}$$

Since $\underline{z}_s^* < 1$, this immediately implies that $e^*(z; \hat{z}) > e(z; \hat{z})$ for all $z \in S_p^*$. □

4.3 Equilibrium when Compute is Abundant Relative to Production Opportunities

In our baseline setting, compute is abundant relative to time but scarce relative to the existing production opportunities. This has two noteworthy implications. First, the equilibrium rental rate of compute is equal to AI's knowledge. Second, all humans earn labor income in the post-AI equilibrium—even those who are less knowledgeable than AI.

In this chapter, we relax the assumption that compute is scarce relative to production opportunities. In this case, the equilibrium price of compute is zero, and AI leads to technological unemployment: All humans who are less knowledgeable than AI become unemployed. Organizations, however, still display a hierarchical structure in that the most knowledgeable humans specialize in tackling problems that AI cannot solve.

4.3.1 Preliminaries

The model is exactly as described in Section 3 of the main text, except that the number of production opportunities is smaller than the compute available to pursue them (but larger than such capacity pre-AI). More precisely, denoting by ϕ the number of production opportunities, we assume that $1 < \phi < \mu$.[1]

As in the baseline model, pursuing each opportunity requires one unit of time or compute and solving a problem whose difficulty is not known ex-ante. To produce, firms must purchase production opportunities—which, given their scarcity, will now command a strictly positive price—and hire the necessary labor or rent the necessary compute to pursue them. Moreover, just as with the owners of compute, we do not explicitly model the "entrepreneurs" who own these opportunities.

Wages, Prices, and Profits.—Let p be the price of production opportunities (note that all opportunities must have the same price since they are all ex-ante identical), and recall that $w(z)$ and r denote the wage of a human with knowledge z and the rental rate of compute, respectively.

The profit of a single-layer organization is as follows:

$$\Pi_1 = \begin{cases} z - w(z) - p & \text{if the firm hires a human with knowledge } z \\ z_{\text{AI}} - r - p & \text{if the firm uses AI} \end{cases}$$

In other words, the profit of a single-layer nonautomated firm is the expected output z of its independent producer net of her wage $w(z)$ and the price of a production opportunity p. The profit of a single-layer automated firm is analogous.

The profit of a two-layer organization as a function of its configuration (tA, bA, or nA) is as follows:

$$\begin{aligned} \Pi_2^{tA}(z) &= n(z)[z_{\text{AI}} - w(z) - p] - r && (\text{where } z \leq z_{\text{AI}}) \\ \Pi_2^{bA}(s) &= n(z_{\text{AI}})[s - r - p] - w(s) && (\text{where } z_{\text{AI}} \leq s) \\ \Pi_2^{nA}(s,z) &= n(z)[s - w(z) - p] - w(s) && (\text{where } z \leq s) \end{aligned}$$

where z and s denote the knowledge of a human worker and a human solver, respectively, and we are using the fact that no two-layer firm hires a solver who is less knowledgeable than its workers.

As in the baseline case, the profit of a two-layer organization is its expected output minus the cost of its resources. For instance, in the case of a tA firm that hires workers with knowledge z, its

[1] To stay as close as possible to the baseline model, we assume that the number of opportunities ϕ is exogenous. In particular, AI cannot be used to generate new production opportunities.

expected output is $n(z)z_{\mathrm{AI}}$, while the cost of resources is $n(z)(w(z)+p)+r$, as the firm hires $n(z)$ workers, rents one unit of compute, and purchases $n(z)$ production opportunities.

Competitive Equilibrium.— Define the sets (W_p, W_a, I, S_p, S_a), the masses μ_i, μ_w, and μ_s, and the matching function $m : W_p \to S_p$ as in the baseline model. Recall that this matching function must satisfy:

(4.7) $$\int_Y h(1-u)dG(u) = \int_{m(Y)} dG(u), \quad \text{for any measurable set } Y \subseteq W_p$$

where $m(Y)$ represents the set of solvers assigned to the set of workers $Y \subseteq W_p$. We denote by U the set of humans that are unemployed and by μ_u the mass of compute not being used. Note that the total mass of opportunities pursued is equal to $\mu_w + \mu_i + \int_{z\in(W\cup I)} dG(u)$.

Definition (Competitive Equilibrium). An equilibrium consists of nonnegative amounts $(\mu_i, \mu_w, \mu_s, \mu_u)$, measurable sets $(W_a, W_p, I, S_p, S_a, U)$, a matching function $m : W_p \to S_p$ satisfying (4.7), a wage schedule $w : [0,1] \to \mathbb{R}_{\geq 0}$, a rental rate of compute $r \in \mathbb{R}_{\geq 0}$, and a price of production opportunities $p \in \mathbb{R}_{\geq 0}$, such that:

1. Firms optimally choose their structure (while earning zero profits).
2. Markets clear: (i) $\mu_i + \mu_w + \mu_s + \mu_u = \mu$, and (ii) the union of the sets $(W_a, W_p, I, S_p, S_a, U)$ is $[0,1]$ and the intersection of any two of these sets has measure zero.

4.3.2 AI Leads to Technological Unemployment

The pre-AI equilibrium is the same as in the baseline, as the number of opportunities is greater than the economy's pre-AI capacity to pursue them (as mentioned in the previous section). The post-AI equilibrium, in turn, is characterized by the next proposition. We index this new post-AI equilibrium with the superscript "†" instead of the superscript "∗" to distinguish it from that of the baseline.

Proposition 4.8. *When $\phi < \mu$, there is a unique equilibrium with autonomous AI. The equilibrium allocations are:*

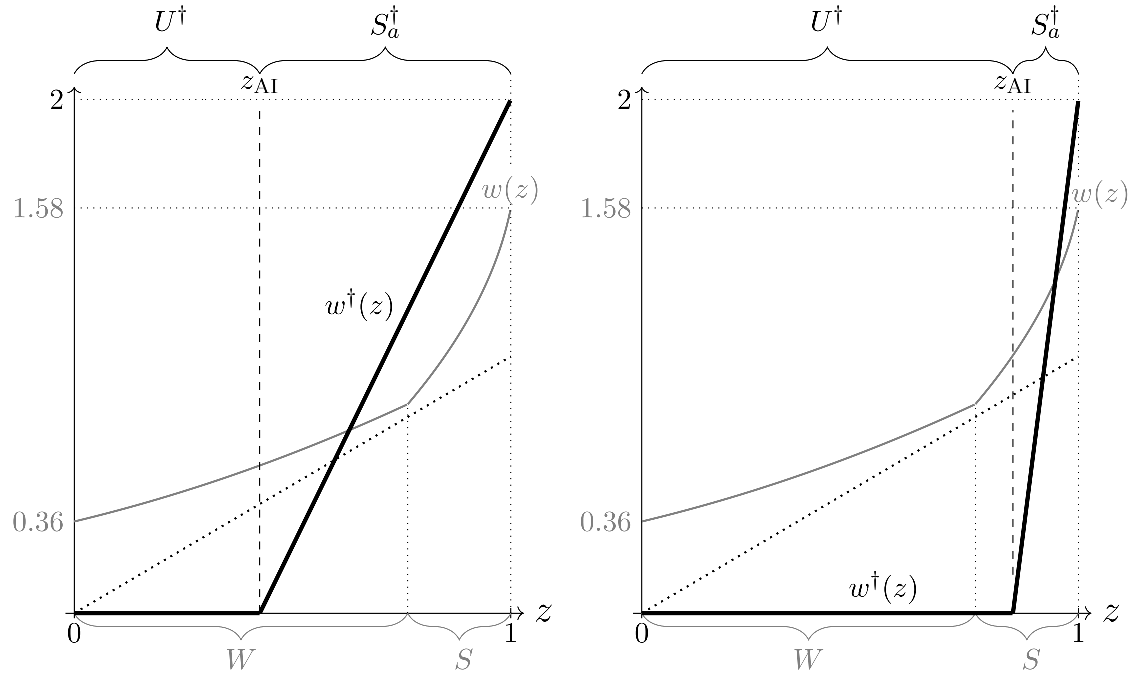
$$U^\dagger = [0, z_{\mathrm{AI}}), \; S_a^\dagger = [z_{\mathrm{AI}}, 1], \; W_a^\dagger = W_p^\dagger = I^\dagger = S_p^\dagger = \emptyset$$
$$\mu_w^\dagger = n(z_{\mathrm{AI}})[1 - G(z_{\mathrm{AI}})], \; \mu_s^\dagger = 0, \; \mu_i^\dagger = \max\left\{0, \phi - \mu_w^\dagger\right\}, \; \mu_u^\dagger = \mu - \mu_w^\dagger - \mu_i^\dagger$$

The equilibrium prices are $p^\dagger = z_{\mathrm{AI}}$, $r^\dagger = 0$, and:

$$w^\dagger(z) = \begin{cases} 0 & \text{if } z \in U^\dagger \\ n(z_{\mathrm{AI}})(z - z_{\mathrm{AI}}) & \text{if } z \in S_a^\dagger \end{cases}$$

Proof. See Chapter 4.3.3 below. □

According to Proposition 4.8, which is illustrated in Figure 4.4, the equilibrium when compute exceeds the available production opportunities differs significantly from that of the baseline. In particular, the price of compute is now zero, and all agents who are less knowledgeable than AI are



(a) Basic AI (i.e., $z_{\text{AI}} \in \text{int}W$) (b) Advanced AI (i.e., $z_{\text{AI}} \in \text{int}S$)

Figure 4.4: The Effects of AI when Compute is Greater than Production Opportunities

Notes. Distribution of knowledge: $G(z) = z$. Parameter values: For both panels, $h = 1/2$. For panel (a), $z_{\text{AI}} = 0.425$, while for panel (b), $z_{\text{AI}} = 0.85$. The thick gray line depicts the pre-AI equilibrium wage function w. The thick black line depicts the post-AI equilibrium wage function $w^\dagger$.

unemployed. Moreover, all those humans who are more knowledgeable than AI become solvers in bottom-automated firms.

Intuitively, given that compute is more abundant than production opportunities, firms use AI to pursue all available production opportunities. This displaces all those humans who are less knowledgeable than AI to unemployment. Furthermore, since some compute is left idle, the rental rate of compute is zero. Knowledge beyond that of AI, however, is still scarce because AI can attempt to solve but not actually solve all problems. Consequently, organizations are still hierarchical: All problems are initially attempted by AI, and humans who are more knowledgeable than AI specialize in solving problems that AI cannot solve. These humans are the only humans who are rewarded for their work.

The scarcity of opportunities also leads to a different distribution of income compared to the baseline. In particular, while in our baseline setting, "capitalists" (i.e., compute owners) earn z_{AI} per unit of compute and "entrepreneurs" (i.e., the owners of production opportunities) earn 0, in this extension, capitalists earn 0, and entrepreneurs earn z_{AI} per opportunity. Moreover, AI need not increase total labor income relative to the pre-AI benchmark, as a significant fraction of the population is displaced toward unemployment. The combination of these results suggests that as compute becomes abundant, there are incentives to reallocate labor and compute from pursuing production opportunities toward generating them.

Nevertheless, despite these differences, many of the results of our baseline continue to hold in this setting. Indeed, if $z_{\text{AI}} \in \text{int}W$, then AI still displaces humans from routine production work to specialized problem-solving, while if $z_{\text{AI}} \in \text{int}S$, then AI still reduces the number of humans doing specialized problem-solving (although the displacement, in this case, is toward unemployment). This immediately implies that our results regarding the distribution of firm productivity and span of control continue to hold without changes. The same applies to the effects of AI on the span of control of those solvers who are not occupationally displaced.

4.3.3 Proof of Proposition 4.8

For ease of exposition, we divide the proof into several smaller claims.

Claim 4.5. *When AI is autonomous and $\phi < \mu$, the equilibrium rental rate of compute is zero, i.e., $r^\dagger = 0$. Hence, so is the wage of the human with knowledge z_{AI}, i.e., $w^\dagger(z_{\text{AI}}) = 0$.*

Proof. This result follows because there is more available compute than production opportunities (i.e., $\phi < \mu$). Hence, some compute must be idle and, thus, not obtain any return. Since all compute is homogenous, this implies that no unit of compute can obtain a strictly positive return in equilibrium, i.e., $r^\dagger = 0$. That $w^\dagger(z_{\text{AI}}) = 0$ then follows because a human with knowledge z_{AI} is indistinguishable from a unit of compute used with AI. □

Claim 4.6. *When AI is autonomous and $\phi < \mu$, in equilibrium, all humans who are strictly less knowledgeable than AI are unemployed, i.e., $[0, z_{\text{AI}}) \subseteq U^\dagger$*

Proof. Suppose that in equilibrium there exists some $z \in [0, z_{\text{AI}})$ that is employed. This implies that all humans with knowledge z must also be employed and obtain a wage $w^\dagger(z) \geq 0$; otherwise, we would contradict the premise that this is an equilibrium. We show that firms hiring such humans can strictly increase their profits by replacing them with AI.

Indeed, if a single-layer firm employs a human with knowledge z, its profits are $z - w^\dagger(z) - p$, which are strictly lower than $z_{\text{AI}} - p$. Similarly, if an nA firm is employing these humans as workers, then its profits are $n(z)(s - w(z) - p) - w(s)$ (where s is the knowledge of the firm's solver), which are strictly lower than $n(z_{\text{AI}})(s - p) - w(s)$. Moreover, if the same type of firm is hiring such a human to be a solver, its profits are $n(z')(z - w(z') - p) - w(z)$ (where z' is the knowledge of the firm's workers), which are strictly lower than $n(z_{\text{AI}})(z - w(z') - p)$. The case of tA and bA firms follow an identical logic to that of nA firms. □

Claim 4.7. *When AI is autonomous and $\phi < \mu$, in equilibrium, a strictly positive mass of firms rent compute.*

Proof. For contradiction, suppose otherwise. Then, only humans are employed in equilibrium. The fact that $\phi > 1$ implies that $p^\dagger = 0$, as there are more production opportunities than humans available to pursue them. Hence, a firm can earn strictly positive profits by renting one unit of compute to pursue a production opportunity. Contradiction. □

Claim 4.8. *When AI is autonomous and $\phi < \mu$, in equilibrium, all humans that are strictly more knowledgeable than AI are employed and receive a strictly positive wage for their work, i.e., $w^\dagger(z) > 0$ for all $z \in (z_{\mathrm{AI}}, 1]$.*

Proof. By Claim 4.7, a strictly positive mass of firms rent compute in equilibrium. Suppose for contradiction that there exists some $z \in (z_{\mathrm{AI}}, 1]$ that is either not employed or receives a wage of $w^\dagger(z) = 0$ for her work. If so, one of the firms renting compute can strictly increase its profits by replacing AI with such a human. Contradiction. □

Given that in this setting the equilibrium still exhibits occupational stratification, the previous claims imply that the equilibrium takes the following form: There exists a cutoff $\zeta^\dagger \in [z_{\mathrm{AI}}, 1]$ such that:

$$U^\dagger = [0, z_{\mathrm{AI}}),\ W_a^\dagger = \emptyset,\ W_p^\dagger = \emptyset,\ I^\dagger = [z_{\mathrm{AI}}, \zeta^\dagger),\ S_p^\dagger = \emptyset,\ S_a^\dagger = [\zeta^\dagger, 1]$$

$$\text{Hence } \mu_w^\dagger = \int_{\zeta^\dagger}^1 n(z_{\mathrm{AI}}) dG(z),\ \mu_s^\dagger = 0,\ \mu_i^\dagger = \max\left\{0, \phi - \mu_w^\dagger - \int_{z_{\mathrm{AI}}}^{\zeta^\dagger} dG(z)\right\},\ \mu_u^\dagger = \mu - \mu_w^\dagger - \mu_i^\dagger$$

We then claim that $\zeta^\dagger = z_{\mathrm{AI}}$, implying that all humans that are strictly more knowledgeable than AI are employed as solvers:

Claim 4.9. *When AI is autonomous and $\phi < \mu$, in equilibrium, the price of problems is equal to AI's knowledge, i.e., $p^\dagger = z_{\mathrm{AI}}$.*

Proof. Suppose first that $I^\dagger \neq \emptyset$. Then, the zero-profit condition of single-layer nonautomated firms implies that $w^\dagger(z) = z - p^\dagger$ for all $z \in I^\dagger$. Since the wage function must be continuous and $w^\dagger(z_{\mathrm{AI}}) = 0$ (by claim 4.5), this immediately implies that $p^\dagger = z_{\mathrm{AI}}$.

Suppose instead that $I^\dagger = \emptyset$. Then, the zero-profit condition of bA firms implies that $w^\dagger(z) = n(z_{\mathrm{AI}})(z - p^\dagger)$. Thus, again, since the wage function must be continuous and $w^\dagger(z_{\mathrm{AI}}) = 0$, this implies that $p^\dagger = z_{\mathrm{AI}}$. □

Claim 4.10. *When AI is autonomous and $\phi < \mu$, in equilibrium, all humans that are strictly more knowledgeable than AI are employed as solvers and supervise the production work of AI, i.e., $\zeta^\dagger = z_{\mathrm{AI}}$.*

Proof. Suppose for contradiction that $\zeta^\dagger > z_{\mathrm{AI}}$. Then, the zero-profit condition of single-layer firms implies that $w^\dagger(z) = z - z_{\mathrm{AI}}$ for all $z \in I^\dagger$. However, if so, then a firm can hire a human with knowledge $z \in I^\dagger$, rent $n(z_{\mathrm{AI}})$ units of compute, and purchase $n(z_{\mathrm{AI}})$ problems to obtain a profit of $n(z_{\mathrm{AI}})(z - p^\dagger) - w^\dagger(z) = [n(z_{\mathrm{AI}}) - 1](z - z_{\mathrm{AI}}) > 0$. Contradiction. □

The last claim describes the equilibrium wages of the set of humans who are indeed employed:

Claim 4.11. *When AI is autonomous and $\phi < \mu$, $w^\dagger(z) = n(z_{\mathrm{AI}})(z - z_{\mathrm{AI}})$ for $z \in S_a^\dagger$.*

Proof. Immediate from the zero-profit condition of bA firms. □

Combining the previous seven claims, we conclude that the unique candidate for the equilibrium has the following form. The candidate allocations are:

$$U^\dagger = [0, z_{\text{AI}}),\ S_a^\dagger = [z_{\text{AI}}, 1],\ W_a^\dagger = W_p^\dagger = I^\dagger = S_p^\dagger = \emptyset$$
$$\mu_w^\dagger = n(z_{\text{AI}})[1 - G(z_{\text{AI}})],\ \mu_s^\dagger = 0,\ \mu_i^\dagger = \max\left\{0, \phi - \mu_w^\dagger\right\},\ \mu_u^\dagger = \mu - \mu_w^\dagger - \mu_i^\dagger$$

while the candidate prices are $p^\dagger = z_{\text{AI}}$, $r^\dagger = 0$, and:

$$w^\dagger(z) = \begin{cases} 0 & \text{if } z \in U^\dagger \\ n(z_{\text{AI}})(z - z_{\text{AI}}) & \text{if } z \in S_a^\dagger \end{cases}$$

From here, verifying that this is indeed an equilibrium, i.e., that no firms have incentives to deviate and all market clearing conditions are satisfied, is straightforward. □

4.3.4 A Reinterpretation of the Price of Production Opportunities

As currently written, the model assumes that unspecified "entrepreneurs" own production opportunities. Firms then purchase these opportunities at the equilibrium price $p^* = z_{\text{AI}}$ and claim the resulting output—priced at 1—if they successfully produce with them.

We now show that an equivalent interpretation is that unspecified "consumers" own production opportunities, which, if successfully pursued, generate a value of 1. Rather than purchasing these opportunities, firms publicly post bids specifying how the surplus from transforming an opportunity into output will be shared between the firm and the consumer providing it.[2]

Formally, let $\alpha(z)$ be the bid posted by a firm whose most knowledgeable employee has knowledge z. These bids vary across firms since their likelihood of transforming opportunities into output differs. The profit of a single-layer organization is equal to:

$$\Pi_1 = \begin{cases} \alpha(z)z - w(z) & \text{if the firm hires a human with knowledge } z \\ \alpha(z_{\text{AI}})z_{\text{AI}} - r & \text{if the firm uses AI} \end{cases}$$

The profit of a two-layer organization as a function of its type (nA, tA, or bA) and the knowledge of its employees is:

$$\Pi_2^{n\text{A}}(s, z) = n(z)[\alpha(s)s - w(z)] - w(s)$$
$$\Pi_2^{t\text{A}}(z) = n(z)[\alpha(z_{\text{AI}})z_{\text{AI}} - w(z)] - r$$
$$\Pi_2^{b\text{A}}(s) = n(z_{\text{AI}})[\alpha(s)s - r] - w(s)$$

It follows that setting $\alpha(z) = 1 - p/z$ recovers the original model, where firms purchase problems from "entrepreneurs" at a price p. Moreover, since $p^* = z_{\text{AI}}$ (Proposition 4.8), equilibrium bids satisfy

[2] For example, in legal services, it is not uncommon for lawyers to obtain a percentage of the settlement or deal they help their clients secure.

$\alpha^*(z) = 1 - z_{\mathrm{AI}}/z$. Thus, single-layer automated firms return all output to consumers, as compute is more abundant than production opportunities, while bottom-automated firms—more likely to succeed by employing human solvers with greater knowledge than AI—claim a positive share of the output.

4.4 AI Heterogeneity

In the main text, we claimed that if multiple AI technologies are available and all of them require similar computational resources to create AI agents, only the most advanced AI will be used in equilibrium. This underpins our assumption of focusing on a single AI technology. In this Appendix, we formalize this claim. We also show that if less advanced AI technologies require substantially less compute than their more advanced counterparts, they could coexist alongside them. This is consistent with the development of "smaller" foundation models like GPT-4o mini and Claude 3 Haiku.

4.4.1 Preliminaries

The model is exactly as described in Section 3 of the main text, except that we introduce an additional AI technology. This technology creates AI agents with knowledge $\xi_{\mathrm{AI}} < z_{\mathrm{AI}}$, but each agent created requires $\rho \leq 1$ units of compute. For brevity, we refer to this new AI as "mini-AI," while we refer to the original technology—the one that produces AI agents with knowledge z_{AI} and requires one unit of compute—by simply "AI."

Note that both AI and mini-AI rely on the same general-purpose compute. As a result, there is a single market price for compute in this economy and the existence of two AI technologies creates competition for this shared resource. The key question in this extension is: what is the equilibrium price of compute, and, at this price, which AI models are used and how?

To start, note that the profit of a single-layer automated firm that uses AI and mini-AI is $z_{\mathrm{AI}} - r$ and $\xi_{\mathrm{AI}} - \rho r$, respectively. This immediately implies that, generically, at most one AI will be used for independent production. Without loss of generality, we assume that z_{AI}, ξ_{AI}, and ρ are such that, if single-layer automated firms form, then these firms strictly prefer to use AI for independent production rather than mini-AI (the opposite case is analogous). For this to be the case, we need $\xi_{\mathrm{AI}} < \rho z_{\mathrm{AI}}$.

Note that, just like AI agents are perfect substitutes with humans with knowledge z_{AI}, mini-AI agents are perfect substitutes with humans with knowledge ξ_{AI}. Hence, by occupational stratification, the fact that AI is used for independent production implies that mini-AI agents cannot be solvers in equilibrium.

Wages, Prices, and Profits.—As in the main text, let $w(z)$ and r denote the wage of a human with knowledge z and the rental rate of compute, respectively. The profit of a single-layer organization is:

$$\Pi_1 = \begin{cases} z - w(z) & \text{if the firm hires a human with knowledge } z \\ z_{\text{AI}} - r & \text{if the firm uses AI} \end{cases}$$

where we are using the fact that no single-layer firms ever rent compute to create mini-AI agents.

The profit of a two-layer organization as a function of its configuration and the type of AI it uses is as follows:

$$\begin{aligned}
&\Pi_2^{t\text{A}}(z) = n(z)[z_{\text{AI}} - w(z)] - r && (\text{where } z \le z_{\text{AI}}) \\
&\Pi_2^{b\text{A}}(s) = n(z_{\text{AI}})[s - r] - w(s) && (\text{where } z_{\text{AI}} \le s) \\
&\Pi_2^{mb\text{A}}(s) = n(\xi_{\text{AI}})[s - \rho r] - w(s) && (\text{where } \xi_{\text{AI}} \le s) \\
&\Pi_2^{n\text{A}}(s,z) = n(z)[s - w(z)] - w(s) && (\text{where } z \le s) \\
&\Pi_2^{f\text{A}} = n(\xi_{\text{AI}})[z_{\text{AI}} - \rho r] - r &&
\end{aligned}$$

where z and s denote the knowledge of a human worker and a human solver, respectively, and we are using the fact that among firms that combine humans and AI, only bottom-automated firms use mini-AI (we call these "mbA" firms). Furthermore, note that—due to AI heterogeneity—two-layer fully automated firms ("fA" firms) can potentially arise in equilibrium.

Competitive Equilibrium.— Define the sets (W_a, W_p, I, S_p) as in the main text. Let S_a be the set of humans who provide assistance to AI, and S_ℓ as the set of humans who provide assistance to mini-AI.[3] Let μ_i and μ_s be the mass of compute rented to do independent production and assist humans, respectively. We define μ_w as the amount of compute rented for assisted production work using AI, and μ_ℓ as the analogous object for mini-AI. Finally, let μ_f denote the total amount of compute rented by fully automated firms and define the matching function $m : W_p \to S_p$ as in the baseline model. Recall that this matching function must satisfy:

$$\int_Y h(1-u)dG(u) = \int_{m(Y)} dG(u), \quad \text{for any measurable } Y \subseteq W_p \tag{4.8}$$

where $m(Y)$ represents the set of solvers assigned to the set of workers $Y \subseteq W_p$.

Since a bA firm that hires a human solver with knowledge $s \in S_a$ rents $n(z_{\text{AI}})$ units of compute, then $\mu_w = n(z_{\text{AI}}) \int_{S_a} dG(z)$. Similarly, since a mbA firm that hires a human solver with knowledge $s \in S_\ell$ rents $\rho n(\xi_{\text{AI}})$ units of compute, we have that $\mu_\ell = \rho n(\xi_{\text{AI}}) \int_{S_\ell} dG(z)$. Finally, since a tA firm that hires human workers with knowledge $z \in W_a$ rents one unit of compute, we have that $\mu_s = \int_{W_a} h(1-z)dG(z)$.

Definition (Competitive Equilibrium). A competitive equilibrium consists of nonnegative amounts $(\mu_i, \mu_w, \mu_\ell, \mu_s, \mu_f)$, measurable sets $(W_a, W_p, I, S_p, S_a, S_\ell)$, a function $m : W_p \to S_p$ satisfying condition (4.8), a wage schedule $w : [0,1] \to \mathbb{R}_{\ge 0}$ and a rental rate of compute $r \in \mathbb{R}_{\ge 0}$, such that:

[3] We use the subscript "ℓ" (for "limited") instead of "m" (for "mini"), to avoid any confusion with the matching function m.

1. Firms optimally choose their structure (while earning zero profits).
2. Markets clear: (i) $\mu_i + \mu_w + \mu_\ell + \mu_s + \mu_f = \mu$, and (ii) the union of the sets $(W_a, W_p, I, S_p, S_a, S_\ell)$ is $[0,1]$ and the intersection of any two of these sets has measure zero.

Note that the human workers in W_p are endogenously matched with the human solvers in S_p according to the pointwise matching function m. In contrast, all the humans in W_a are matched with an AI solver with knowledge z_{AI}, all humans in S_a are matched with AI workers with knowledge z_{AI}, and all the humans in S_ℓ are matched with mini-AI workers with knowledge ξ_{AI}.

For future reference, we use the superscript "$**$" to denote equilibrium outcomes in this setting, distinguishing them from those in the main text, which are marked by "$*$."

4.4.2 Equilibrium

We begin with the following lemma:

Lemma 4.5. *There exists $\bar{\rho} \in [\xi_{\text{AI}}/z_{\text{AI}}, 1)$ such that if $\rho > \bar{\rho}$, then mini-AI is not used in equilibrium.*

Proof. Let $\bar{\rho} = \max\{w^*(\xi_{\text{AI}})/z_{\text{AI}}, 1 - n(\xi_{\text{AI}})^{-1}\}$ where w^* is the wage function of the equilibrium with one AI characterized in the main text. That $\bar{\rho} < 1$ follows from $w^*(\xi_{\text{AI}}) < w^*(z_{\text{AI}}) = z_{\text{AI}}$. That $\bar{\rho} \geq \xi_{\text{AI}}/z_{\text{AI}}$ follows from $w^*(\xi_{\text{AI}}) \geq \xi_{\text{AI}}$.

We now show that if $\rho > \bar{\rho}$, then no fully-automated two-layer firm emerges in equilibrium. Note that $r^* \geq z_{\text{AI}}$, since AI can always be used as an independent producer. Hence, $\Pi_2^{f\text{A}} \leq n(z_{\text{AI}})(z_{\text{AI}} - \rho z_{\text{AI}}) - z_{\text{AI}} \leq 0$, where the last inequality follows because $r^* \geq z_{\text{AI}}$ and $\rho > \bar{\rho} \geq 1 - n(\xi_{\text{AI}})^{-1}$.

Hence, if $\rho > \bar{\rho}$, the only firms potentially using mini-AI are mbA firms. We now show that these firms cannot arise either. Note that if $\rho > \bar{\rho}$, then the AI equilibrium described in the main text—which only considers AI—continues to be the equilibrium when both AI and mini-AI are present. This is because each mini-AI agent is a perfect substitute for a human with knowledge ξ_{AI}, but the price of a mini-AI agent is $\rho z_{\text{AI}} > w^*(\xi_{\text{AI}})$ (as $\rho > \bar{\rho} \geq w^*(\xi_{\text{AI}})/z_{\text{AI}}$). Hence, no firm has incentives to deploy a mini-AI agent. That this is the unique equilibrium when both AI and mini-AI are present follows because any equilibrium maximizes total output, and there is a unique allocation that does so. □

Lemma 4.5 formalizes our claim that only a single AI technology is used in equilibrium if all available AI technologies have similar compute requirements. The intuition was already explained in the main text: If multiple AI technologies are available and all of them require similar computational resources to create AI agents, compute naturally flows toward the highest-performing model.

We close this chapter by illustrating an equilibrium where mini-AI is used. In particular, we introduce mini-AI into a Type 1 equilibrium where AI is used as an independent producer and as a worker (but not as a solver). For simplicity, fully automated firms do not arise in the equilibrium we construct; only mbA-firms use mini-AI. For this to be the case we need that:

$$1 - n(\xi_{\text{AI}})^{-1} \leq \rho \leq w^*(\xi_{\text{AI}})/z_{\text{AI}} = \bar{\rho} \tag{4.9}$$

To start, note that if (4.9) holds, then the equilibrium price of compute is the same as when only AI is present, i.e., $r^{**} = r^* = z_{\text{AI}}$. This follows because there are no fully automated firms, and single-layer automated firms use AI—not mini-AI—for independent production. Note also that since a mini-AI agent costs $\rho z_{\text{AI}} < w^*(\xi_{\text{AI}})$, then the AI equilibrium described in the main text—which only considers AI—cannot be an equilibrium when both AI and mini-AI are both present. This also implies that in a world where both AI and mini-AI are present, the wage of $z = \xi_{\text{AI}}$ has to go down to ρz_{AI}, i.e., $w^{**}(\xi_{\text{AI}}) = \rho z_{\text{AI}}$.

Given that the equilibrium continues to satisfy occupational stratification and positive assortative matching, an equilibrium where AI is used as an independent producer and as a worker (but not as a solver) and mini-AI is used as a worker must take the following form:

$$W_a^{**} = \emptyset,\ W_p^{**} = [0, \xi_{\text{AI}}] \cup [\xi_{\text{AI}}, z_{\text{AI}}],\ I^{**} = (z_{\text{AI}}, \underline{\underline{z}}_1^{**})$$
$$S_p^{**} = [\underline{\underline{z}}_1^{**}, \underline{z}_1^{**}] \cup [\bar{z}_1^{**}, \bar{\bar{z}}_1^{**}],\ S_\ell^{**} = [\underline{z}_1^{**}, \bar{z}_1^{**}],\ S_a^{**} = [\bar{\bar{z}}_1^{**}, 1]$$

where the cutoffs $z_{\text{AI}} < \underline{\underline{z}}_1^{**} < \underline{z}_1^{**} < \bar{z}_1^{**} < \bar{\bar{z}}_1^{**}$ satisfy:

$$\begin{aligned}
&\textstyle\int_0^{\xi_{\text{AI}}} h(1-u)dG(u) = \int_{\underline{\underline{z}}_1^{**}}^{\underline{z}_1^{**}} dG(u) \\
&\textstyle\int_{\xi_{\text{AI}}}^{z_{\text{AI}}} h(1-u)dG(u) = \int_{\bar{z}_1^{**}}^{\bar{\bar{z}}_1^{**}} dG(u) \\
&n(\xi_{\text{AI}})(\underline{z}_1^{**} - \rho z_{\text{AI}}) = \underline{\underline{z}}_1^{**} + \textstyle\int_{\underline{\underline{z}}_1^{**}}^{\underline{z}_1^{**}} n(e_-(z; \underline{\underline{z}}_1^{**}))dz \\
&n(z_{\text{AI}})(\bar{\bar{z}}_1^{**} - z_{\text{AI}}) = n(\xi_{\text{AI}})(\underline{z}_1^{**} - \rho z_{\text{AI}}) + \textstyle\int_{\bar{z}_1^{**}}^{\bar{\bar{z}}_1^{**}} n(e_+(z; \bar{z}_1^{**}))dz
\end{aligned}$$

and $e_- : [\underline{\underline{z}}_1^{**}, \underline{z}_1^{**}] \to [0, \xi_{\text{AI}}]$ and $e_+ : [\bar{z}_1^{**}, \bar{\bar{z}}_1^{**}] \to [\xi_{\text{AI}}, z_{\text{AI}}]$ are given by:

$$\textstyle\int_0^{e_-(z;\underline{\underline{z}}_1^{**})} h(1-u)dG(u) = \int_{\underline{\underline{z}}_1^{**}}^{z} dG(u) \ \text{ and } \ \int_{\xi_{\text{AI}}}^{e_+(z;\bar{z}_1^{**})} h(1-u)dG(u) = \int_{\bar{z}_1^{**}}^{z} dG(u)$$

The equilibrium wages can be constructed following a similar logic as in the case of a single AI. They are given as follows:

$$w^*(z) = \begin{cases}
m_-(z; \underline{\underline{z}}_1^{**}) - w(m_-(z; \underline{\underline{z}}_1^{**}))/n(z) & \text{if } z \in [0, \xi_{\text{AI}}] \subset W_p^{**} \\
m_+(z; \bar{z}_1^{**}) - w(m_+(z; \bar{z}_1^{**}))/n(z) & \text{if } z \in [\xi_{\text{AI}}, z_{\text{AI}}] \subset W_p^{**} \\
z & \text{if } z \in (z_{\text{AI}}, \underline{\underline{z}}_1^{**}) = I^{**} \\
\underline{\underline{z}}_1^{**} + \int_{\underline{\underline{z}}_1^{**}}^{z} n(e_-(u; \underline{\underline{z}}_1^{**}))du & \text{if } z \in [\underline{\underline{z}}_1^{**}, \underline{z}_1^{**}] \subset S_p^{**} \\
n(\xi_{\text{AI}})(\underline{z}_1^{**} - \rho z_{\text{AI}}) & \text{if } z \in [\underline{z}_1^{**}, \bar{z}_1^{**}] = S_\ell^{**} \\
n(\xi_{\text{AI}})(\underline{z}_1^{**} - \rho z_{\text{AI}}) + \int_{\bar{z}_1^{**}}^{z} n(e_+(u; \bar{z}_1^{**}))du & \text{if } z \in [\bar{z}_1^{**}, \bar{\bar{z}}_1^{**}] \subset S_p^{**} \\
n(z_{\text{AI}})(z - z_{\text{AI}}) & \text{if } z \in [\bar{\bar{z}}_1^{**}, 1] = S_a^{**}
\end{cases}$$

where $m_-(z; \cdot) = (e_-)^{-1}(z; \cdot)$ and $m_+(z; \cdot) = (e_+)^{-1}(z; \cdot)$.

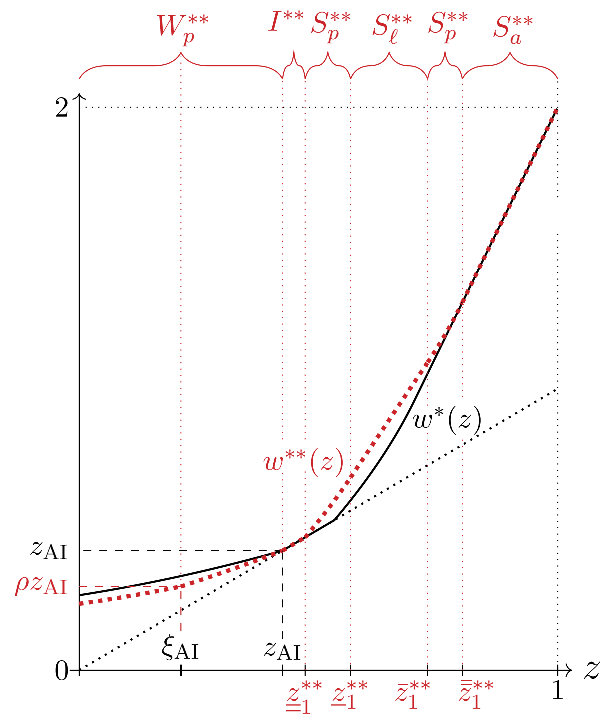


Figure 4.5: The Equilibrium with One vs Two AI Technologies

Notes. Distribution of knowledge: $G(z) = z$. Parameter values: $h = 1/2$, $\xi_{\mathrm{AI}} = 0.2125$, $z_{\mathrm{AI}} = 0.425$, and $\rho = 0.7$. The thick black line represents the equilibrium wage function when only one AI is present, w^*. The thick red dotted line represents the equilibrium wage function when both AI and mini-AI are present, w^{**}.

Figure 4.5 illustrates the equilibrium in this case when $h = 1/2$, $\xi_{\mathrm{AI}} = 0.2125$, $z_{\mathrm{AI}} = 0.425$, and $\rho = 0.7$. The figure also compares this equilibrium with the one considered in the main text, where only AI is present. As it is evident from this figure, the introduction of mini-AI reduces the equilibrium wage of those humans with knowledge around ξ_{AI}, while benefitting the least knowledgeable solvers, who now have access to cheaper workers. This is also the reason why the introduction of mini-AI reduces the mass of human independent producers compared to the equilibrium with one AI.

4.5 Autonomous AI vs. Offshoring/Immigration

As explained in the main text, the introduction of an autonomous AI diverges from classical labor shocks like offshoring or immigration. The key distinction lies in AI's scalability. Once knowledge is encoded into an AI system, it can be deployed across all units of compute, which are widely available. This scalability amounts to the introduction of a large population of homogeneous agents.

In contrast, large human populations are inherently heterogeneous. Consequently, offshoring or immigration typically involve either the introduction of a relatively small population of homogeneous agents, or the introduction of a large population of heterogeneous agents. As discussed in this appendix, this difference leads to qualitatively different outcomes.

4.5.1 AI vs. Offshoring

The canonical model of offshoring in a knowledge economy is due to Antràs et al. (2006), who consider a two-country model in which countries differ only in their knowledge distributions. In particular, one country, the North, has a distribution of knowledge with a relatively high mean, while the other country, the South, has a distribution of knowledge with a relatively low mean. They show that allowing the formation of international teams shifts humans from routine work to specialized problem solving in the North, while it shifts humans in the opposite direction in the South.

Given the similarity between the occupational displacement effects of AI and offshoring, one might conjecture that the effects of AI when $z_{\mathrm{AI}} \in \mathrm{int}W$ are qualitatively similar to the effects of offshoring from the North's perspective and that the effects of AI when $z_{\mathrm{AI}} \in \mathrm{int}S$ are qualitatively similar to the effects of offshoring from the South's perspective. This, however, is not the case. For instance, while offshoring increases the productivity of the best workers who remain workers in the North (Antràs et al., 2006, Proposition 1), AI reduces the productivity of all non-occupationally displaced workers when $z_{\mathrm{AI}} \in \mathrm{int}W$. Similarly, while offshoring decreases the span of control of all southern solvers who remain solvers (Antràs et al., 2006, Proposition 1), AI increases the span of control of all non-occupationally displaced solvers when $z_{\mathrm{AI}} \in \mathrm{int}S$.

As we mentioned above, the key difference between AI and offshoring is the capacity of AI to operate at scale. This implies that, although both offshoring and AI induce the best northern workers pre-offshoring/pre-AI to switch to specialized problem solving, the best northern solvers switch to supervising the best non-occupationally displaced northern workers in the case of offshoring, while they switch to supervising AI in the case of artificial intelligence.

Similarly, when $z_{\mathrm{AI}} \in \mathrm{int}S$, both offshoring and AI induce the worst southern solvers to switch to routine work, improving the overall pool of southern workers. However, in the case of offshoring, the best southern workers are matched with the best northern solvers, leaving a worse pool of workers for the non-occupationally displaced southern solvers. In contrast, in the case of AI, the worst workers end up being supervised by AI, leaving the best workers for the solvers who are not occupationally displaced.

4.5.2 AI vs. Targeted Immigration

Immigration can be targeted to specific knowledge levels, leading to an influx of a population of homogeneous humans. However, the restriction to such specific knowledge typically implies that the influx is small relative to the existing population.

Such targeted immigration is analogous to the introduction of AI when the amount of compute μ is small. We describe the effects of such a shock in Chapter 4.2. As described in that chapter, beyond the obvious quantitative differences, there are two main qualitative differences.

First, the effects on the quality of matches of non-occupationally displaced humans are different (see Proposition 4.7 in Chapter 4.2.1). Second, when μ is small, the individuals who see their wage increase due to the shock (i.e., those complemented by it) need not be at the extremes of the knowledge distribution (see Figure 4.3 in Chapter 4.2.1).

4.6 More Layers

We now show that several insights of our model that generalize to a more general model that allows for an arbitrary number of layers. We follow the literature on knowledge hierarchies (e.g., Garicano and Rossi-Hansberg, 2006; Caliendo and Rossi-Hansberg, 2012) in assuming that each organization requires one agent at the top. This implies that firms cannot have infinite layers. As in the main text, we continue to assume that compute is abundant relative to time.

No worker is more knowledgeable than AI, and no solver is less knowledgeable than AI.— Because occupational stratification holds regardless of the number of layers and AI agents are used for independent production, the result that no worker is more knowledgeable than AI and no solver is less knowledgeable than AI continues to hold with an arbitrary number of layers. This implies that if AI is used for assistance, it provides assistance to the least knowledgeable individuals.

The Most Knowledgeable Always Win from AI's Introduction.— To begin, consider the pre-AI equilibrium. As shown in Garicano and Rossi-Hansberg (2006, p. 1416), the wage of a solver in an arbitrary layer can be expressed as follows: for each of the $1/h$ problems she encounters, she earns the conditional probability that she can solve the problem given that those below her could not. Additionally, she pays a positive fee to receive each of these problems and earns a positive fee when she cannot solve the problem and passes it along to the next layer.

Now, the most knowledgeable individuals are necessarily at the top layer and, therefore, do not pass any problems to higher layers. This implies that their pre-AI wage is strictly less than $1/h$. However, post-AI, individuals with knowledge $z = 1$ earn at least $1/h$. This result follows because they can always rent $n(z_{\text{AI}})$ units of compute to perform production work on their behalf, yielding earnings of $n(z_{\text{AI}})(1-z_{\text{AI}}) = 1/h$. Hence, by continuity, there exists $\epsilon > 0$ such that, for all $z \in (1-\epsilon, 1]$, $w^*(z) > w(z)$.

The Least Knowledgeable Individuals Lose when AI's Knowledge is Low.— Irrespective of the number of layers, it is clear that in the pre-AI economy, workers earn strictly more than their output as independent producers. The intuition is the same as in the case of two-layer organizations, and follows from the fact that the marginal value of knowledge is strictly lower than 1 for workers, exactly equal to 1 for independent producers, and strictly greater than 1 for solvers.[4] Consequently, when z_{AI} is sufficiently low, the introduction of AI inevitably harms the least knowledgeable individuals by driving

[4]To be more precise, suppose there exists an interval of workers $[z', z'']$ who earn their expected output as independent producers. In this case, no hierarchical firm would hire workers with knowledge $z \in (z', z'')$, as the firm could strictly increase its profits by maintaining the same number of layers, employing the same solvers, and hiring workers with knowledge z' instead. This is a violation of market clearing.

their wages closer to their expected output as independent producers, as AI directly competes with them in production work.

The Least Knowledgeable Individuals Win when AI's Knowledge is Sufficiently High.— Post-AI, an individual with knowledge $z = 0$ can always earn $(1-h)z_{\mathrm{AI}}$ by renting one unit of compute to create an AI solver. Therefore, $w^*(0) \geq (1-h)z_{\mathrm{AI}}$. We claim that, pre-AI, the wage of $z = 0$ is strictly less than $1-h$, i.e., $w(0) < 1-h$. Consequently, for z_{AI} sufficiently close to 1, $w^*(0) > w(0)$, meaning there exists $\varepsilon > 0$ such that, for all $z \in [0, \varepsilon)$, $w^*(z) > w(z)$.

To demonstrate that $w(0) < 1-h$, assume for contradiction that $w(0) \geq 1-h$. As mentioned above, Garicano and Rossi-Hansberg (2006, p. 1416) show that the wage of an individual in an arbitrary layer can be expressed as follows: for each of the $1/h$ problems she encounters, she earns the conditional probability that she can solve the problem given that those below her could not. Additionally, she pays a positive fee to receive each of these problems and earns a positive fee if she cannot solve a problem and passes it to the next layer. Since workers belong to the first layer of the organization (i.e., layer zero), they do not pay to receive problems. Conversely, the top solver in an organization does not earn a positive fee for unresolved problems, as she does not pass problems to higher layers.

Now consider the least knowledgeable individual, $z = 0$. This individual is necessarily a worker. Since the probability that an individual with knowledge 0 solves the problem is 0, $w(0)$ is equal to the fee t_0 that she receives for passing the problem on to the next layer. Hence, the premise that $w(0) \geq 1-h$, implies that $t_0 \geq 1-h$.

Let t_n be the fee that her superior in layer n—with knowledge z_n—receives for passing the problem up the hierarchy. We then have that:

$$w(z_{n+1}) = \frac{z_{n+1} - z_n - t_n + t_{n+1}}{h(1-z_n)}$$

We will show that:

$$t_0 \geq 1-h \Longrightarrow t_n \geq (1-h)(1-z_n) + h\sum_{i=0}^{n} z_{i-1}(1-z_i) > 0, \text{ for every } n = 1, 2, \ldots \tag{4.10}$$

Hence, if $t_0 \geq 1-h$, then every one of 0's superiors must be sending problems up to higher layers (t_n is strictly positive), so 0 is in a firm with infinitely many layers. This contradicts the fact that each organization requires one agent at the top.

We prove (4.10) by induction. Suppose that for any $n = 0, 1, \cdots$, we have that:

$$t_n \geq (1-h)(1-z_n) + h\sum_{i=0}^{n} z_{i-1}(1-z_i) \tag{4.11}$$

We prove that:

$$t_{n+1} \geq (1-h)(1-z_{n+1}) + h\sum_{i=0}^{n+1} z_{i-1}(1-z_i) \tag{4.12}$$

Indeed, note that since z_{n+1} can always do independent production, $w(z_{n+1}) \geq z_{n+1}$, so:

$$w(z_{n+1}) = \frac{z_{n+1} - z_n - t_n + t_{n+1}}{h(1-z_n)} \geq z_{n+1} \tag{4.13}$$

Combining (4.11) and (4.13) gives:

$$\frac{z_{n+1} - z_n - (1-h)(1-z_n) - h\sum_{i=0}^{n} z_{i-1}(1-z_i) + t_{n+1}}{h(1-z_n)} \geq z_{n+1}$$

which rearranging gives (4.12). □

Chapter 5

Non-Autonomous AI: Further Results

5.1 Occupational Choice, Firm Distribution, and Worker-Solver Matching

In this chapter, we describe the effects of non-autonomous AI on (i) human occupational choices, (ii) the distribution of firm size, productivity, and decentralization, and (iii) the productivity and span of control of workers and solvers who remain in their positions, compared to the pre-AI equilibrium. As in the main text, we assume that compute is abundant relative to time.

Proposition 5.1. *When $z_{\text{AI}} > w(0)$, the effects of a non-autonomous AI relative to the pre-AI equilibrium are the following:*

- *Occupational Displacement:*
  - *AI always displaces humans from specialized problem-solving to routine work, i.e., $W^\star \supset W$ and $S^\star \subset S$.*
- *Distribution of Firm Productivity, Size, and Span of Control:*
  - *When $z_{\text{AI}} \in \text{int}W$, AI decreases the minimum firm productivity and the minimum firm size, while increasing the maximum span of control and the maximum firm size.*
  - *When $z_{\text{AI}} \in \text{int}S$, AI increases the maximum span of control, the minimum firm productivity, and both the minimum and maximum firm size.*
  - *AI never leads to the creation of superstar firms that have scale but no mass.*
- *Productivity of Non-Occupationally Displaced Workers:*
  - *When $z_{\text{AI}} \in \text{int}W$, the productivity of all workers is lower post-AI than pre-AI.*
  - *When $z_{\text{AI}} \in \text{int}S$, the productivity of the least knowledgeable workers is higher post-AI than pre-AI. The productivity of the most knowledgeable workers is lower post-AI than pre-AI.*
- *Span of Control of the Non-Occupationally Displaced Solvers.*
  - *The span of control of all solvers is larger post-AI than pre-AI.*

We now provide the proof of this proposition. For clarity, we have divided this proof into smaller claims. For all the claims that follow, we assume that $z_{\mathrm{AI}} > w(0)$ (the case considered in the proposition).

Claim 5.1. *Non-autonomous AI always displaces humans from specialized problem-solving to routine work, i.e., $W^\star \supset W$ and $S^\star \subset S$.*

Proof. We first show that $S^\star \subset S$. Pre-AI, we have that $S = [\hat{z}, 1]$, while post-AI we have either $S^\star = [\bar{z}_1^\star, 1]$ (in a Type 1a configuration) or $S^\star = [\bar{z}_2^\star, 1]$ (in a Type 2a configuration), where $\bar{z}_1^\star$ and $\bar{z}_2^\star$ are given as in Lemma 2.8. We claim that $\bar{z}_1^\star > \hat{z}$ and $\bar{z}_2^\star > \hat{z}$, which implies that $S^\star \subset S$.

Indeed, recall that if the equilibrium follows a Type 1a configuration, then $\underline{z}_1^\star = \kappa$ and $\bar{z}_1^\star = \hat{z}(\kappa = \underline{z}_1^\star)$, where $\hat{z}(\kappa)$ is the equilibrium cutoff of the auxiliary κ-economy when $h < \tilde{h}_0(\kappa)$ characterized in Claim 2.5. Similarly, if the equilibrium follows a Type 2a configuration, then $\underline{z}_2^\star = \kappa$, $\hat{z}_2^\star = \underline{z}(\kappa = \underline{z}_2^\star)$, and $\bar{z}_2^\star = \bar{z}(\kappa = \underline{z}_2^\star)$, where $\underline{z}(\kappa)$ and $\bar{z}(\kappa)$ are the equilibrium cutoff of the auxiliary κ-economy when $h \geq \tilde{h}_0(\kappa)$. Moreover, as argued in the proof of Lemma 2.8, $\hat{z}(\kappa)$ and $\bar{z}(\kappa)$ are strictly increasing in κ, and coincide with each other when $h = \tilde{h}_0(\kappa)$. Since the pre-AI cutoff of the 'true' economy is $\hat{z} = \hat{z}(\kappa = 0)$, we then have that for $\kappa > 0$:

$$\hat{z} = \hat{z}(\kappa = 0) < \hat{z}(\kappa) \text{ and } \hat{z} = \hat{z}(\kappa = 0) < \bar{z}(\kappa)$$

Thus, we conclude that $\bar{z}_1^\star = \hat{z}(\kappa = \underline{z}_1^\star) > \hat{z}$ and $\bar{z}_2^\star = \bar{z}(\kappa = \underline{z}_2^\star) > \hat{z}$.

Next, we show that $W^\star \supset W$. Pre-AI, we have that $W = [0, \hat{z}]$, while post-AI we have either $W^\star = [0, \bar{z}_1^\star]$ (in a Type 1a configuration) or $W^\star = [0, \hat{z}_2^\star]$ (in a Type 2a configuration), where $\bar{z}_1^\star$ and $\hat{z}_2^\star$ are given as in Lemma 2.8. We already established that $\bar{z}_1^\star > \hat{z}$. Hence, if the post-AI equilibrium follows a Type 1a configuration, we immediately have $W^\star \supset W$. Thus, it only remains to show that $\hat{z}_2^\star > \hat{z}$.

To do this, let $\alpha(x, y) \equiv \frac{1}{h} - x - \int_x^1 n(e_2^\star(z; y))dz$, where $e_2^\star(z; y)$ is given by $\int_{e_2^\star(z;y)}^y h(1-u)dG(u) = \int_z^1 dG(u)$. It is not difficult to prove that $\alpha(x, y)$ is strictly increasing in x and strictly decreasing in y. Furthermore, note that the employee matching function in a Type 2a equilibrium is $e_2^\star(z; \hat{z}_2^\star)$. This implies that:

$$\alpha(\bar{z}_2^\star, \hat{z}_2^\star) = \tfrac{1}{h} - \bar{z}_2^\star - \int_{\bar{z}_2^\star}^1 n(e_2^\star(z; \hat{z}_2^\star))dz = 0$$

Moreover, notice that $e_2^\star(z; \hat{z}) = e(z; \hat{z})$, where $e(z; \hat{z})$ is the employee matching function of the pre-AI equilibrium. Hence:

$$\alpha(\hat{z}, \hat{z}) = \tfrac{1}{h} - \hat{z} - \int_{\hat{z}}^1 n(e(z; \hat{z}))dz > 0$$

Suppose then for contradiction that $\hat{z}_2^\star \leq \hat{z}$, so $\alpha(x, \hat{z}) \leq \alpha(x, \hat{z}_2^\star)$. Because $\bar{z}_2^\star > \hat{z}$, we have $\alpha(\hat{z}, \hat{z}) \leq \alpha(\hat{z}, \hat{z}_2^\star) < \alpha(\bar{z}_2^\star, \hat{z}_2^\star) = 0$. Hence, $\alpha(\hat{z}, \hat{z}) < 0$, which contradicts the fact that $\alpha(\hat{z}, \hat{z}) > 0$. $\square$

Claim 5.2. *When $z_{\mathrm{AI}} \in \mathrm{int}W$, non-autonomous AI decreases the minimum firm productivity and the minimum firm size, while increasing the maximum span of control and the maximum firm size.*

Proof. Consider first span of control. The most decentralized firm of the pre-AI economy has a span of control $n(\sup W)$, while the most decentralized firm of the post-AI world has a span of control of $n(\sup W^\star)$. That AI increases the maximum span of control follows from the fact that $W^\star \supset W$.

Consider next productivity. The least productive firm of the pre-AI economy has productivity $\inf S$, while the least productive firm of the post-AI world has productivity $z_{\text{AI}} \in \text{int}W$. This immediately implies that AI decreases the minimum firm productivity.

Finally, consider size. As noted in the main text, the size of the smallest firm is $n(0)$ times the minimum firm productivity, while the size of the largest firm is 1 times the maximum span of control. Consequently, the results regarding the maximum span of control and minimum firm productivity immediately imply that AI decreases the minimum firm size and increases the maximum firm size. □

Claim 5.3. *When $z_{\text{AI}} \in \text{int}S$, non-autonomous AI increases the maximum span of control, the minimum firm productivity, and both the minimum and maximum firm size.*

Proof. Consider first span of control. The most decentralized firm of the pre-AI economy has a span of control $n(\sup W)$, while the most decentralized firm of the post-AI world has a span of control of $n(\sup W^\star)$. That AI increases the maximum span of control follows from the fact that $W^\star \supset W$.

Consider next productivity. The least productive firm of the pre-AI economy has productivity $\inf S$, while the least productive firm of the post-AI world has productivity $z_{\text{AI}} \in \text{int}S$. This immediately implies that AI increases the minimum firm productivity.

Finally, consider size. Due to positive assortative matching, the size of the smallest firm is $n(0)$ times the minimum firm productivity, while the size of the largest firm is 1 times the maximum span of control. Consequently, the results regarding the maximum span of control and minimum firm productivity immediately imply that AI increases both the minimum and maximum firm size. □

Claim 5.4. *Non-autonomous AI never leads to the creation of superstar firms that have scale but no mass.*

Proof. This follows directly from the fact that AI cannot do routine production work. Hence, there cannot be bottom-automated firms in equilibrium. □

Claim 5.5. *When $z_{\text{AI}} \in \text{int}W$ and AI is non-autonomous, the productivity of all non-occupationally displaced workers is lower post-AI than pre-AI.*

For the proof that follows, it is useful that we make explicit that the pre-AI matching and employee functions depend on $\hat{z}$ (i.e., the knowledge threshold that separates workers and solvers in the pre-AI equilibrium). For this reason, we write $m(z;\hat{z})$ and $e(z;\hat{z})$ instead of $m(z)$ and $e(z)$, respectively.

Proof. As noted in the main text, a worker's productivity increases if and only if her match improves. Hence, it is enough to show that all non-occupationally displaced workers have a worse match post-AI than pre-AI. Note that if $z \in W_a^\star$, then such a worker matches with AI in the post-AI equilibrium. Hence, in this case $m(z;\hat{z}) \geq \hat{z} > z_{\text{AI}}$, as $z_{\text{AI}} \in \text{int}W$.

Proving that every $z \in W_p^\star$ also has a worse solver is more involved. The post-AI equilibrium can be either Type 1a or Type 2a. Suppose first that it is Type 1a. Then, the matching functions pre- and post-AI are given by:

$$\int_z^{\bar{z}_1^\star} h(1-u)dG(u) = \int_{m_1^\star(z;\bar{z}_1^\star)}^1 dG(u) \text{ for } z \in W_p^\star = [\underline{z}_1^\star, \bar{z}_1^\star]$$
$$\int_z^{\hat{z}} h(1-u)dG(u) = \int_{m(z;\hat{z})}^1 dG(u) \text{ for } z \in W = [0, \hat{z}]$$

Thus, if $z \in W_p^\star \cap W = [\underline{z}_1^\star, \hat{z}]$, then substracting both conditions we get $\int_{\hat{z}}^{\bar{z}_1^\star} h(1-u)dG(u) = \int_{m_1^\star(z;\bar{z}_1^\star)}^{m(z;\hat{z})}$. Since $\bar{z}_1^\star > \hat{z}$, the left-hand side of this last expression is strictly positive, which implies that $m_1^\star(z;\bar{z}_1^\star) < m(z;\hat{z})$ for all $z \in W_p^\star \cap W = [\underline{z}_1^\star, \hat{z}]$.

Suppose instead that post-AI equilibrium is Type 2a. Then, the matching functions pre- and post-AI are given by:

$$\int_z^{\hat{z}_2^\star} h(1-u)dG(u) = \int_{m_2^\star(z;\hat{z}_2^\star)}^1 dG(u) \text{ for } z \in W_p^\star = [\underline{z}_2^\star, \hat{z}_2^\star]$$
$$\int_z^{\hat{z}} h(1-u)dG(u) = \int_{m(z;\hat{z})}^1 dG(u) \text{ for } z \in W = [0, \hat{z}]$$

Thus, if $z \in W_p^\star \cap W = [\underline{z}_2^\star, \hat{z}]$, then substracting both conditions we get:$\int_{\hat{z}}^{\hat{z}_2^\star} h(1-u)dG(u) = \int_{m_2^\star(z;\hat{z}_2^\star)}^{m(z;\hat{z})}$. Since $\hat{z}_2^\star > \hat{z}$, the left-hand side of this last expression is strictly positive, which implies that $m_2^\star(z;\hat{z}_2^\star) < m(z;\hat{z})$ for all $z \in W_p^\star \cap W = [\underline{z}_2^\star, \hat{z}]$. □

Claim 5.6. *When $z_{\text{AI}} \in \text{int}S$ and AI is non-autonomous, the productivity of the least knowledgeable non-occupationally displaced workers is higher post-AI than pre-AI. The productivity of the most knowledgeable non-occupationally displaced workers is lower post-AI than pre-AI.*

Proof. The least knowledgeable non-occupationally displaced workers are those with knowledge $z = 0$. These workers are clearly better matched post-AI than pre-AI, as post-AI they are assisted by $z_{\text{AI}} \in \text{int}S$. The most knowledgeable non-occupationally displaced workers are those with knowledge $\hat{z}$ (since $W^\star \supset W$). Pre-AI, these workers match with solvers with knowledge 1. Post-AI, these workers match with solvers with knowledge $z < 1$, since post-AI, the most knowledgeable individuals assist those workers with knowledge $z \in W_p^\star \cap S$. We conclude that the productivity of the most knowledgeable non-occupationally displaced workers is lower post-AI than pre-AI. □

Claim 5.7. *When AI is non-autonomous, the span of control of all non-occupationally displaced solvers is larger post-AI than pre-AI.*

Proof. As noted in the main text, the span of control of a solver increases if and only if her workers' knowledge increases. Hence, it is enough to show that all non-occupationally displaced solvers match with more knowledgeable workers post-AI than pre-AI.

The post-AI equilibrium can be either Type 1a or Type 2a. Suppose first that it is Type 1a. In this case, the employee matching functions pre- and post-AI are given by:

$$\int_{e_1^\star(z;\bar{z}_1^\star)}^{\bar{z}_1^\star} h(1-u)dG(u) = \int_z^1 dG(u) \text{ for } z \in S^\star = S_p^\star = [\bar{z}_1^\star, 1]$$
$$\int_{e(z;\hat{z})}^{\hat{z}} h(1-u)dG(u) = \int_z^1 dG(u) \text{ for } z \in S = [\hat{z}, 1]$$

Thus, if $z \in S^\star \cap S = [\bar{z}_1^\star, 1]$, then $\int_{e_1^\star(z;\bar{z}_1^\star)}^{\bar{z}_1^\star} h(1-u)dG(u) = \int_{e(z;\hat{z})}^{\hat{z}} h(1-u)dG(u)$. Since $\bar{z}_1^\star > \hat{z}$, this immediately implies that $e_1^\star(z;\bar{z}_1^\star) > e(z;\hat{z})$ for all $z \in S^\star \cap S = [\bar{z}_1^\star, 1]$.

Suppose instead that the post-AI equilibrium is Type 2a. In this case, the employee matching functions pre- and post-AI are given by:

$$\int_{e_2^\star(z;\hat{z}_2^\star)}^{\hat{z}_2^\star} h(1-u)dG(u) = \int_z^1 dG(u) \text{ for } z \in S^\star = S_p^\star = [\bar{z}_2^\star, 1]$$
$$\int_{e(z;\hat{z})}^{\hat{z}} h(1-u)dG(u) = \int_z^1 dG(u) \text{ for } z \in S = [\hat{z}, 1]$$

Thus, if $z \in S^\star \cap S = [\bar{z}_2^\star, 1]$, then $\int_{e_2^\star(z;\hat{z}_2^\star)}^{\hat{z}_2^\star} h(1-u)dG(u) = \int_{e(z;\hat{z})}^{\hat{z}} h(1-u)dG(u)$. Since $\hat{z}_2^\star > \hat{z}$, this immediately implies that $e_2^\star(z;\hat{z}_2^\star) > e(z;\hat{z})$ for all $z \in S^\star \cap S = [\bar{z}_2^\star, 1]$. $\square$

5.2 Negligible Amount of Compute: $\mu \to 0$

We now characterize the equilibrium with non-autonomous AI when $\mu \to 0$ and compare it to the case where compute is abundant relative to time. This comparison allows us to disentangle in a clean way the role that abundant compute is playing in our results. As in the main text, we focus on the case $h < h_0$.

5.2.1 Overview

Proposition 5.2. *Regarding non-autonomous AI:*

- *When $z_{\mathrm{AI}} \in W$, there exists $\bar{\mu}_w > 0$ such that if $\mu \in (0, \bar{\mu}_w)$, then there is a unique equilibrium with a non-autonomous AI. It is given as follows:*
  - *If $z_{\mathrm{AI}} \leq w(0)$, then the equilibrium allocation follows a pre-AI configuration and $r^\star = 0$.*
  - *If $z_{\mathrm{AI}} > w(0)$, then the equilibrium is as follows. There exist cutoffs $0 < \underline{z}_1^\star < \bar{z}_1^\star < 1$ such that:*

$$W_a^\star = [0, \underline{z}_1^\star],\ W_p^\star = [\underline{z}_1^\star, \bar{z}_1^\star],\ I^\star = \emptyset,\ S_p^\star = [\bar{z}_1^\star, 1],\ S_a^\star = \emptyset$$

These cutoffs satisfy:

$$\mu = \int_0^{\underline{z}_1^\star} h(1-z)dG(z),\ m_1^\star(\underline{z}_1^\star;\bar{z}_1^\star) = \bar{z}_1, \text{ and } \tfrac{1}{h} > \bar{z}_1^\star + \int_{\bar{z}_1^\star}^1 n(e_1^\star(z;\bar{z}_1^\star))dz$$

where $m_1^\star : [\underline{z}_1^\star, \bar{z}_1^\star] \to [\bar{z}_1^\star, 1]$ satisfies $\int_z^{\bar{z}_1^\star} h(1-u)dG(u) = \int_{m_1^\star(z;\bar{z}_1^\star)}^1 dG(u)$, $e_1^\star(z;\cdot) = (m_1^\star)^{-1}(z;\cdot)$. The equilibrium wage function, $w^\star$, and rental rate of compute, $r^\star$, are then given by:

$$w^\star(z) = \begin{cases} z_{\mathrm{AI}} - r^\star/n(z) & \text{if } z \in W_a^\star \\ m_1^\star(z;\bar{z}_1^\star) - w^\star(m_1^\star(z;\bar{z}_1^\star))/n(z) & \text{if } z \in W_p^\star \\ C^\star + \int_{\bar{z}_1^\star}^z n(e_1^\star(u;\bar{z}_1^\star))du & \text{if } z \in S_p^\star \end{cases}$$

where:

$$C^\star = \bar{z}_1^\star + \left(\frac{1}{1+n(\bar{z}_1^\star)}\right)\left(\frac{1}{h} - \bar{z}_1^\star - \int_{\bar{z}_1^\star}^1 n\big(e_1^\star(z;\bar{z}_1^\star)\big)dz\right) \text{ and } r^\star = C^\star - n(\underline{z}_1^\star)(\bar{z}_1^\star - z_{\mathrm{AI}}) \geq 0$$

- *When $z_{\text{AI}} \in \text{int}S$, there exists $\bar{\mu}_s > 0$ such that if $\mu \in (0, \bar{\mu}_s)$, then the unique equilibrium with a non-autonomous AI coincides with the equilibrium with an autonomous AI. That is, there exist cutoffs $0 < \underline{z}_w^\star < \bar{z}_w^\star < \hat{z}^\star$ such that:*

$$W_a^* = [\underline{z}_w^\star, \bar{z}_w^\star],\ W_p^* = [0, \underline{z}_w^\star] \cup [\bar{z}_w^\star, \hat{z}^\star],\ I^* = \emptyset,\ S_p^* = [\hat{z}^\star, 1] \ni z_{\text{AI}},\ S_a^* = \emptyset$$

where $\hat{z}^\star > \hat{z}$ and satisfies $\int_0^{\hat{z}^\star} h(1-z)dG(z) = \mu + \int_{\hat{z}^\star}^1 dG(z)$ and the cutoffs $(\underline{z}_w^\star, \bar{z}_w^\star)$, the wage $w^\star$, and the equilibrium price of compute $r^\star$ are as in Lemma 4.3 of Chapter 4.2.

Proof. See Chapter 5.2.2 below. □

As the proposition states, the equilibrium when $\mu \to 0$ shares both similarities and differences with the case where compute is abundant relative to time. In terms of similarities, in both scenarios, non-autonomous AI is used only when $z_{\text{AI}} > w(0)$; otherwise, the introduction of AI has no impact on the pre-AI equilibrium. Furthermore, when $z_{\text{AI}} \in W$, it remains the case that the least knowledgeable individuals are the ones who adopt AI.

Regarding differences, when $\mu \to 0$, the equilibrium price of compute is not necessarily zero. Additionally, when $z_{\text{AI}} \in \text{int}S$, it is no longer the least knowledgeable individuals who use AI, because AI is no longer the least knowledgeable solver. In this case, the equilibrium with non-autonomous AI is identical to the equilibrium with autonomous AI.

As established in the main text (Proposition 6), when compute is abundant, non-autonomous AI always benefits the least knowledgeable individuals relative to the pre-AI equilibrium and these benefits exceed those these individuals could obtain from autonomous AI (if the latter also raises their wages). In contrast, for the most knowledgeable individuals, a non-autonomous AI may lead to gains or losses compared to the pre-AI scenario, but, in any case, their wages with non-autonomous AI are always lower than with autonomous AI. As the following proposition shows, only the results regarding the least knowledgeable individuals generalize to the case where $\mu \to 0$, and only when $z_{\text{AI}} \in W$.

Proposition 5.3. *Suppose $\mu > 0$ but arbitrarily small. If $z_{\text{AI}} \in \text{int}W$, then $\exists\, \epsilon > 0$ such that $w^\star(z) \geq \max\{w(z), w^*(z)\}$ for all $z \in [0, \epsilon)$ (with strict inequality if $z_{\text{AI}} > w(0)$).*

Proof. See Chapter 5.2.2 below. □

To understand why the results regarding the wages of the most knowledgeable individuals do not extend to the case $\mu \to 0$, consider Figure 4.3 of Chapter 4.2. As illustrated by this figure, when $z_{\text{AI}} = 0.01$, $z_{\text{AI}} = 0.73$, and $\mu = 0.01$, the most knowledgeable individuals end up worse off with autonomous AI than with no AI. Moreover, at such low values of z_{AI}, non-autonomous AI is not used in equilibrium. Thus, here we have an example where the most knowledgeable individuals are strictly better off with non-autonomous AI (since it is not used in equilibrium) than with autonomous AI.

Intuitively, the result that wages of the most knowledgeable individuals with non-autonomous AI are always lower than with autonomous AI relies critically on the fact that with autonomy, the most knowledgeable individuals are necessarily winners from AI. However, as we explain in detail in Chapter 4.2, that result relies on compute being abundant relative to time.

To understand why the results regarding the wages of the least knowledgeable individuals do not necessarily hold when $z_{\text{AI}} \notin \text{int}W$, note that when z_{AI} is very high, only the most knowledgeable pre-AI workers use AI, and their wages increase as a result. This can negatively affect the wages of the least knowledgeable individuals because they match with solvers who are very similar to the most knowledgeable pre-AI workers.

5.2.2 Proof of Proposition 5.2

First, note that when $\mu \to 0$ and $z_{\text{AI}} \in \text{int}S$, the unique equilibrium with autonomous AI involves using AI exclusively as a solver (see Lemma 4.3 of Chapter 4.2). Since the equilibrium is efficient and autonomous AI can be used in more ways than non-autonomous AI, we conclude that the unique equilibrium with non-autonomous AI coincides with the one with autonomous AI in this case.

Hence, we focus on showing that (i) when $z_{\text{AI}} \leq w(0)$, there is a unique equilibrium that follows the pre-AI configuration, and (ii) when $z_{\text{AI}} > w(0)$ and $z_{\text{AI}} \in W$, the unique equilibrium is given as in the statement of the proposition.

• *When $z_{\text{AI}} \leq w(0)$, there is a unique equilibrium that follows the pre-AI configuration.*— First, let us show that if $z_{\text{AI}} \leq w(0)$, then there exists an equilibrium that follows a pre-AI configuration. Given that nobody uses AI, in equilibrium $r^\star = 0$. This immediately implies that, at the pre-AI wages, no firm has incentives to deviate. In particular, if a top-automated firm tried to hire a human with knowledge z, the most that firm would earn in profits is $n(z)[\max\{z, z_{\text{AI}}\} - w(z)] \leq 0$, where the inequality follows because $w(z) > z$ and $z_{\text{AI}} \leq w(0) < w(z)$ (as $w(z)$ is strictly increasing in z).

Now, let us show that this is the unique equilibrium. Note that $w(0) = \hat{z} - hw(\hat{z}) < \hat{z}$. Hence, if $z_{\text{AI}} \leq w(0)$, then $z_{\text{AI}} \in \text{int}W$. The fact that $z_{\text{AI}} \in \text{int}W$ coupled with $\mu > 0$ but arbitrarily small, immediately implies that if AI is used in equilibrium, the least knowledgeable individuals are the ones using AI; otherwise, there would be a violation of positive assortative matching. Furthermore, because $\mu > 0$ but arbitrarily small and there are no independent producers in the pre-AI equilibrium (as $h < h_0$), by continuity, there cannot be independent producers in the post-AI equilibrium either. Thus, if another equilibrium exists, it must have the following form:

$$W_a^\star = [0, \underline{z}_1^\star],\ W_p^\star = [\underline{z}_1^\star, \bar{z}_1^\star],\ I^\star = \emptyset,\ S_p^\star = [\bar{z}_1^\star, 1],\ S_a^\star = \emptyset,\ \text{where } 0 < \underline{z}_1^\star < \bar{z}_1^\star < 1. \tag{5.1}$$

Now, as explained in Chapter 2.6, the fact that human occupations follow the form in (5.1), also implies that the equilibrium wages and occupations of the humans who do not work with AI—those in $W_p^\star$ and $S_p^\star$—must coincide with the pre-AI equilibrium of the auxiliary κ-economy with a wage schedule $\tilde{w}(z; \underline{z}_1^\star)$ when $\kappa = \inf W_p^\star = \underline{z}_1^\star$ (characterized in Claim 2.5). Moreover, because the equilibrium wage function must be continuous in z, it must be that $\tilde{w}(\underline{z}_1^\star;\underline{z}_1^\star) = z_{\mathrm{AI}} - r^\star/n(\underline{z}_1^\star)$, where the right-hand side of this last equation comes from the zero profit condition of top-automated firms. Rearraging terms, we then obtain $r^\star = n(\underline{z}_1^\star)(z_{\mathrm{AI}} - \tilde{w}(\underline{z}_1^\star;\underline{z}_1^\star))$. However, this also implies that $r^\star < 0$ given that $\tilde{w}(\underline{z}_1^\star;\underline{z}_1^\star) > \tilde{w}(0;0) = w(0) \geq z_{\mathrm{AI}}$. Since $r^\star$ cannot be strictly negative, we obtain the desired contradiction. □

• *When $z_{\mathrm{AI}} > w(0)$ and $z_{\mathrm{AI}} \in W$, the unique equilibrium is given as in the statement of the proposition.*— First, it is clear that the pre-AI configuration cannot be an equilibrium in this case. If so, then $r^\star = 0$ (as nobody would be using AI), so a top-automated firm could hire an individual with $z = 0$ and obtain $(1/h)(z_{\mathrm{AI}} - w(0)) > 0$.

Now, following an identical reasoning as in the previous step, we have that $z_{\mathrm{AI}} \in W$ and $\mu > 0$ but arbitrarily small imply that if AI is used in equilibrium, the least knowledgeable individuals are the ones using AI. Furthermore, because $\mu > 0$ but arbitrarily small and there are no independent producers in the pre-AI equilibrium (as $h < h_0$), by continuity, there cannot be independent producers in the post-AI equilibrium either. Thus, any equilibrium must have the following form:

$$W_a^\star = [0, \underline{z}_1^\star],\ W_p^\star = [\underline{z}_1^\star, \bar{z}_1^\star],\ I^\star = \emptyset,\ S_p^\star = [\bar{z}_1^\star, 1],\ S_a^\star = \emptyset,\ \text{where } 0 < \underline{z}_1^\star < \bar{z}_1^\star < 1.$$

where the cutoffs satisfy:

$$\mu = \int_0^{\underline{z}_1^\star} h(1-z)dG(z),\ m_1^\star(\underline{z}_1^\star;\bar{z}_1^\star) = \bar{z}_1,\ \text{and } \tfrac{1}{h} > \bar{z}_1^\star + \int_{\bar{z}_1^\star}^1 n(e_1^\star(z;\bar{z}_1^\star))dz \tag{5.2}$$

and $m_1^\star : [\underline{z}_1^\star, \bar{z}_1^\star] \to [\bar{z}_1^\star, 1]$ is given by $\int_z^{\bar{z}_1^\star} h(1-u)dG(u) = \int_{m_1^\star(z;\bar{z}_1^\star)}^1 dG(u)$, and $e_1^\star(z;\cdot) = (m_1^\star)^{-1}(z;\cdot)$. The optimality and zero-profit condition of the different firms then yield the candidate wage function:

$$w^\star(z) = \begin{cases} w^\star(z) = z_{\mathrm{AI}} - r^\star/n(z) & \text{if } z \in W_a^\star \\ m_1^\star(z;\bar{z}_1^\star) - w^\star(m_1^\star(z;\bar{z}_1^\star))/n(z) & \text{if } z \in W_p^\star \\ C^\star + \int_{\bar{z}_1^\star}^z n(e_1^\star(u;\bar{z}_1^\star))du & \text{if } z \in S_p^\star \end{cases}$$

Because the equilibrium wage function must be continuous, it must be that $\lim_{z\uparrow\underline{z}_1^\star} w^\star(z) = \lim_{z\downarrow\underline{z}_1^\star} w^\star(z)$ and $\lim_{z\uparrow\bar{z}_1^\star} w^\star(z) = \lim_{z\downarrow\bar{z}_1^\star} w^\star(z)$. These two conditions together imply that:

$$C^\star = \bar{z}_1^\star + \left(\tfrac{1}{1+n(\bar{z}_1^\star)}\right)\left(\tfrac{1}{h} - \bar{z}_1^\star - \int_{\bar{z}_1^\star}^1 n\big(e_1^\star(z;\bar{z}_1^\star)\big)dz\right) \text{ and } r^\star = C^\star - n(\underline{z}_1^\star)(\bar{z}_1^\star - z_{\mathrm{AI}}) \geq 0$$

It is not difficult to prove that there exists a unique pair of cutoffs $(\underline{z}_1^\star, \bar{z}_1^\star)$ that satisfies the equality conditions in (5.2). Hence, if the equilibrium exists, it is unique.

Finally, to show existence, we simply need to verify that the equilibrium cutoffs $(\underline{z}_1^\star, \bar{z}_1^\star)$ satisfy $\frac{1}{h} > \bar{z}_1^\star + \int_{\bar{z}_1^\star}^1 n(e_1^\star(z;\bar{z}_1^\star))dz$, and that the rental rate of compute $r^\star = C^\star - n(\underline{z}_1^\star)(\bar{z}_1^\star - z_{\mathrm{AI}})$ is weakly positive. However, we know that $\underline{z}_1^\star \to 0$ as $\mu \to 0$, which immediately implies that $\bar{z}_1^\star \to \hat{z}$ (where $\hat{z}$ is the knowledge cutoff between workers and solvers in the pre-AI world), and $e_1^\star(z;\bar{z}_1^\star) \to e(z;\hat{z})$. Thus, as $\mu \to 0$, condition $\frac{1}{h} > \bar{z}_1^\star + \int_{\bar{z}_1^\star}^1 n(e_1^\star(z;\bar{z}_1^\star))dz$ becomes $\frac{1}{h} > \hat{z} + \int_{\hat{z}}^1 n(e(z;\hat{z}))dz$ (which we know is satisfied since $h < h_0$). Furthermore, condition $r^\star = C^\star - n(\underline{z}_1^\star)(\bar{z}_1^\star - z_{\mathrm{AI}})$ becomes $r^\star = (1/h)(z_{\mathrm{AI}} - w(0)) > 0$, where the last inequality follows because $z_{\mathrm{AI}} > w(0)$. □

Proof of Proposition 5.3

For ease of exposition, we split the proof into two bullets.

• *If $z_{\text{AI}} \in \text{int}W$, then $\exists\ \epsilon > 0$ such that $w^\star(z) \geq w(z)$ for all $z \in [0, \epsilon)$ (with strict inequality if $z_{\text{AI}} > w(0)$).*— If $z_{\text{AI}} \leq w(0)$, then $w^\star(z) = w(z)$ for all $z \in [0, 1]$, so the statement holds trivially. Thus, it only remains to show that if $z_{\text{AI}} > w(0)$ and $z_{\text{AI}} \in \text{int}W$, then $w^\star(0) > w(0)$. The statement of the proposition then follows by continuity.

To show that $w^\star(0) > w(0)$, note that according to Proposition 5.2, $r^\star(\mu) = C^\star(\mu) - n(\underline{z}_1^\star(\mu))(\bar{z}_1^\star(\mu) - z_{\text{AI}})$, which can be written as $r^\star(\mu) = n(\underline{z}_1^\star(\mu))(z_{\text{AI}} - \tilde{w}(\underline{z}_1^\star(\mu); \underline{z}_1^\star(\mu)))$, where $\tilde{w}(\kappa; \kappa)$ is the equilibrium wage function of the pre-AI equilibrium of the auxiliary κ-economy (characterized in Claim 2.5).[1] Thus,

$$w^\star(0; \mu) = z_{\text{AI}} - \tfrac{n(\underline{z}_1^\star(\mu)))}{n(0)}(z_{\text{AI}} - \tilde{w}(\underline{z}_1^\star(\mu); \underline{z}_1^\star(\mu)))$$

Moreover, we know that $\underline{z}_1^\star(\mu) \to 0$ as $\mu \to 0$. Hence,

$$w^\star(0; 0) = z_{\text{AI}} - \tfrac{n(0)}{n(0)}(z_{\text{AI}} - \tilde{w}(0; 0)) = \tilde{w}(0; 0) = w(0)$$

where $w(0)$ is the wage of individual $z = 0$ in the pre-AI equilibrium. Thus, to show $w^\star(0; \mu) > w(0)$ when $\mu \to 0$, it suffices to show that $(d/d\mu)w^\star(0; \mu) > 0$ when evaluated at $\mu = 0$. Differentiating:

$$\left(\tfrac{d}{d\mu} w^\star(0; \mu)\right)\Big|_{\mu=0} = \left(\left[\tfrac{d}{d\kappa}\tilde{w}(\kappa; \kappa)\right]\Big|_{\kappa=0} - z_{\text{AI}} + w(0)\right)\left(\tfrac{d}{d\mu}\underline{z}_1^\star\right)\Big|_{\mu=0}$$

It is then not difficult to prove that $\underline{z}_1^\star(\mu)$ is strictly increasing in μ and that $\frac{d}{d\kappa}\tilde{w}(\kappa; \kappa)|_{\kappa=0} > \hat{z} - w(0)$. Thus,

$$\left(\tfrac{d}{d\mu} w^\star(0; \mu)\right)\Big|_{\mu=0} = \left(\left[\tfrac{d}{d\kappa}\tilde{w}(\kappa; \kappa)\right]\Big|_{\kappa=0} - z_{\text{AI}} + w(0)\right)\left(\tfrac{d}{d\mu}\underline{z}_1^\star\right)\Big|_{\mu=0} > (\hat{z} - z_{\text{AI}})\left(\tfrac{d}{d\mu}\underline{z}_1^\star\right)\Big|_{\mu=0} \geq 0$$

given that $z_{\text{AI}} \in W$ (so $z_{\text{AI}} \leq \hat{z}$).

• *If $z_{\text{AI}} \in \text{int}W$, then $\exists\ \epsilon > 0$ such that $w^\star(z) > w^*(z)$ for all $z \in [0, \epsilon)$.*— Note that as $\mu \to 0$, we have that $w^*(0; \mu) \to w(0)$ and that $w^\star(0; \mu) = w(0)$. Furthermore, by the previous bullet, we know that $w^\star(0; \mu)$ is strictly increasing in μ. Hence, to show the desired result, it suffices to show that when $z_{\text{AI}} \in \text{int}W$, $w^*(0; \mu)$ is strictly decreasing in μ. This immediately implies that $w^*(0; \mu) < w^\star(0; \mu)$ for $\mu > 0$ but $\mu \to 0$, so statement follows by continuity.

To show that $w^*(0; \mu)$ is strictly decreasing in μ, note that according to Lemma 4.2, $w^*(0; \mu) = \hat{z}^*(\mu) - hw^*(\hat{z}^*(\mu); \mu) = \hat{z}^*(\mu) - hC^*(\mu)$, where:

$$\begin{aligned} C^*(\mu) = \hat{z}^*(\mu) + \left(\tfrac{1}{1+n(\hat{z}^*(\mu))}\right)\Big[\tfrac{1}{h} - \hat{z}^*(\mu) - n(z_{\text{AI}})(\bar{z}_s^*(\mu) - \underline{z}_s^*(\mu)) \\ - \textstyle\int_{\hat{z}^*(\mu)}^{\underline{z}_s^*(\mu)} n(e_-^*(z; \hat{z}^*(\mu)))dz - \int_{\bar{z}_s^*(\mu)}^{1} n(e_+^*(z; \hat{z}^*(\mu)))dz\Big] \end{aligned}$$

[1]Here we are making explicit that the equilibrium variables of the post-AI world (e.g., $r^\star(\mu)$, $C^\star(\mu)$, $\underline{z}_1^\star(\mu)$) depends on the available compute μ.

Using that (i) $\hat{z}^*(\mu) \to \hat{z}$, $\underline{z}_s^*(\mu) - \overline{z}_s^*(\mu) \to 0$ as $\mu \to 0$ (where $\hat{z}$ is the cutoff between workers and solvers of the pre-AI equilibrium), and (ii) $e_-(\hat{z}^*; \hat{z}^*) = 0$ and $e_-(\underline{z}_s^*; \hat{z}^*) = e_+(\overline{z}_s^*; \hat{z}^*) = z_{\text{AI}}$, we obtain that:

$$\left(\frac{d}{d\mu} w^*(0;\mu)\right)\Big|_{\mu=0} = \left[\frac{1-h-\hat{z}+hw(\hat{z})}{(1-\hat{z})(1+h(1-\hat{z}))} + h\left(\frac{1}{1+n(\hat{z}^*(\mu))}\right)\int_{\hat{z}}^{1}\left(\frac{\partial e_+(z;\hat{z}^*)/\partial \hat{z}^*}{h(1-e_+(z;\hat{z}^*))^2}\right)\Big|_{\hat{z}^*=\hat{z}} dz\right]\left(\frac{d}{d\mu}\hat{z}^*(\mu)\right)\Big|_{\mu=0} < 0$$

where the last inequality follows because (i) $\hat{z}^*(\mu)$ is strictly decreasing in μ, (ii) $\partial e_+(z;\hat{z}^*)/\partial \hat{z}^* > 0$ for all feasible z, and:

$$\frac{1-h-\hat{z}+hw(\hat{z})}{(1-\hat{z})(1+h(1-\hat{z}))} > \frac{1-h}{1+h(1-\hat{z})} > 0$$

where the first inequality follows because $w(\hat{z}) > \hat{z}$. $\square$

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